# Innovations in Chemical Engineering & Technology

## Course Content – Part I

- General outline of innovations in the chemical industry
- Innovation – meaning and concept, business case, opportunities for innovation in CPI, barriers and challenges, how to innovate, innovative chemical engineering solutions, examples
- Innovation – types, fields, stages, strategies, metrics, phases, categories, levers, framework, protection of innovations, types of intellectual property, introduction to patents
- Technology readiness levels (TRL)
- Examples of innovations in chemical and allied industry (in concept, process, catalyst, strategy, product, reactor, separator, business model and application)

# Innovation in Chemical Engineering

www.aiche.org/cep February 2020

- ❑ **Innovation – what does it mean?**
  - Innovation satisfies the customer's need in a new way
  - Innovation is not just about new ideas
  - Innovation is not always expensive

- ❑ **Is the CPI innovative?**
  - Scale-up, plant design, operation and distribution
  - Industry 4.0, IIoT, data analytics, cloud connectivity
  - Robotics, machine-to-machine communications

- ❑ **Companies must innovate to remain competitive**

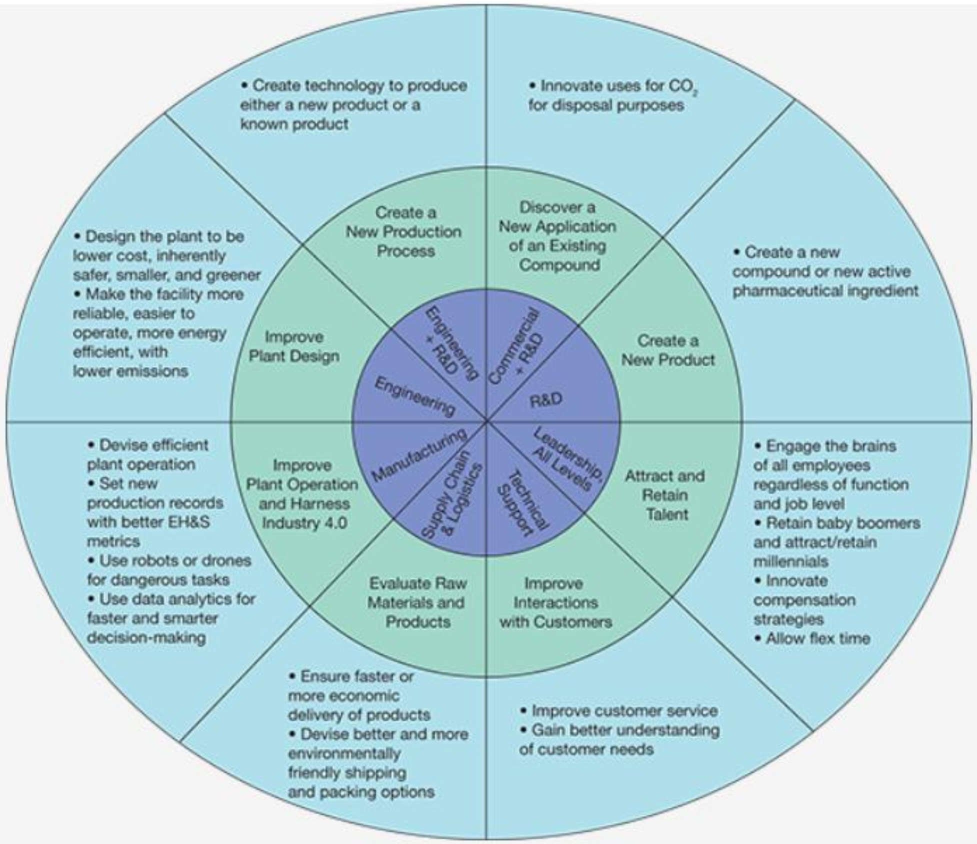
- ❑ **Companies must understand customer needs**

- ❑ **What are barriers to innovation?**
  - No risk-taking, no foresight, no collaboration
  - Excessive use of tools can stifle innovation

- **Examples of innovative chemical engineering solutions**
  - Process Intensification
  - Alternative/Clean energy systems
  - Green chemistry/technology/engineering
  - Automation systems
  - Nanotechnology and nanomaterials
  - Advanced pharmaceuticals
  - Sustainable construction materials
  - Innovations in process safety

- **Examples of innovations in the energy sector**
  - Alternative fuels, batteries, solar panels, materials

- **Examples of innovations in food and agriculture**
  - Ammonia production, food waste valorisation
  - Sustainable agriculture

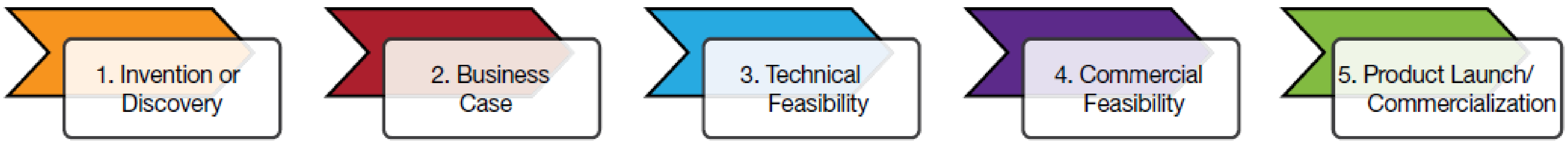



**▲Figure 1.** Opportunities abound for innovation in the chemical process industries (CPI). In the innermost (purple) circle are a variety of job functions a chemical engineer may fulfill at their company. The next (green) circle shows innovation opportunities. The outermost circle (blue) gives more specific examples of how to go about innovating.

# Integrating Process Safety and Innovation

www.aiche.org/cep October 2016

## Phased Gate Process for Innovation Projects




1. Invention or discovery — Something new has been found!

2. Business case — What unmet customer or market needs will this product fulfill?

3. Technical feasibility — By what route, mechanism, or process can this product be made?

4. Commercial feasibility — Can this product be made profitably? Will people buy it for a price that makes it commercially attractive?

5. Product launch/commercialization — Where will this product be made, and in what quantities?

# How to Innovate During a Crisis

www.aiche.org/cep August 2020

# The Innovation Process

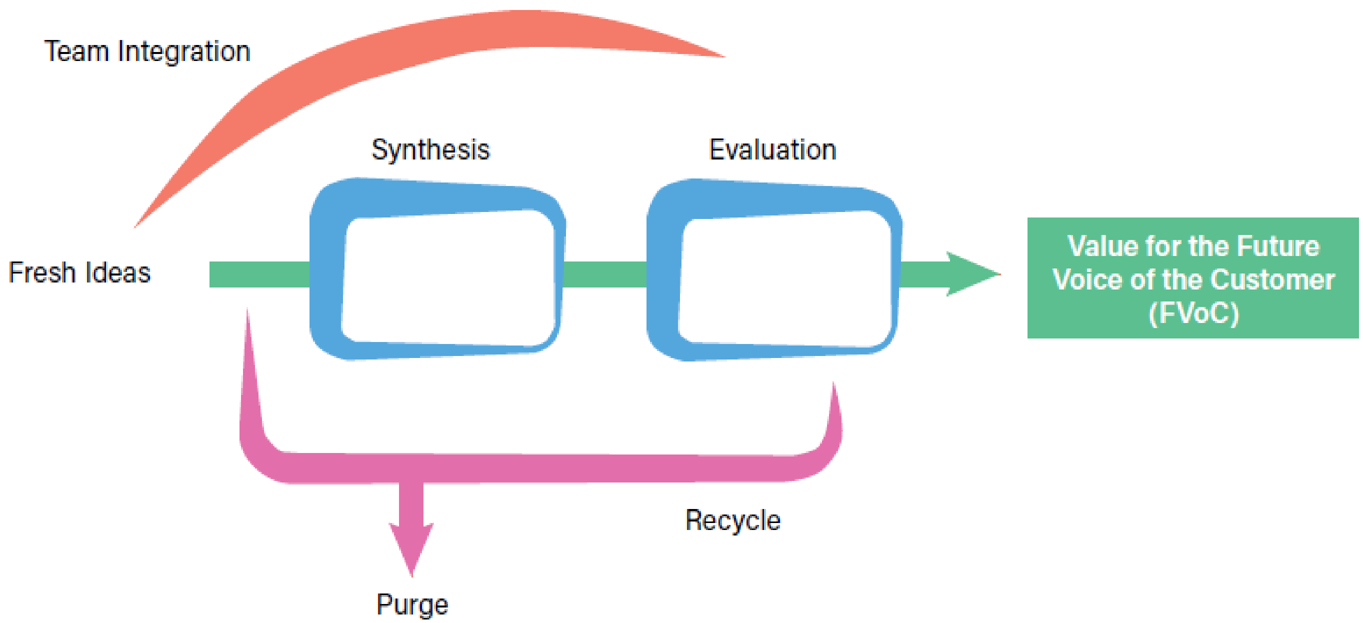



**Figure 1.** The innovation process is like a basic chemical engineering process. In chemical engineering, to create the desired product, we send a feed stream through a reactor. This is akin to sending a feed of ideas through a creative synthesis and evaluation process. Rejected ideas can be recycled to the start of the innovation process, just as unreacted material can be recycled back to the reactor. In a chemical process, the goal is to create a valuable product. In the innovation process, the goal is to create a product or service that meets the demand of the future voice of the customer (FVoC).

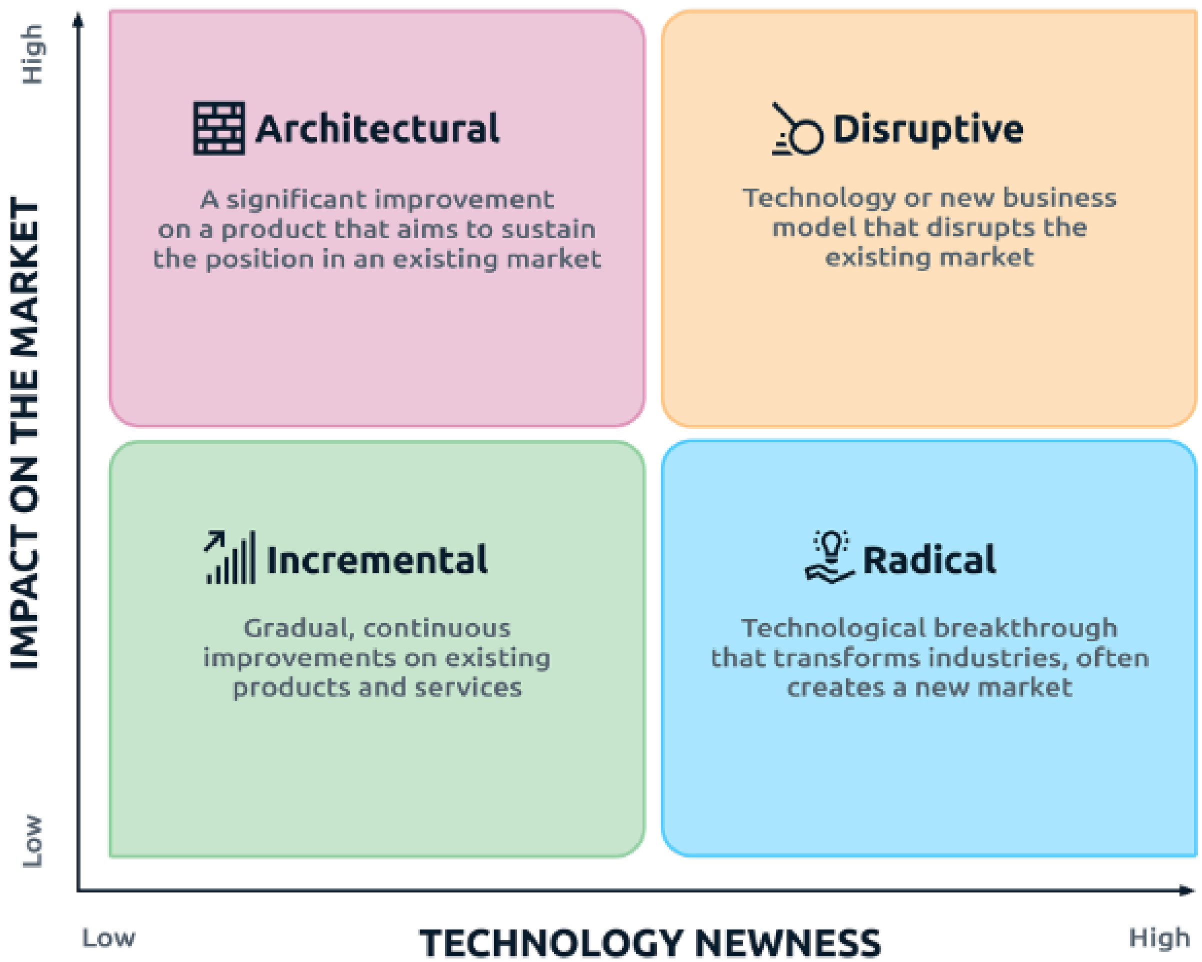
# Innovation Explained – Definition, Types & Meaning

Innovation is the specific function of entrepreneurship, whether in an existing business, a public service institution, or a new venture started by a lone individual in the family kitchen. It is the means by which the entrepreneur either creates new wealth-producing resources or endows existing resources with enhanced potential for creating wealth. – **Peter Drucker**

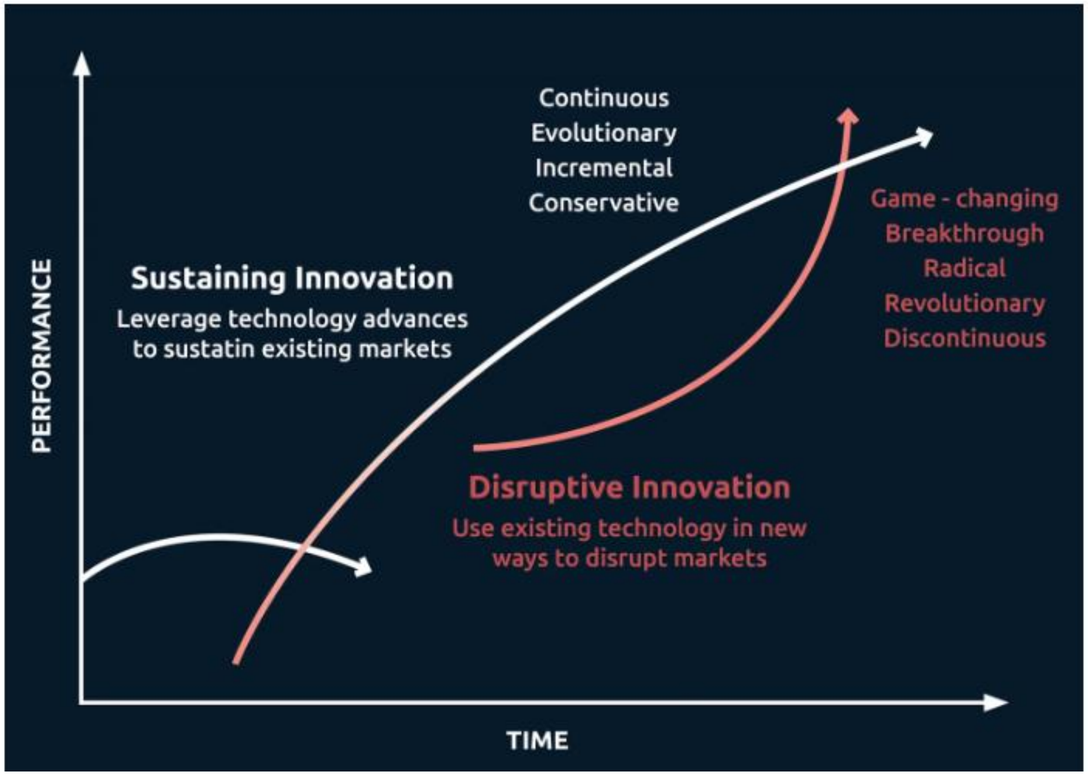
## 8 Fields of Innovation

1. Product & Product Performance Innovation
2. Technology Innovation
3. Business Model Innovation
4. Organizational Innovation
5. Process Innovation
6. Marketing / Sales – New Channel Innovation
7. Network Innovation
8. Customer Engagement / Retention

# 4 Types of Innovation



# Disruptive Innovation



## 10 Types of Innovation (Based on Applications)

1- Product Innovation

2- Service Innovation

3- Process Innovation

4- Business Model Innovation

5- Open Innovation

6- Sustaining Innovation

7- Technology Innovation

8- Value Innovation

9- Breakthrough Innovation

10- Digital Innovation

## 5 Stages of Corporate Innovation

1. Ideation & Idea generation
2. Evaluation
3. Testing and Experimenting
4. Development and Implementation
5. Optimization & Scaling

## 4 Types of Innovation Strategies

**Proactive**

**Active**

**Reactive**

**Passive**

## How to Measure Innovations?

1. Return on innovation investment
2. Time to market
3. Customer feedback and satisfaction
4. Employee engagement

## How to Protect Innovations?

1. Patents
2. Trademarks
3. Copyrights
4. Trade Dress
5. Trade Secrets

# Innovation in the Chemical Industry

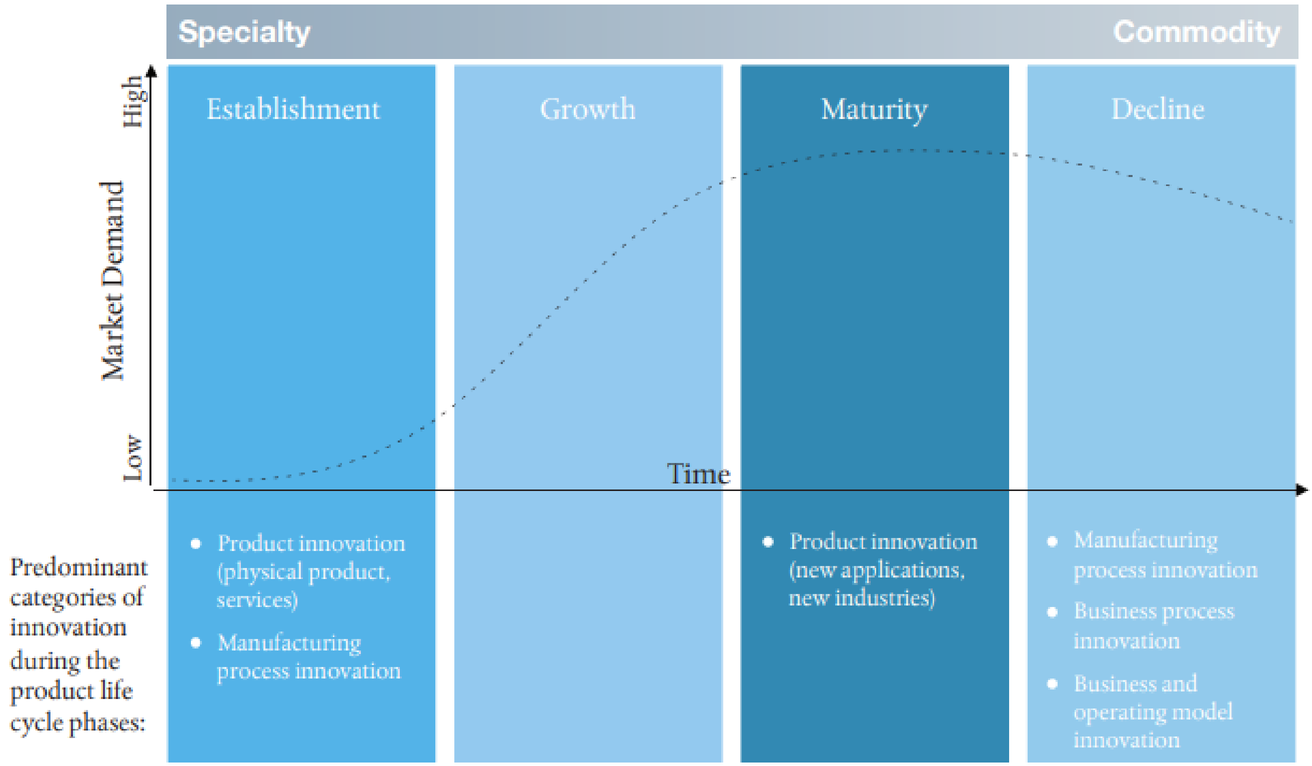
## Innovation in the chemical industry can be clustered into four main categories: Product innovation, manufacturing process innovation, business process innovation and business & operating model innovation.

Source: Stratley

| Categories of innovation | | Examples |
|---|---|---|
| Product and its application | Physical | Improved chemical properties of a material |
| | Service | Customized material selection support |
| Manufacturing process | | New synthesis route for chemical product which saves 30% of energy |
| Business process | | Improved human resource process to select better talents |
| Business and operating model | | Separate commodity business (besides specialty business) |

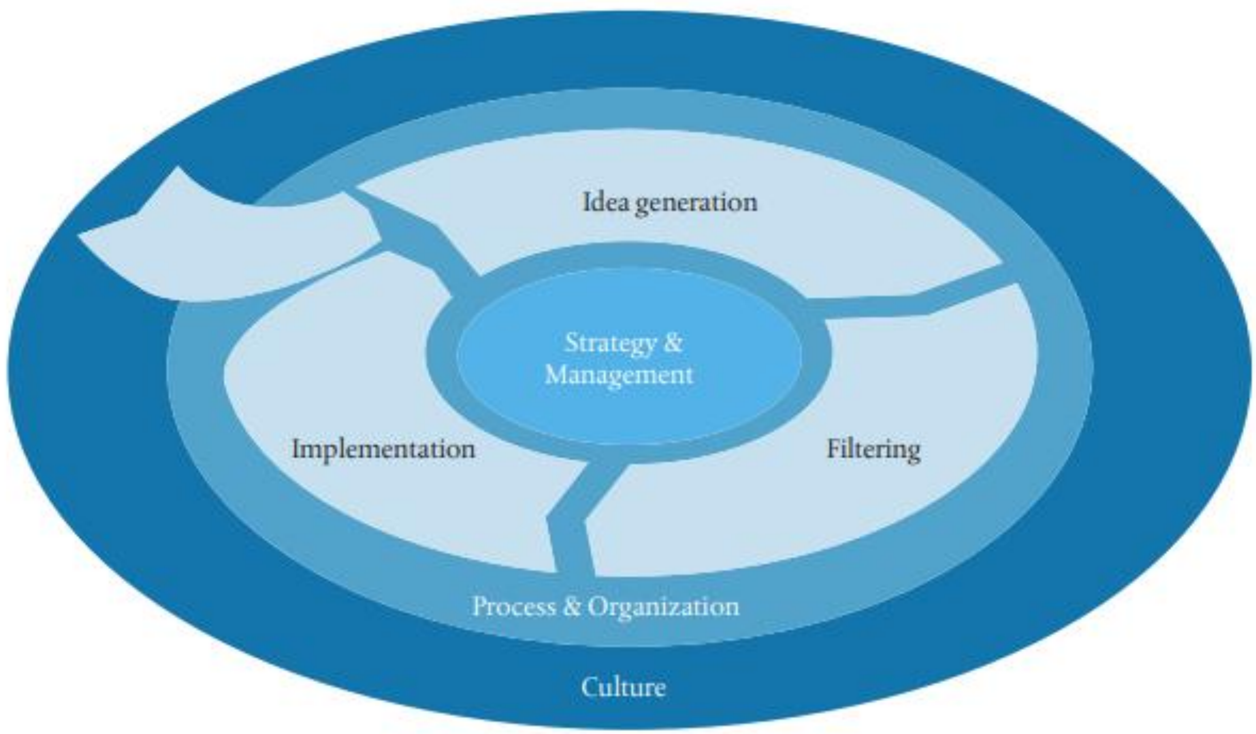
**During the four phases of the product life cycle, different innovation categories are leveraged to boost the business.**

Source: Stratley



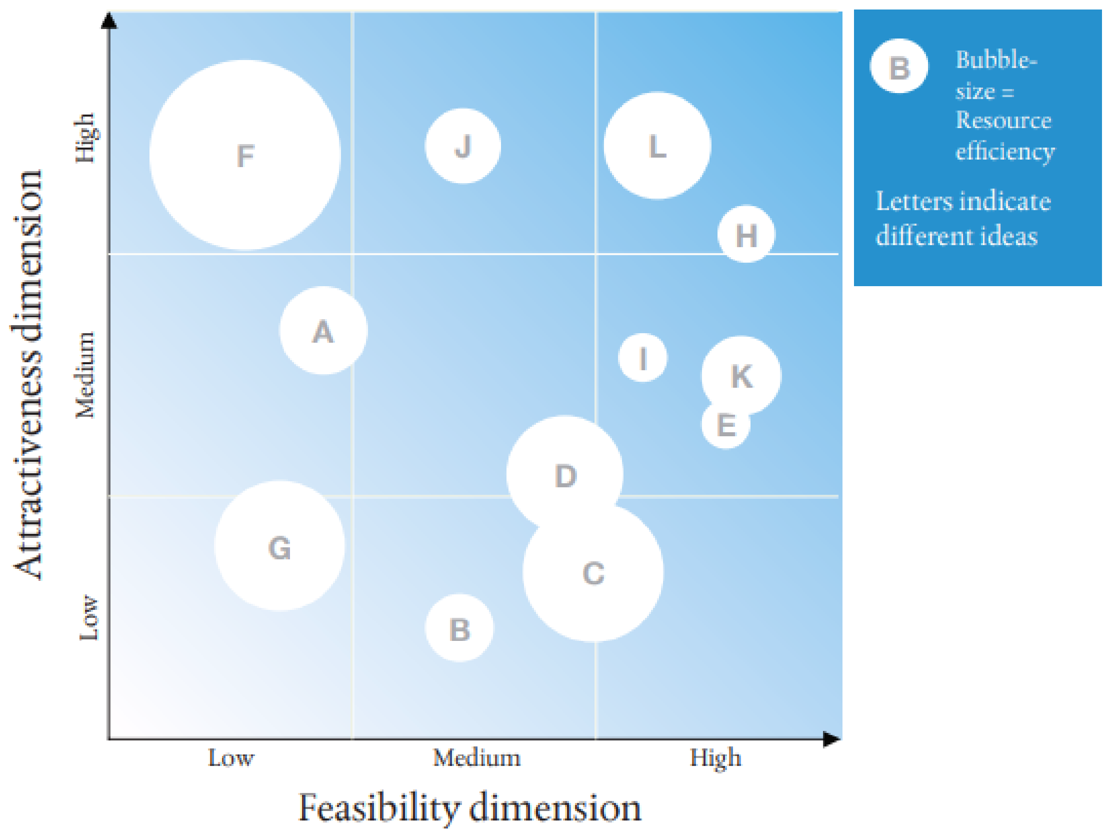
**Successful innovation depends on three major innovation levers: Strategy & Management, Process & Organization (with the 3 phases: idea generation, filtering and implementation) and Culture.**

Source: Stratley



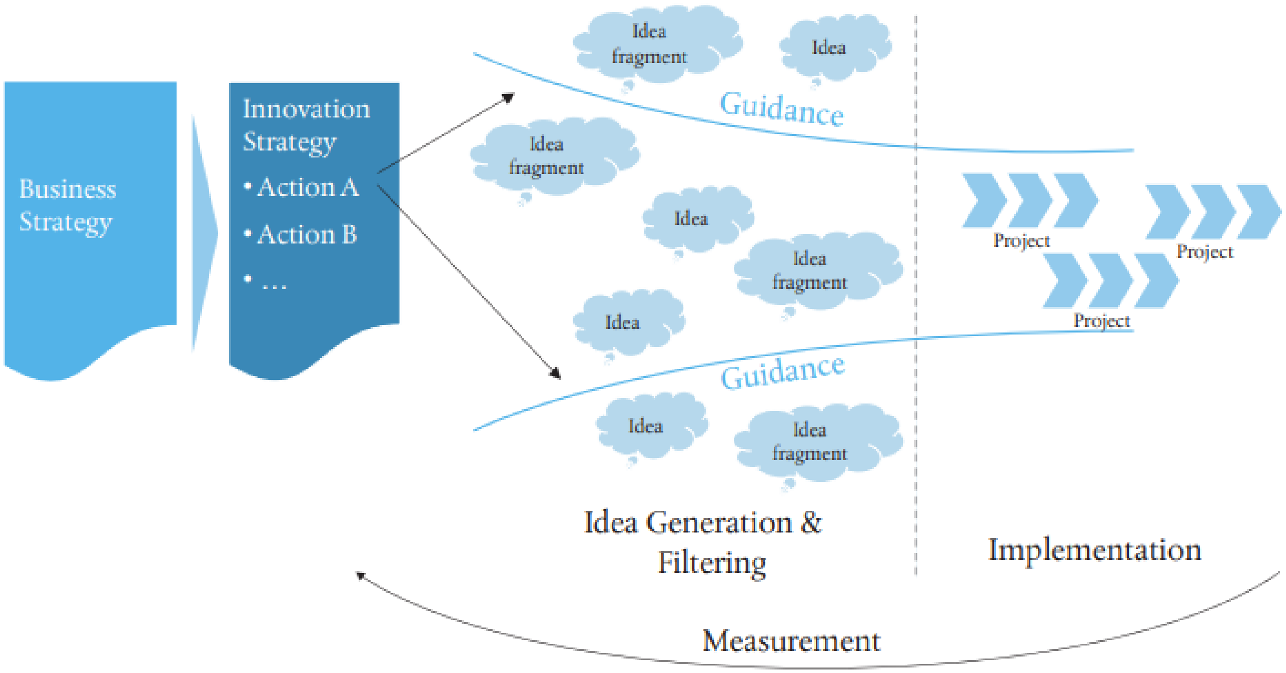
**Filter dimension matrix to comprehensively visualize the filter dimensions “attractiveness”, “feasibility” and “resource efficiency” of different ideas and thereby help to decide which projects to pursue.**

Source: Stratley



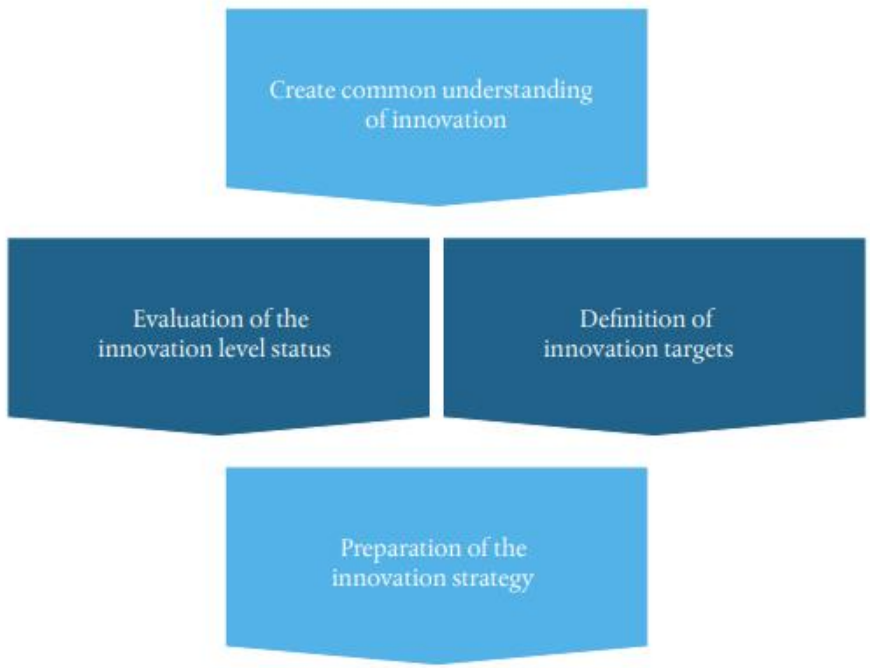
**The innovation strategy is based on and aligned to the business strategy. The innovation strategy defines actions and guiding principles for innovation.**

Source: Stratley



**When setting up an innovation strategy, a common understanding on the definition of innovation is key. After evaluating the status and defining the innovation targets, the innovation strategy can be prepared.**

Source: Stratley

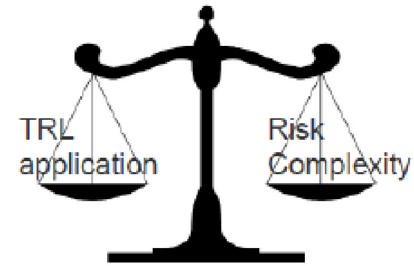


# Technology Readiness Levels

# Technology Readiness Level (TRL)

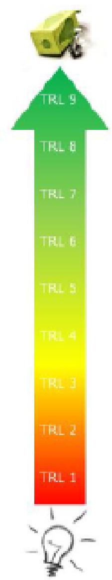
- Technology readiness levels (TRLs) - Measurement of the maturity level of a particular technology
- TRLs are based on a scale from 1 to 9, with 9 being the most mature technology.
- Systematic addressing of TRLs is required, allowing a technology to evolve from conception through to research, development and deployment.
- Universities, along with government funding sources, focus on TRLs 1-4, while the private sector focuses on TRLs 7-9.
- The term 'Valley of Death' represents the often neglected addressing of TRLs 4 through to 7, where neither academia nor the private sector prioritize investment.

# Purpose of TRL



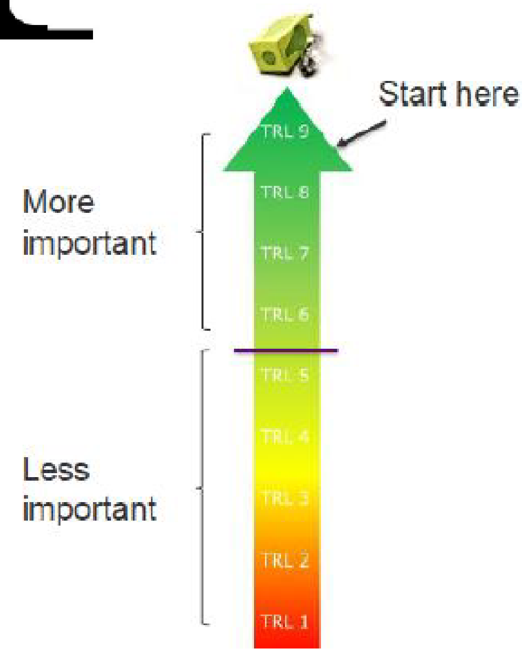

**Communication tool**

More objective assessment of the development level between stakeholders



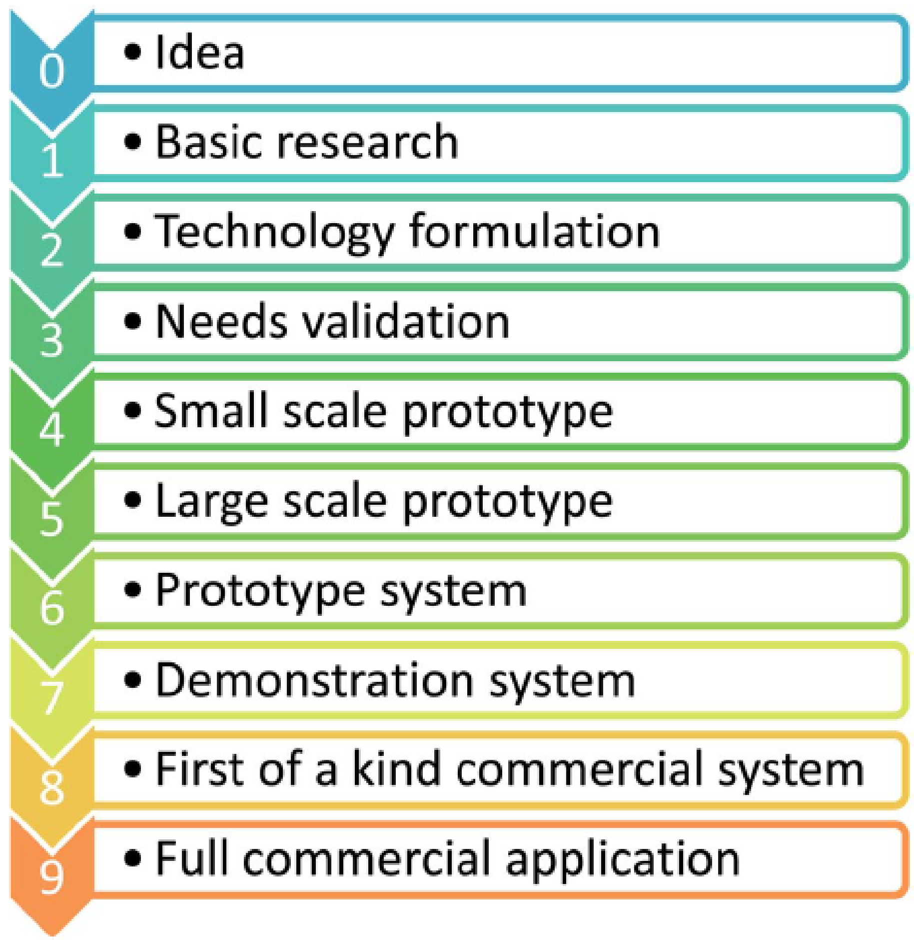

**Development roadmap**

- Minimize risk in the development
- Develop products that are fit for purpose
- Encourage real-world testing and iteration
- Introduce "reality checks" in the development process




- Provides a common understanding of technology status
- Used to make decisions concerning technology funding
- Used to make decisions concerning transition of technology

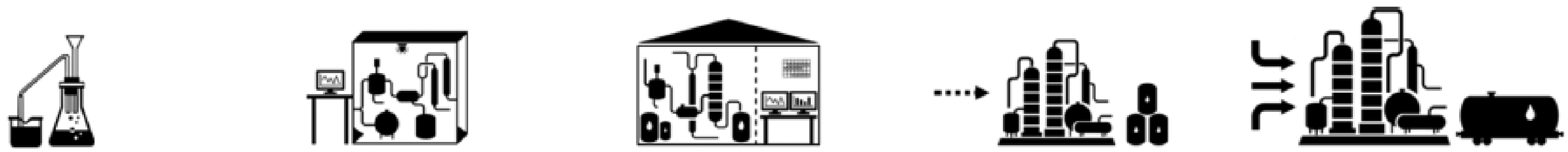
Technology Readiness Levels for Chemical Engineers



# Technology Readiness Levels for the Chemical Industry

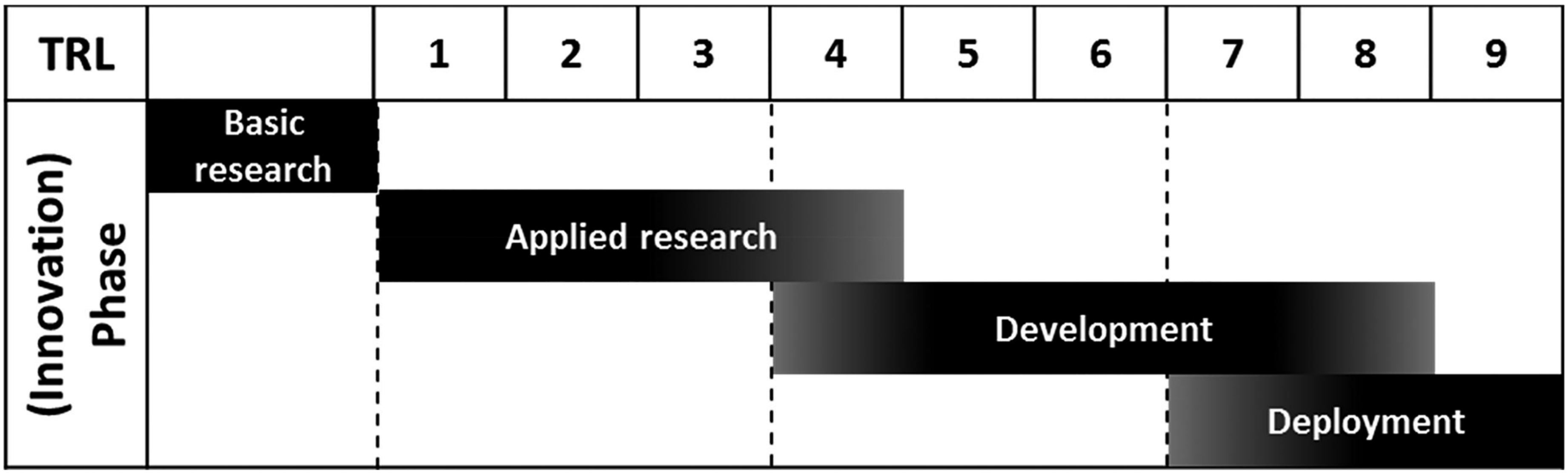
Ind. Eng. Chem. Res. 2019, 58, 6957−6969

# Tasks, goals and future characteristics of plant types in chemical (or process) industry

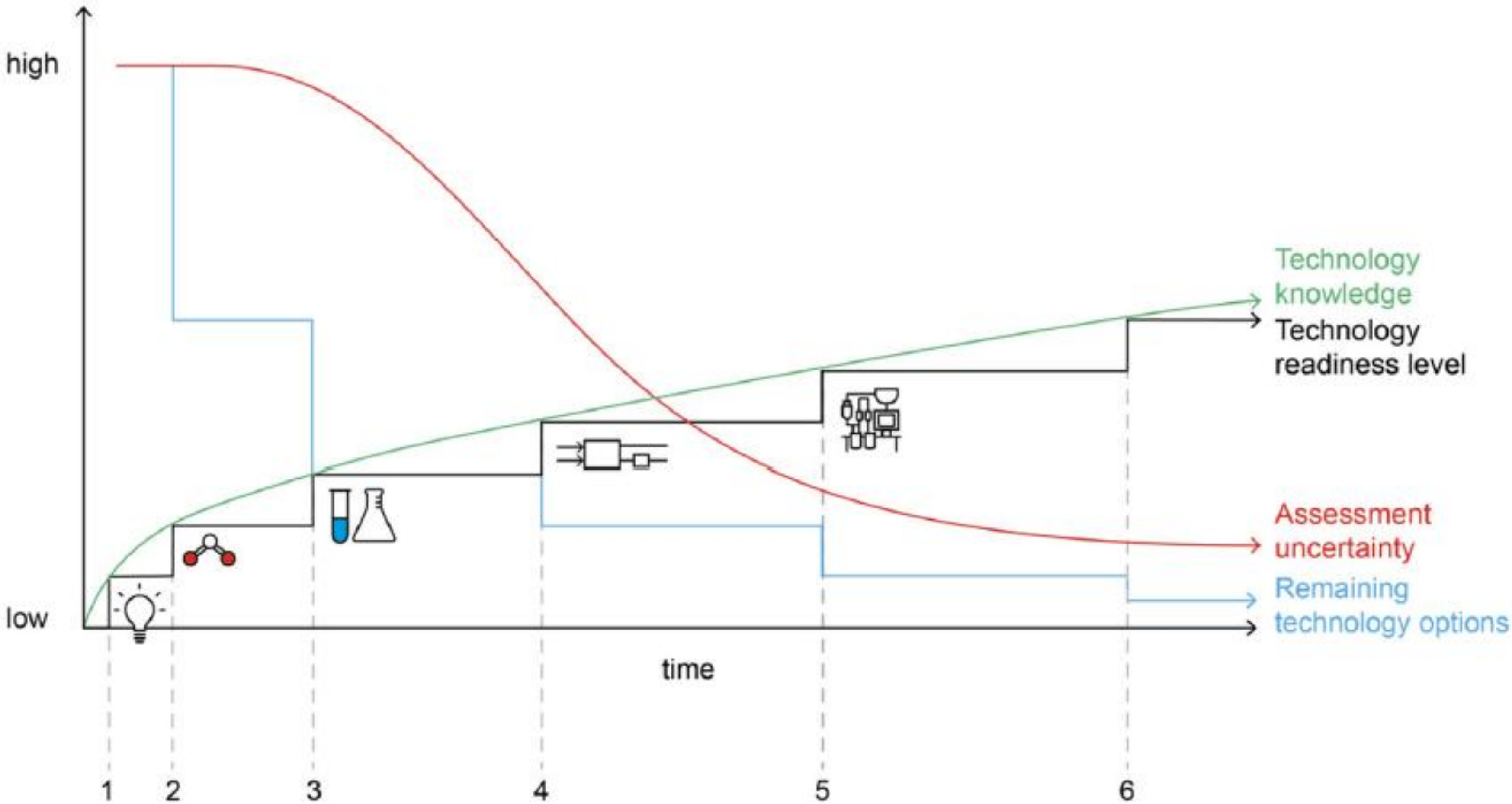


| | Laboratory bench | Mini plant | Pilot plant | Demo plant | Full-scale plant |
|---|---|---|---|---|---|
| Typical tasks and goals | • Analysis of (basic) kinetic behavior<br>• Analysis of thermodynamic properties | • Connecting multiple process elements<br>• Implementation of circular flows, product separation | • Analysis of dynamic behavior (start-up, shutdown)<br>• Analysis of catalyst lifetime<br>• Process optimization<br>• Logistics of feed, product, samples | • Obtain general proof that a technology is suitable for the targeted (economic) purpose<br>• Producing product for an initial market | • Producing product that is sold, answering to market demands |
| Further characteristic elements | • Using (small-scale) laboratory equipment<br>• Operations often include a substantial degree of manual labor | • Often includes switching from laboratory glass to steel equipment<br>• Certain degree of process automation | • Usually includes a significant increase of equipment size and plant capacity<br>• Often needs change of location | • Integration in an industrial value-chain | • Contributes to the economic sustainability of an economic entity, (*i.e.* making profit) |

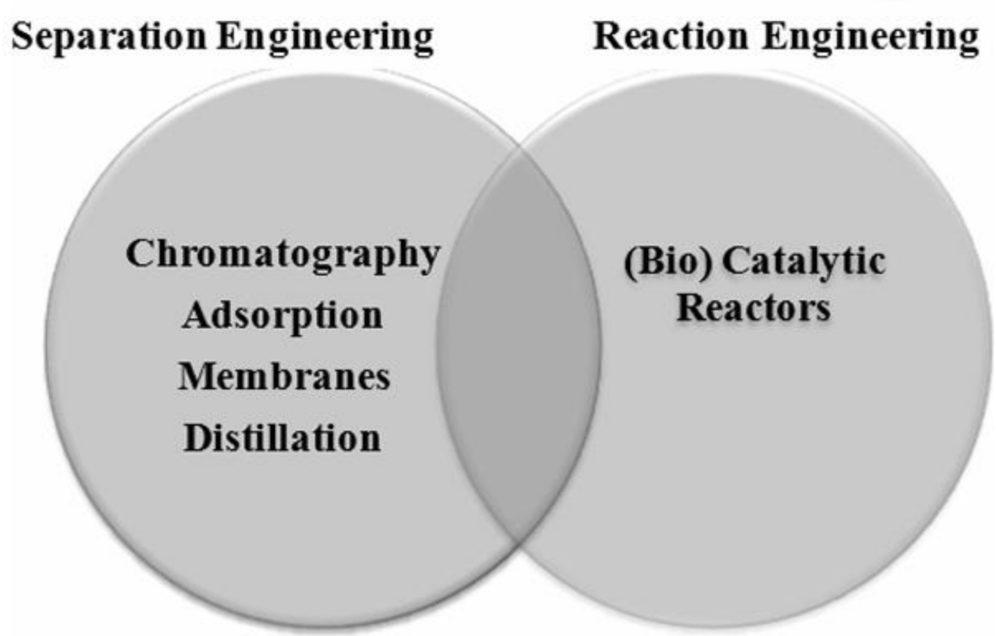
## TRLs Attributed to Innovation Phases



Relations between Technology knowledge, TRLs, Assessment uncertainty & Remaining Technology Options



# INNOVATIONS IN SEPARATION BY PROCESS INTENSIFICATION



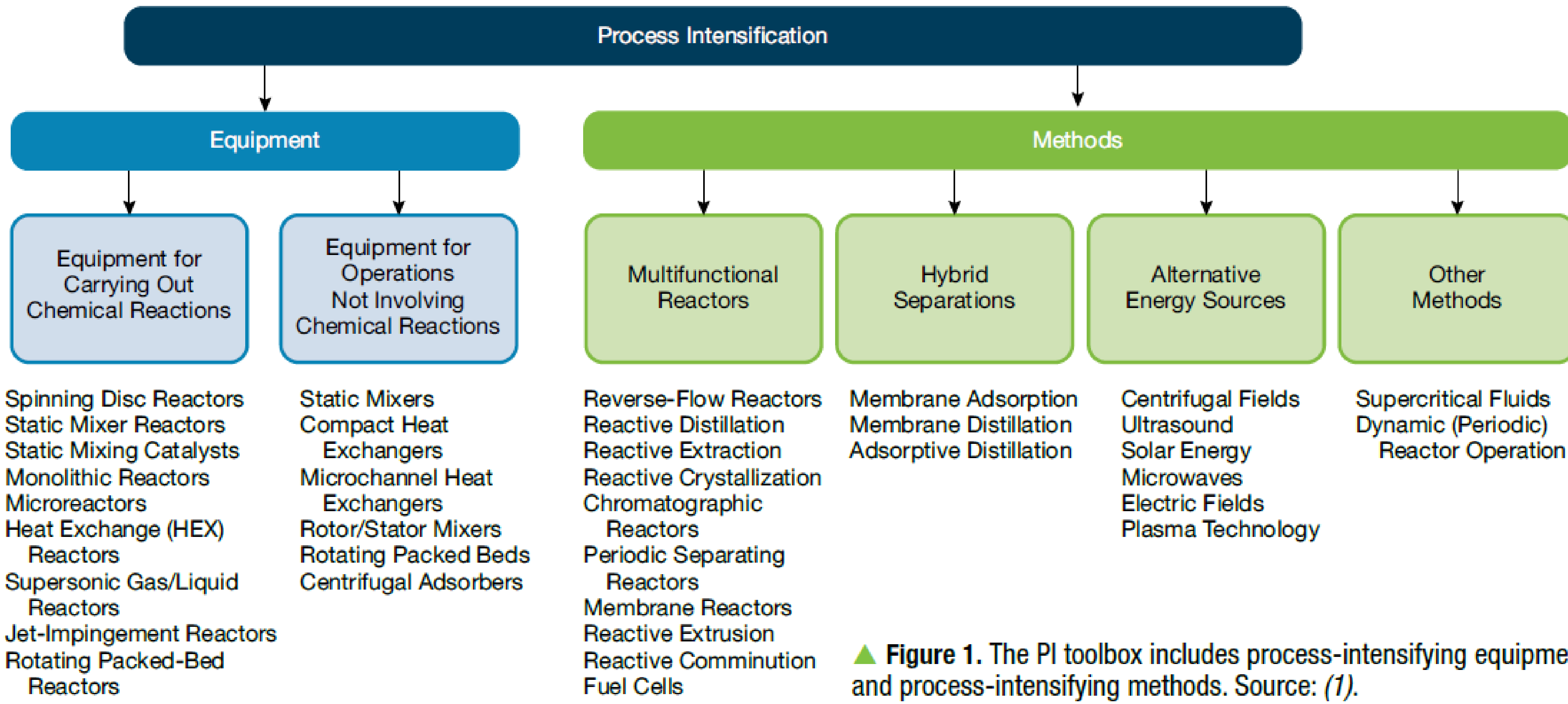



**Figure 1.** The PI toolbox includes process-intensifying equipment and process-intensifying methods. Source: *(1)*.

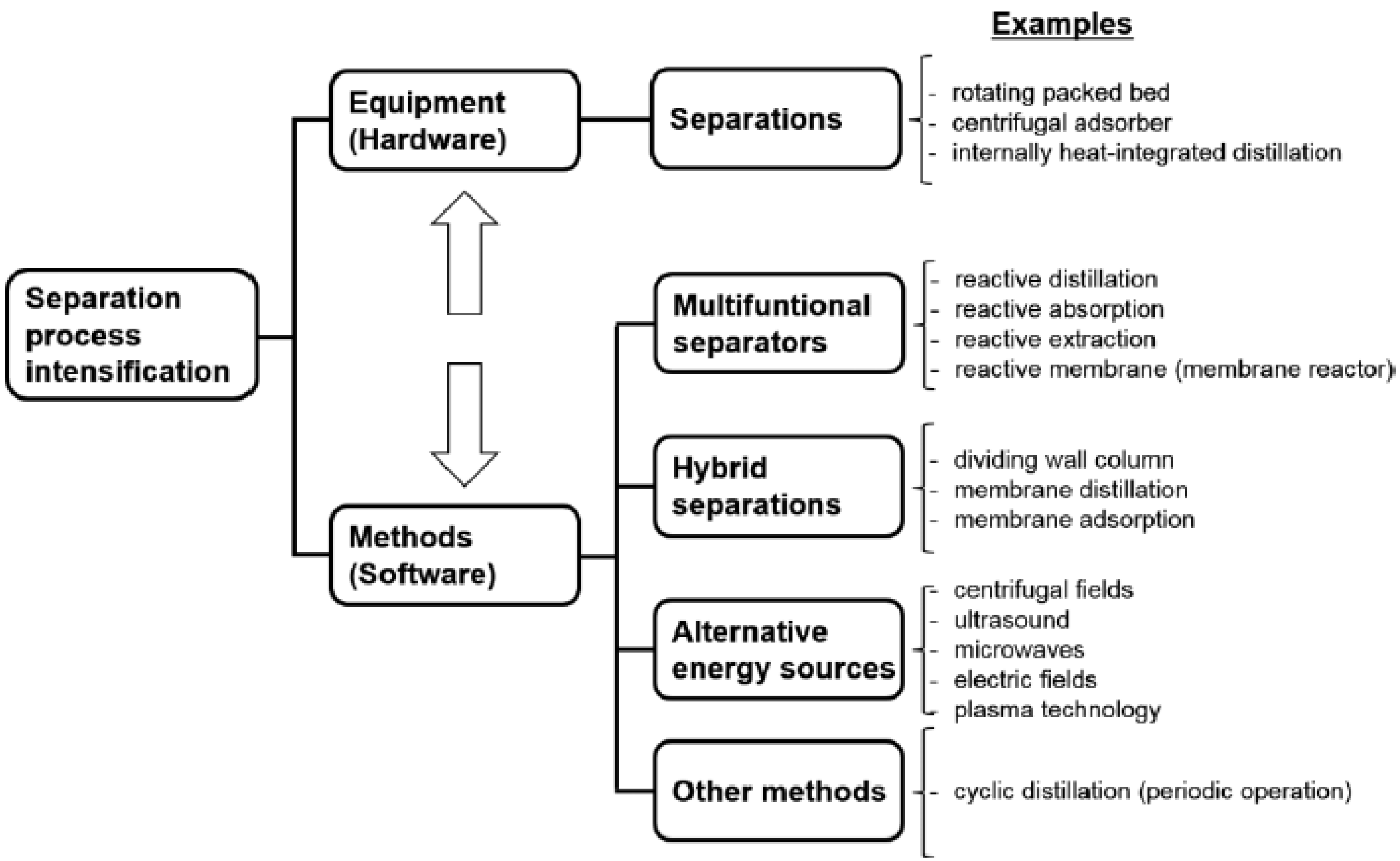



**Figure 1.** Separation process intensification toolbox.

The ultimate goal of PI is an ideal, intensified processing system in which

- Reactions proceed at a maximum achievable efficiency
- All molecules undergo the same processing history
- Hydrodynamic, heat- and mass-transfer limitations are removed
- Synergies resulting from interrelations between various operations and steps are fully utilized

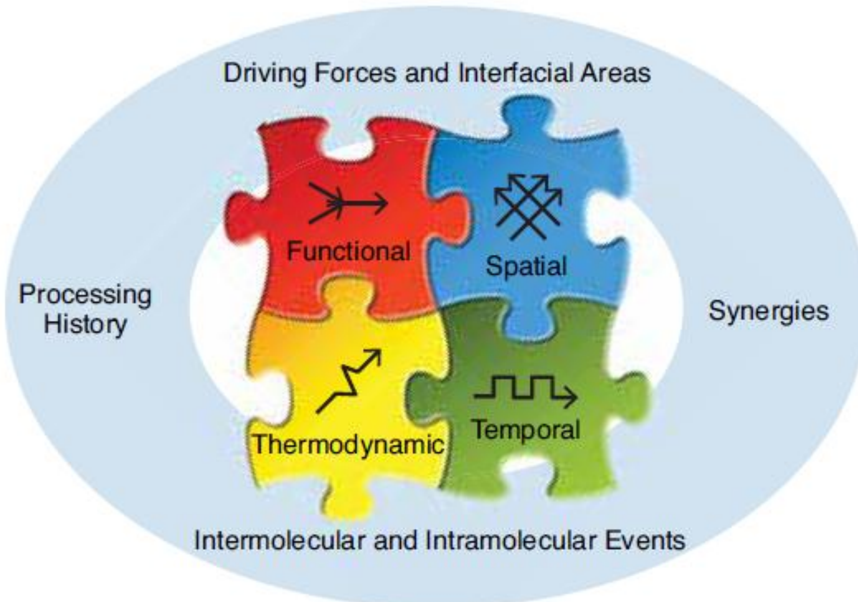



**Figure 2.** The four principles of PI (outer circle) are realized in the four domains of PI: spatial, thermodynamic, functional, and temporal.

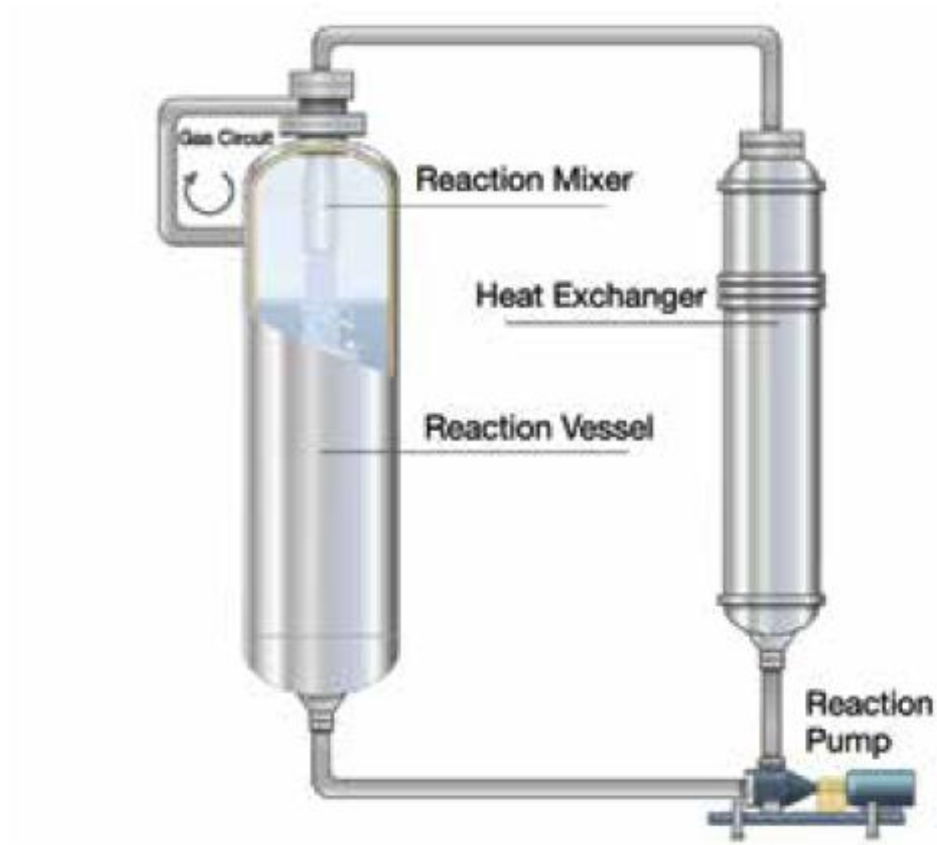
Guiding principles of PI

- Maximize the effectiveness of intermolecular and intramolecular events
- Give each molecule the same processing experience
- Optimize the driving forces and/or resistances at every scale and maximize the specific surface areas to which these forces or resistances apply
- Maximize the synergistic effects of partial processes

Avoid randomness in chemical equipment and replace it with structure.

▲ **Figure 3.** This millichannel reactor is an example of PI in the spatial domain. Image courtesy of Ehrfeld Mikrotechnik GmbH.

Transfer energy from the source to the recipient in the optimal way, including the best form of energy and energy transfer mechanism.




▲ **Figure 4.** This loop reactor is an example of PI in the thermodynamic domain. Image courtesy of Buss ChemTech AG.

Achieve synergy by combining one or more functions (e.g., heat transfer and mixing) in a single device or process step.

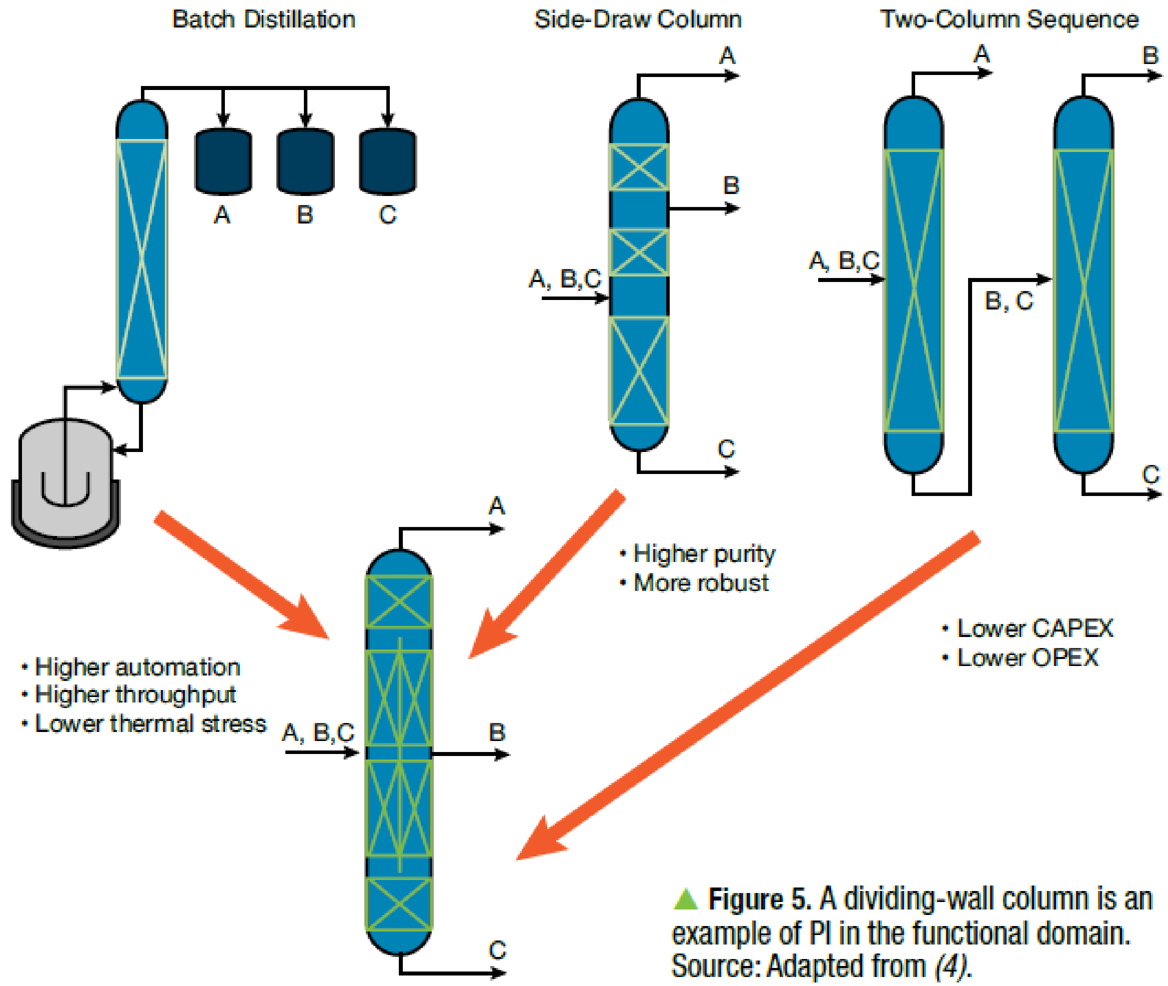



**Figure 5.** A dividing-wall column is an example of PI in the functional domain. Source: Adapted from *(4)*.

Manipulate time by

- Introducing an controlled unsteady state in a steady state process to attain operating conditions that improve process performance
- Change process duration dramatically by changing process conditions

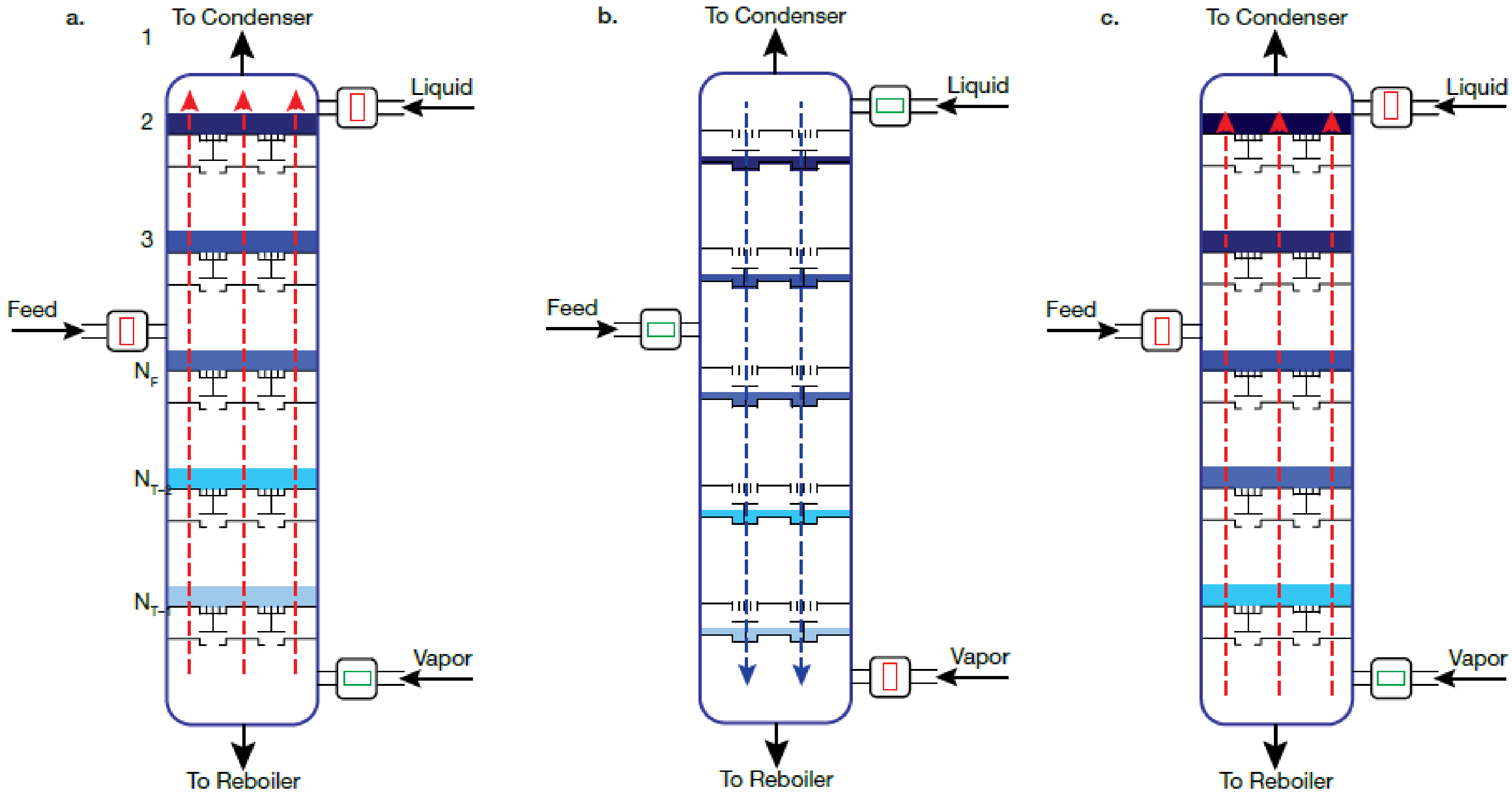
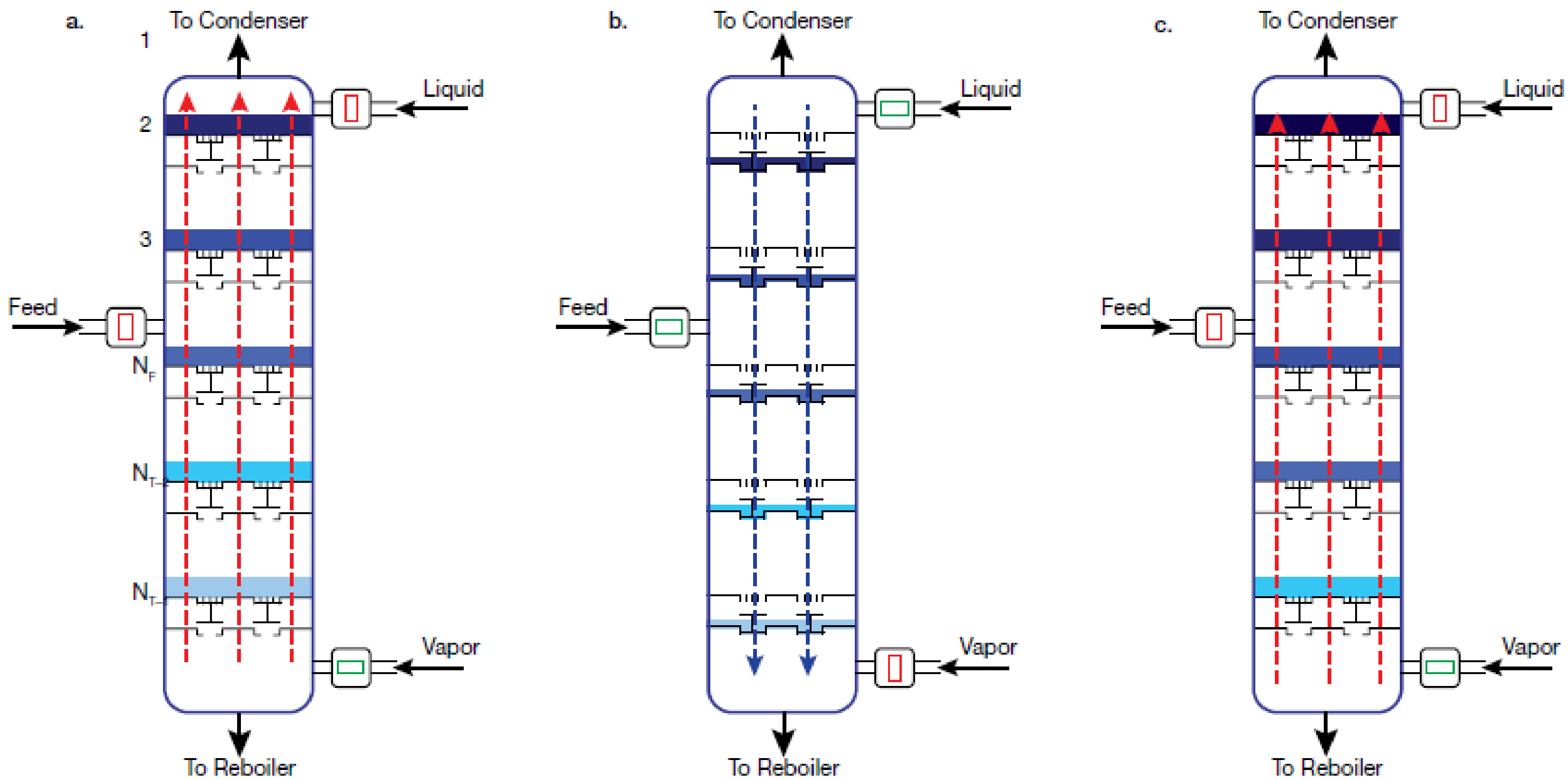
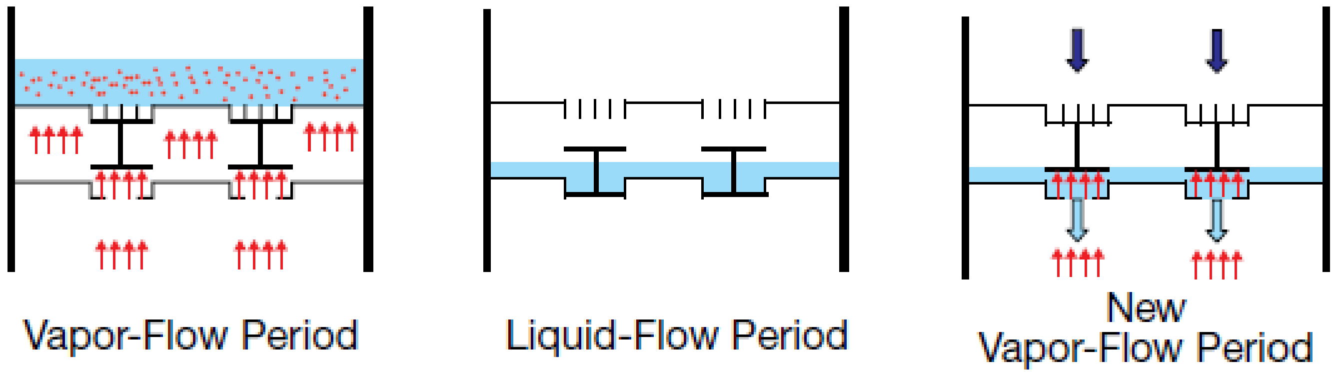



**Figure 6.** Cyclic distillation is an example of PI in the temporal domain. Source: Adapted from *(5)*.

# CYCLIC DISTILLATION




▲ **Figure 1.** Cyclic distillation works by alternating a vapor-flow period (a) with a liquid flow-period (b) and then repeating with another vapor-flow period (c). The darkening color from the bottom to the top indicates the liquid-phase concentration gradient across the plates of the column. Source: Adapted from *(12)*.




▲ **Figure 2.** Trays designed for cyclic distillation feature valves and a sluice chamber below the tray. During vapor flow, the valves are closed and the liquid remains on the tray. During liquid flow, the valves open and the liquid flows from the tray to the chamber below. When the next vapor-flow period begins, the sluice chamber opens and the liquid flows to the empty tray below. Source: Adapted from *(22)*.

# DIVIDING WALL COLUMN

# DIVIDING WALL COLUMN

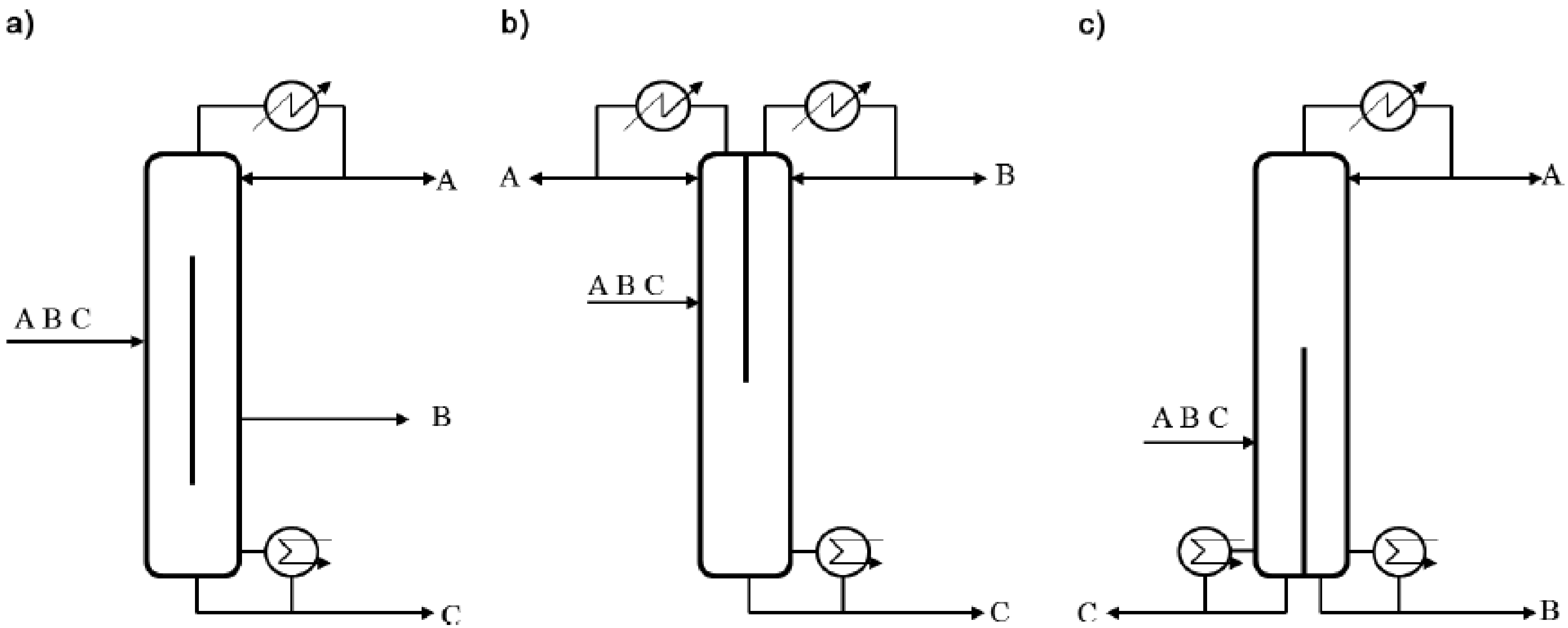



Figure 2. Schematic diagram of the (a) DWC, (b) TDWC, and (c) BDWC.

# REACTIVE DIVIDING WALL COLUMN

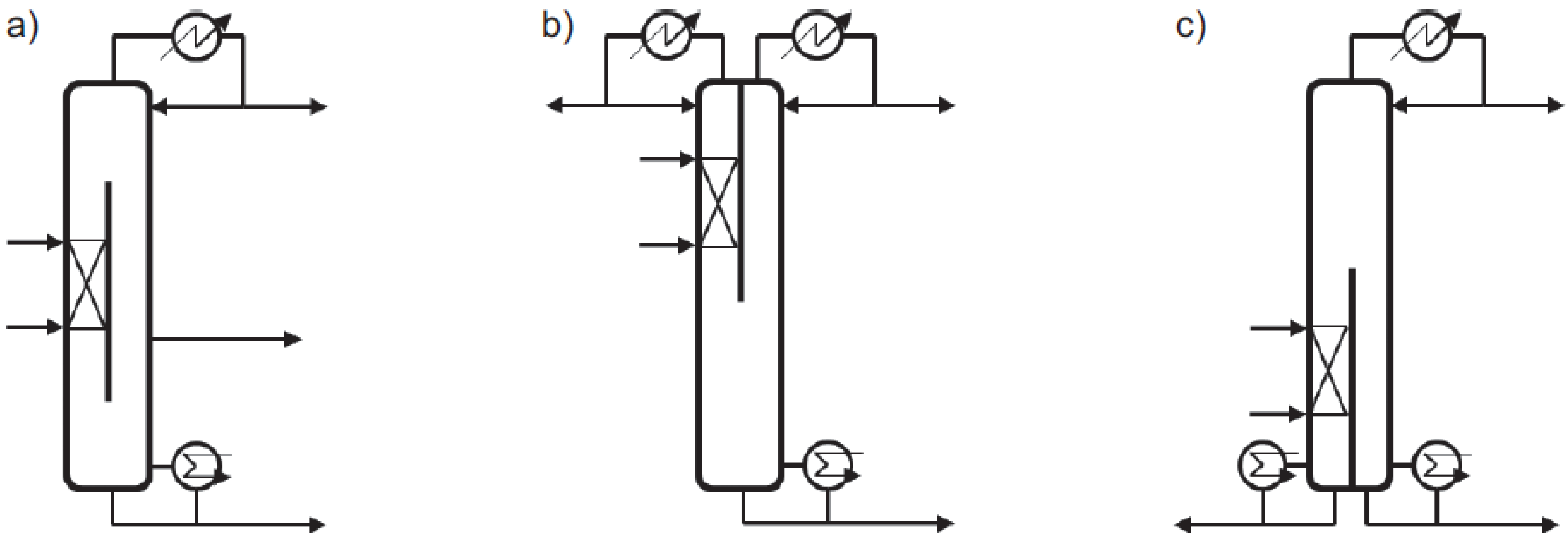



**Figure 9.** Schematic diagram of the (a) conventional reactive dividing-wall column, (b) top reactive dividing-wall column, and (c) bottom reactive dividing-wall column.

# EXAMPLES OF INDUSTRIAL APPLICATIONS

**Table 2.** Industrial applications of dividing-wall columns[6,82]

| Applied research and industrial applications | Additional information/remarks |
|---|---|
| ***Ternary separations*** | |
| Benzene-toluene-xylene fractionation | ExxonMobil |
| Separation of hydrocarbons from Fischer-Tropsch synthesis unit | Linde AG, Tray column, $H = 107$ m, $D = 5$ m |
| Separation of benzene from pyrolysis gasoline | Uhde, 170 000 mt yr$^{-1}$ feed capacity |
| Separation of C7+ aromatics from C7+ olefin /paraffin | UOP, five DWC, Trap tray |
| Mostly undisclosed systems | BASF / Montz, over 70 columns, $D = 0.6 - 4$ m, $P = 2$ mbar to 10 bar |
| Undisclosed systems | Sumitomo Heavy Ind., Kyowa Yuka, Sulzer Chemtech Ltd. (20 DWC), Koch Glitsch (10 DWC) |
| ***Multi-component separations*** | |
| Recovery of four component mixtures of fine chemical intermediates | BASF/Montz, Single wall column, $H = 34$ m, $D = 3.6$ m, deep vacuum |
| Integration of a product separator and an HPNA stripper | UOP, 5 product streams |
| ***Retrofit of conventional columns to DWC*** | |
| Recovers mixed xylenes from reformate motor gasoline | Koch-Glitsch, $D = 3.8 - 4.3$ m, tray column, over 50% energy savings |
| Separation of (iso)paraffins. Production of isohexane | Koch-Glitsch |
| Separation and purification of 2-ethylhexanol (2-EH) | Dual operation possible |
| Undisclosed systems | MW Kellogg |

| | |
|---|---|
| ***Reactive DWC*** | |
| MeOH + BuOH + AcOH ↔ AcOMe + AcOBu + $H_2O$ | Rate-based model/AspenTech ACM |
| DMC + EtOH ↔ DEC + MeOH | Aspen Plus® or PRO/II |
| Methyl Acetate + Water ↔ Methanol + Acetic Acid | |
| Isoamylenes + Ethanol ↔ Tertiary Amyl Ethyl Ether | Aspen DISTIL and HYSYS, CSTR |
| Ethanol + Acetic Acid ↔ Ethyl Acetate + Water | Aspen Plus® and Aspen Dynamics |
| Reactive distillation with 10 chemical species | Aspen Plus (RADFRAC) |
| Isobutane + Ethanol ↔ Ethyl Tertiary Butyl Ether | Aspen HYSYS |
| Ethanol + Acetic Acid ↔ Ethyl Acetate + Water | Aspen Plus® |
| Methanol + Isobutylene ↔ Methyl Tertiary Butyl Ether | Aspen Plus® |
| Ethanol + Ethylene ↔ Ethoxyethanol | |
| Methanol ↔ Dimethyl Ether + Water | Aspen Plus (RADFRAC) |
| ***Azeotropic DWC*** | |
| Ethanol dehydration | Entrainer: cyclohexane, n-pentane |
| ***Extractive DWC*** | |
| Separation of toluene and non-aromatics with N-formyl-morpholine | Uhde, 28000 mt $yr^{-1}$ feed capacity |
| Crude butadiene from a crude C4 using N-methyl-pyrrolidone (NMP) as solvent | BASF, Both trays and packing |
| Bioethanol dehydration | Ethylene glycol used as solvent |

**Table 3.** Main industrial applications of reactive distillation[6]

| Reaction type | Catalyst/internals |
|---|---|
| ***Alkylation*** | |
| Alkyl benzene from ethylene/propylene and benzene | Zeolite $\beta$, molecular sieves |
| ***Amination*** | |
| Amines from ammonia and alcohols | $H_2$ and hydrogenation catalyst |
| ***Carbonylation*** | |
| Acetic acid from CO and methanol / dimethyl ether | Homogeneous |
| ***Condensation*** | |
| Diacetone alcohol from acetone | Heterogeneous |
| Bisphenol-A from phenol and acetone | N/A |
| Trioxane from formaldehyde | Strong acid catalyst, zeolite ZSM-5 |
| ***Esterification*** | |
| Methyl acetate from methanol and acetic acid | $H_2SO_4$, Dowex 50, Amberlyst-15 |
| Ethyl acetate from ethanol and acetic acid | N/A |
| 2-methyl propyl acetate from 2-methyl propanol and acid | Katapak-S |
| Butyl acetate from butanol and acetic acid | Cation exchange resin |
| Fatty acid methyl esters from fatty acids and methanol | $H_2SO_4$, Amberlyst-15, Metal oxides |

| | |
|---|---|
| ***Etherification*** | |
| MTBE from isobutene and methanol | Amberlyst-15 |
| ETBE from isobutene and ethanol | Amberlyst-15/pellets, structured |
| TAME from isoamylene and methanol | Ion exchange resin |
| DIPE from isopropanol and propylene | ZSM 12, Amberlyst-36, zeolite |
| ***Hydration / Dehydration*** | |
| Mono ethylene glycol from ethylene oxide and water | Homogeneous |
| ***Hydrogenation / Dehydrogenation*** | |
| Cyclohexane from benzene | Alumina supported Ni catalyst |
| MIBK from benzene | Cation exchange resin with Pd/Ni |
| ***Hydrolysis*** | |
| Acetic acid and methanol from methyl acetate + water | Ion exchange resin bags |
| Acrylamide from acrylonitrile | Cation exchanger, copper oxide |
| ***Isomerization*** | |
| Iso-parafins from n-parafins | Chlorinated alumina and $H_2$ |
| ***Nitration*** | |
| 4-Nitrochlorobenzene from chlorobenzene + nitric acid | Azeotropic removal of water |
| ***Transesterification*** | |
| Ethyl acetate from ethanol and butyl acetate | Homogeneous |
| Diethyl carbonate from ethanol and dimethyl carbonate | Heterogeneous |
| Vinyl acetate from vinyl stearate and acetic acid | N/A |
| ***Unclassified reactions*** | |
| Monosilane from trichlorsilane | Heterogeneous |
| Methanol from syngas | $Cu/Zn/Al_2O_3$ and inert solvent |
| DEA from monoethanolamine and ethylene oxide | N/A |
| Polyesterification | Autocatalytic |

# CENTRIFUGAL FIELD FOR PI – HIGEE DISTILLATION

The essence of HiGee technology is replacing the gravitational field by a high centrifugal field achieved by rotating a specially shaped rigid bed, typically a disk with an eye in the center. The higher mass-transfer coefficients and higher flooding limits allow the use of high surface-area packing. In this way, the momentum, heat and mass transfer can be tremendously intensified.

HiGee distillation uses the rotating packed bed (RPB) concept in a high-gravity field (100–1000 *g*) technology – claiming HETP values as low as 1–2 cm, about 3–6 times higher throughput and an equipment volume reduction of 2–3 orders of magnitude lower compared with that of conventional packed columns

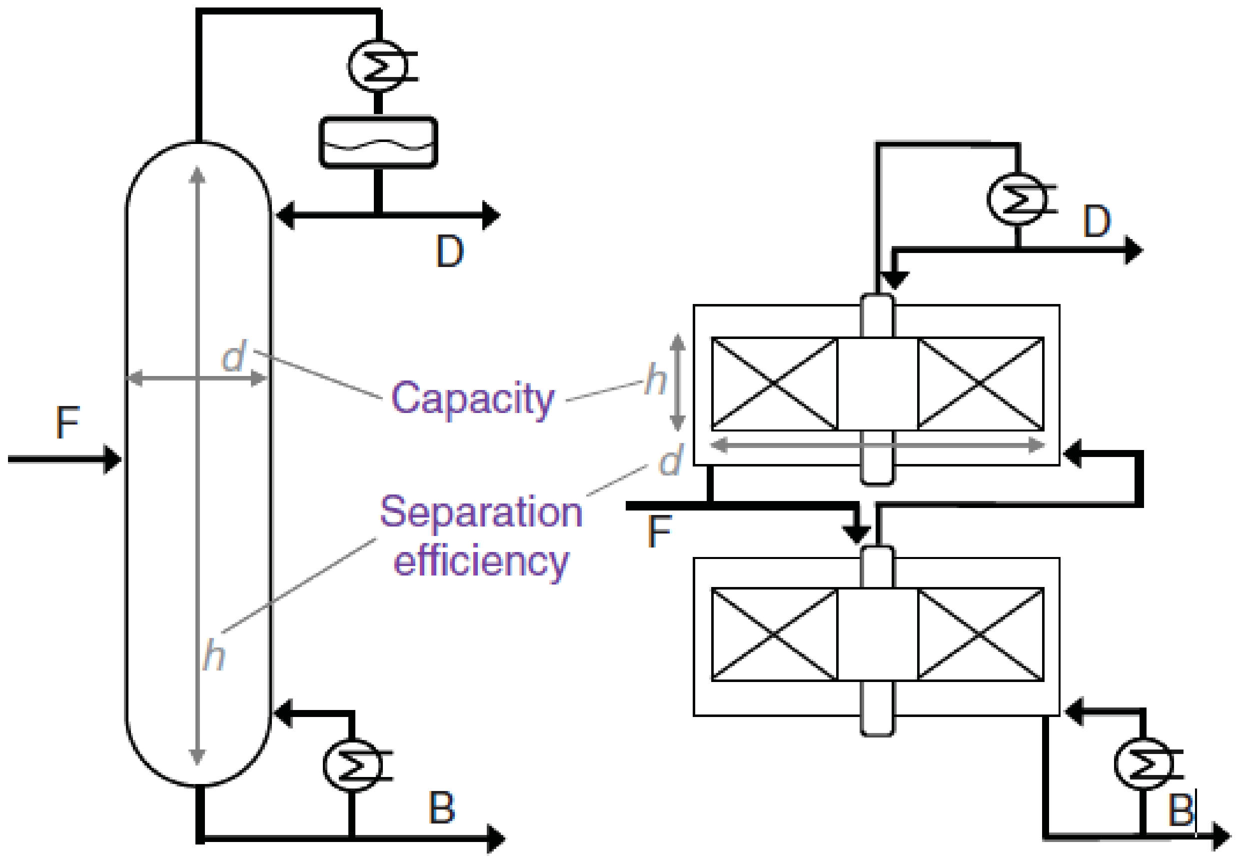



**Figure 8.** Design analogy between conventional and HiGee distillation.

## HIGEE DISTILLATION – ROTATING ZIGZAG BED

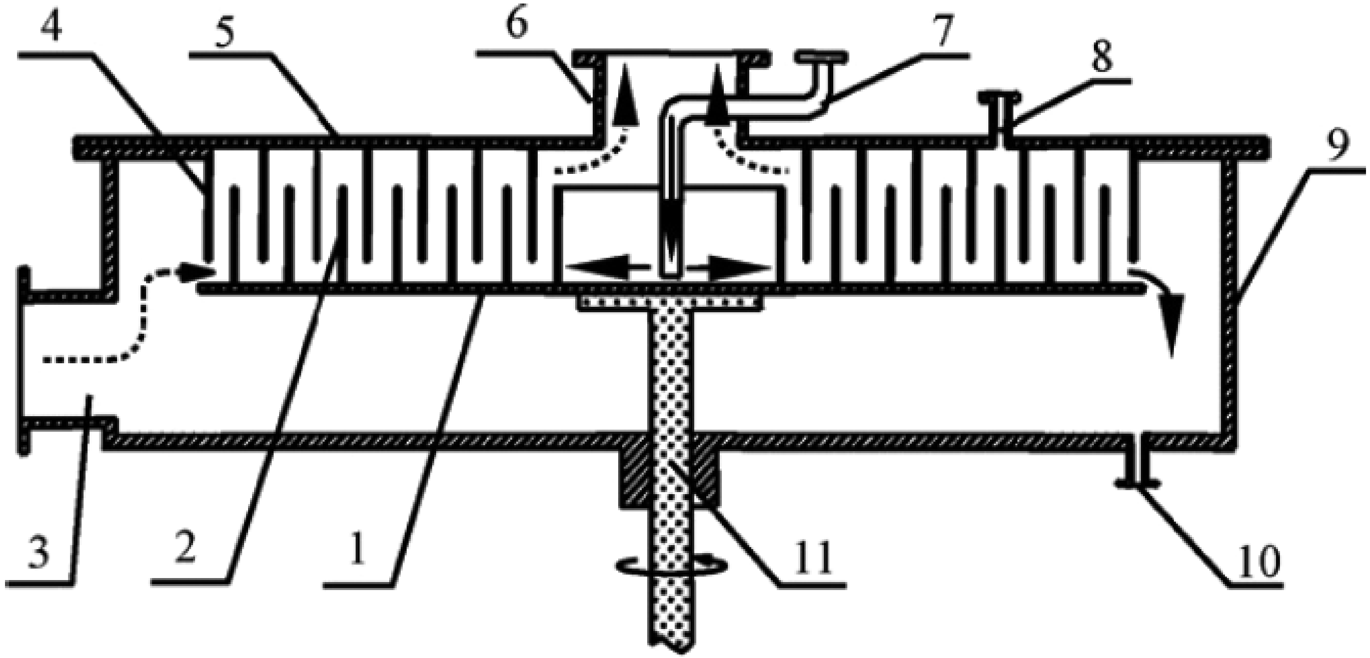



**Figure 6.** Simplified sketch of the RZB and its rotor [40]. (1) Rotational disc, (2) rotational baffle, (3) gas inlet, (4) stationary baffle, (5) stationary disc, (6) gas outlet, (7) liquid inlet, (8), middle feed, (9) liquid outlet, (10) casing, (11) shaft

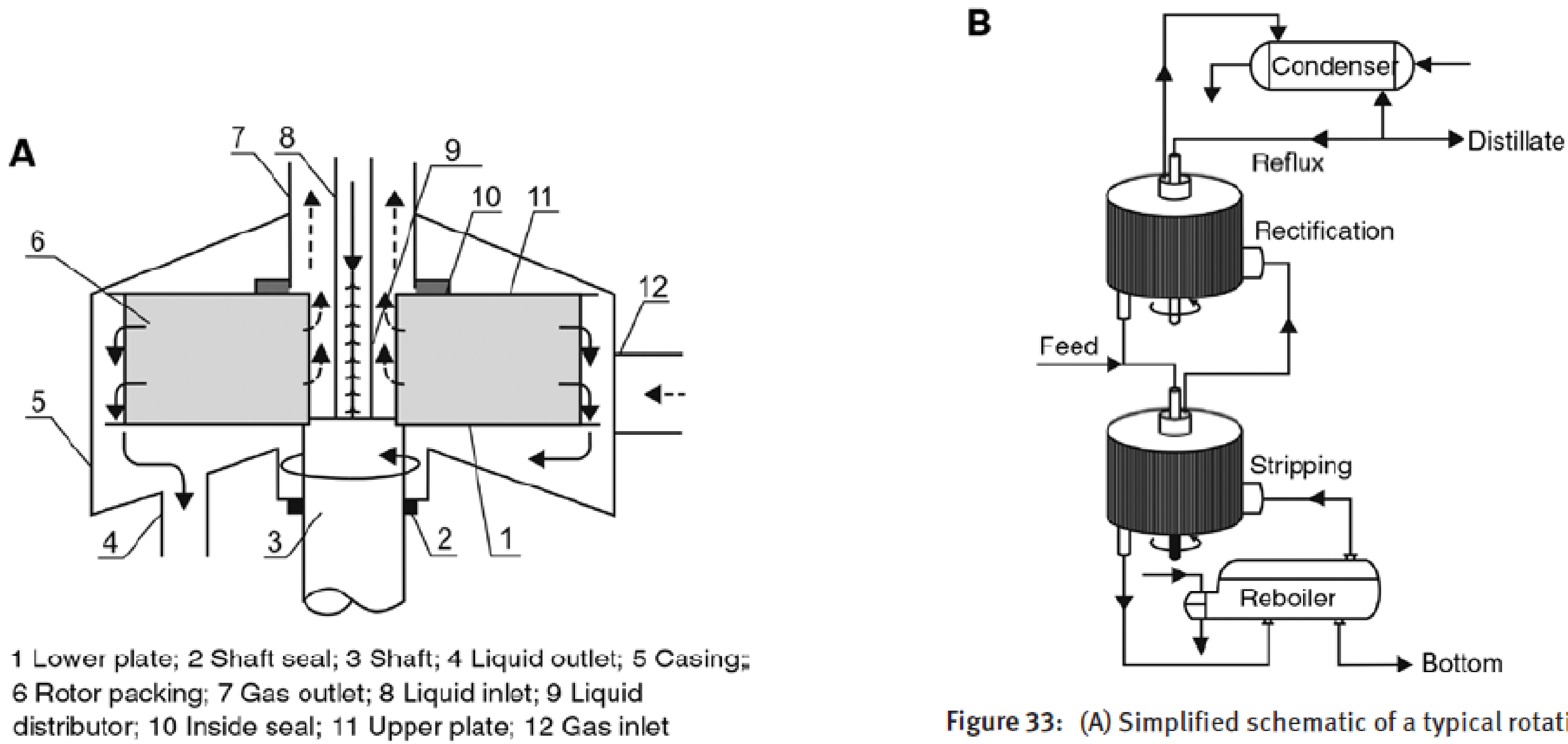



**Figure 33:** (A) Simplified schematic of a typical rotating packed bed (RPB); (B) continuous distillation process using rotating packed bed (Wang et al. 2011).

# HEAT-INTEGRATED DISTILLATION

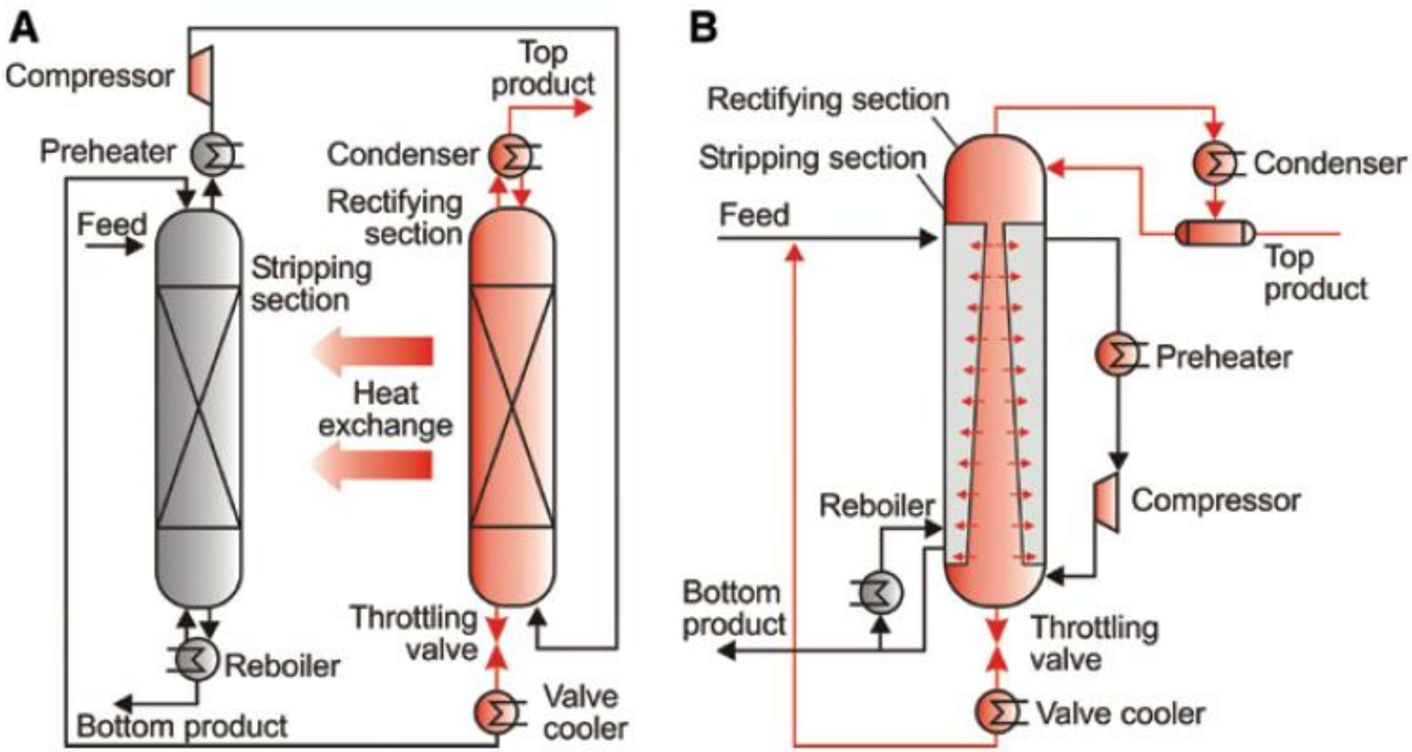



**Figure 15:** (A) Principles of a heat-integrated distillation column; (B) schematic drawing of a pilot plant (Górak and Stankiewicz 2011).

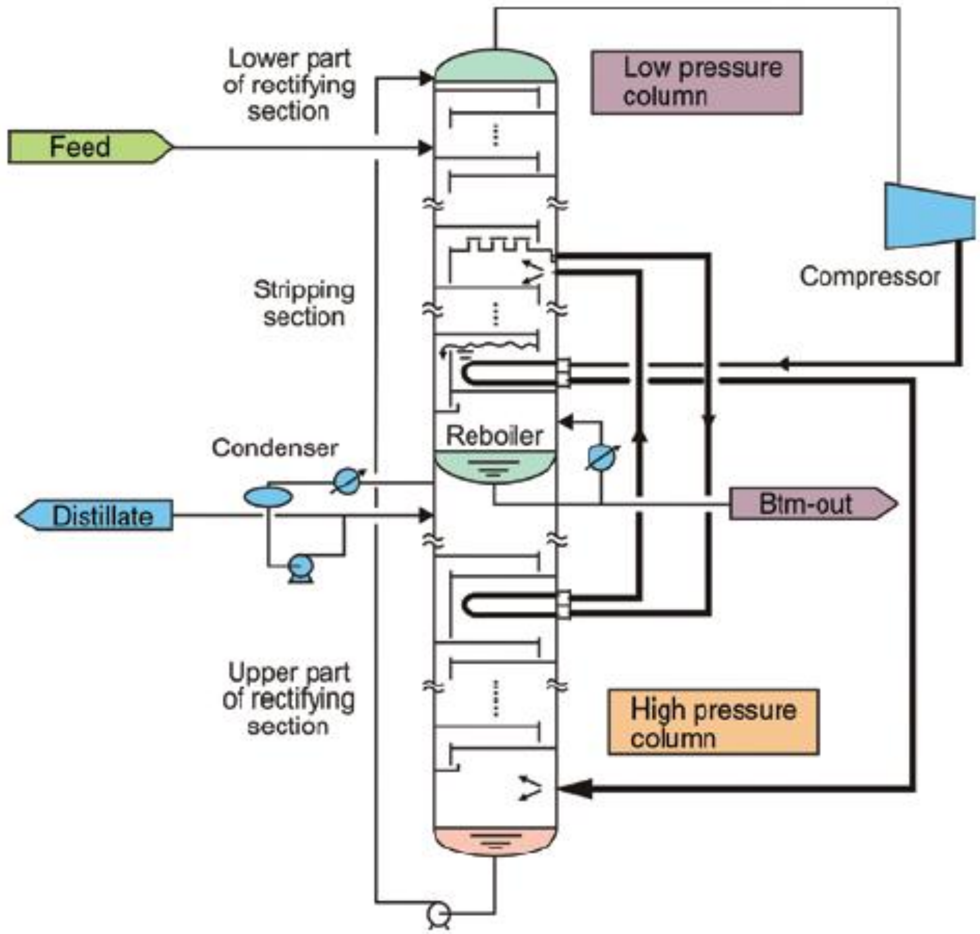



**Figure 16:** SuperHIDiC configuration developed by Toyo Engineering Corp. in collaboration with the National Institute of Advanced Industrial Science and Technology (Tokyo, JP).

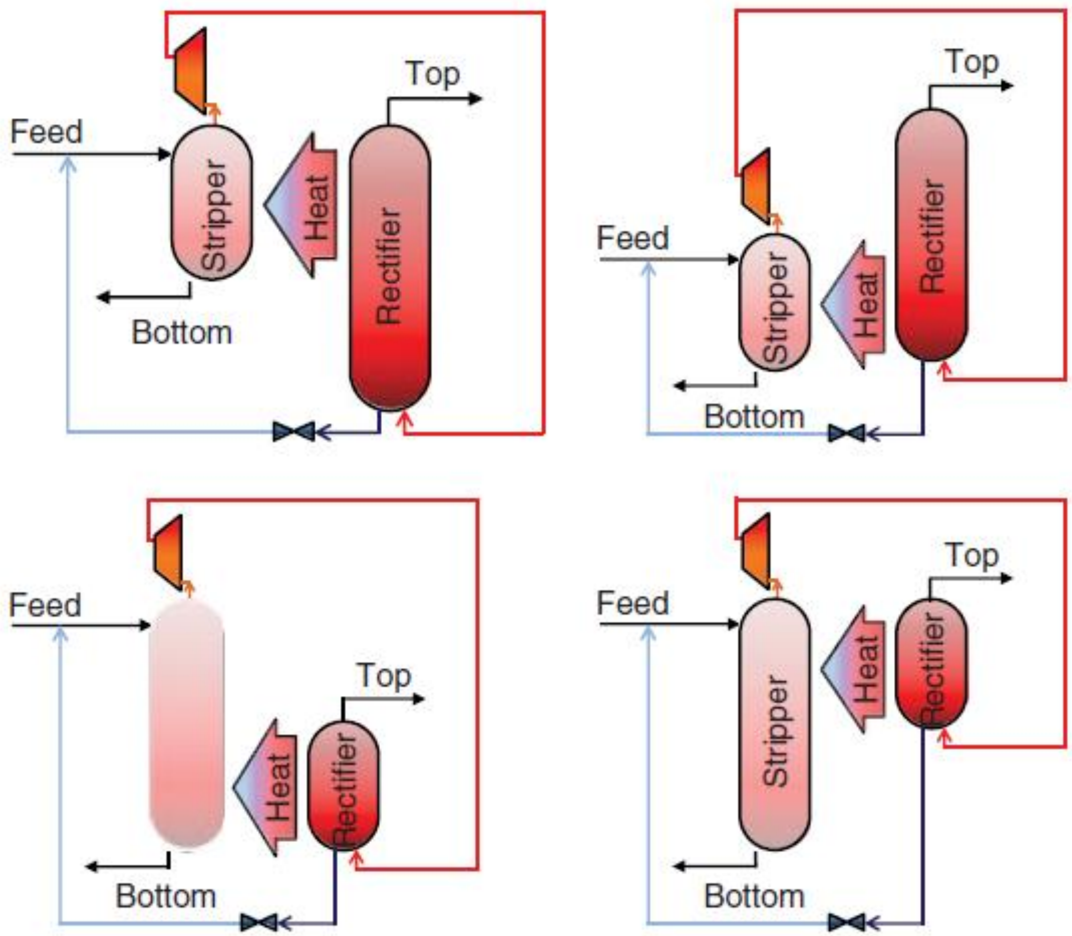
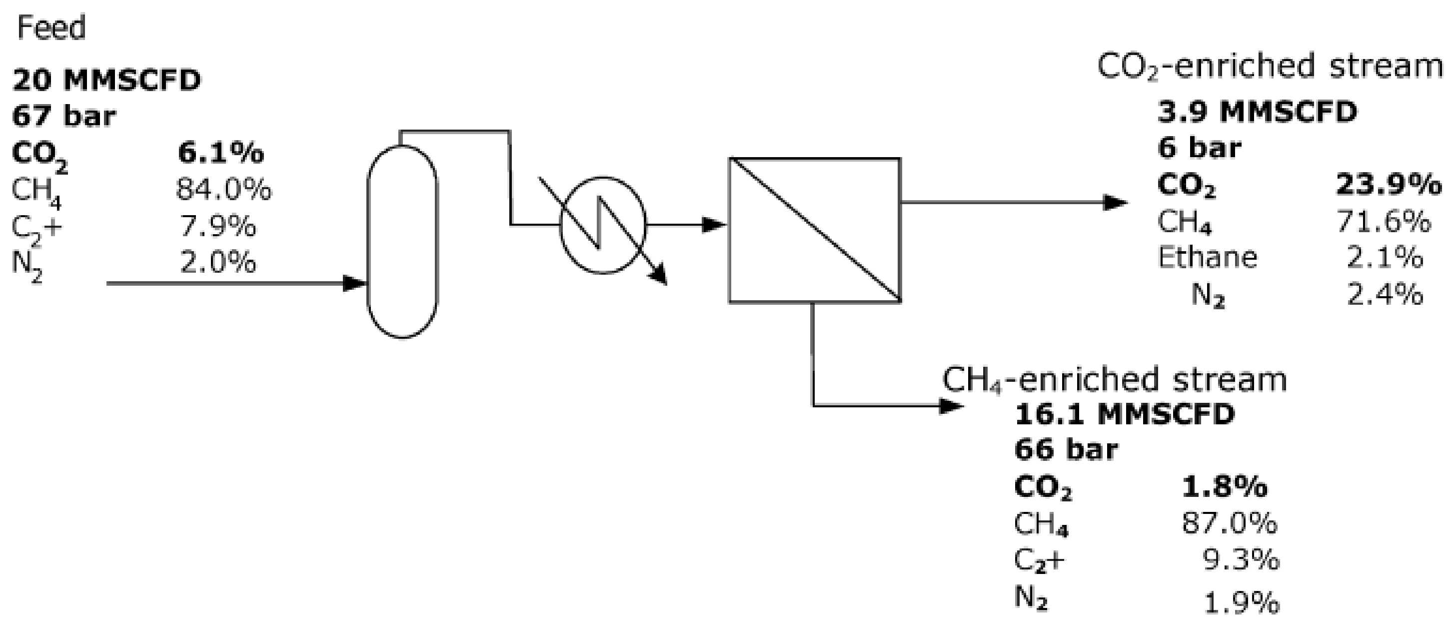



**Figure 4.** Heat integrated distillation column (HIDiC) – alternative configurations.

# MEMBRANE SEPARATIONS

## a. Natural gas treatment



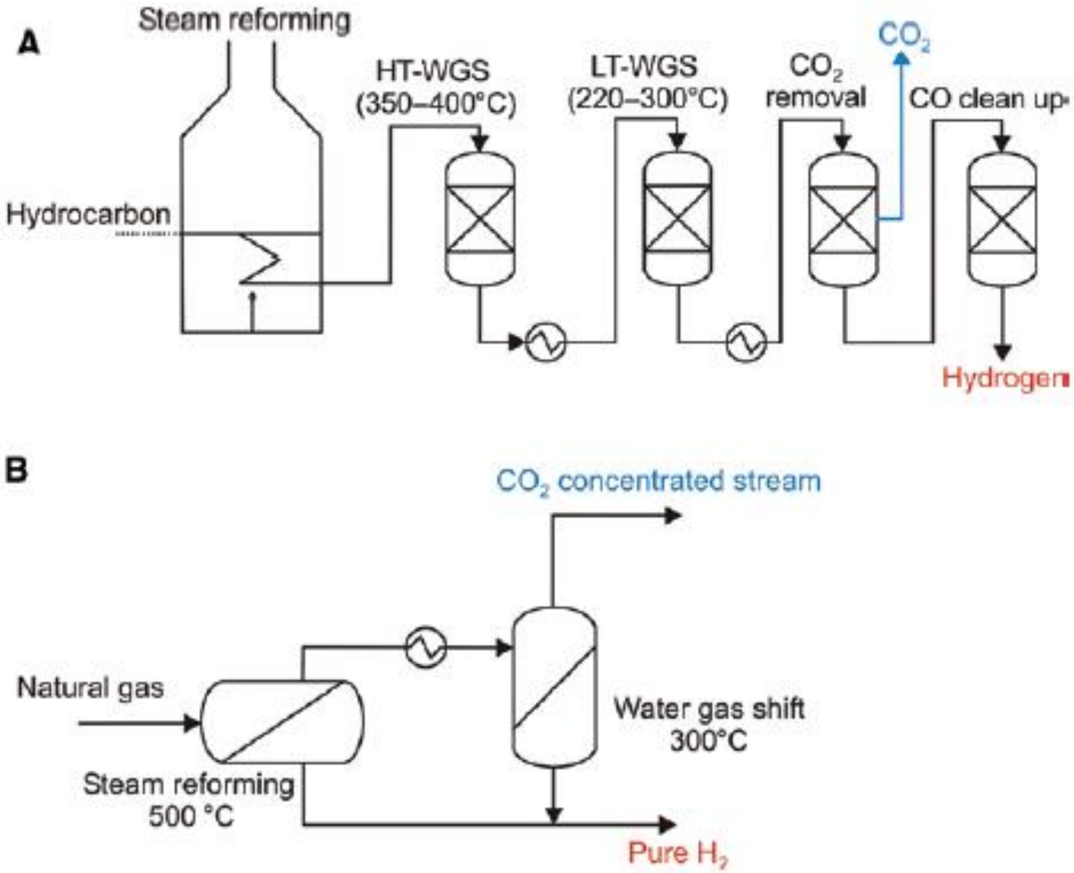



**Figure 24:** Hydrogen production from natural gas; (A) conventional plant; (B) integrated membrane plant (Drioli et al. 2012).

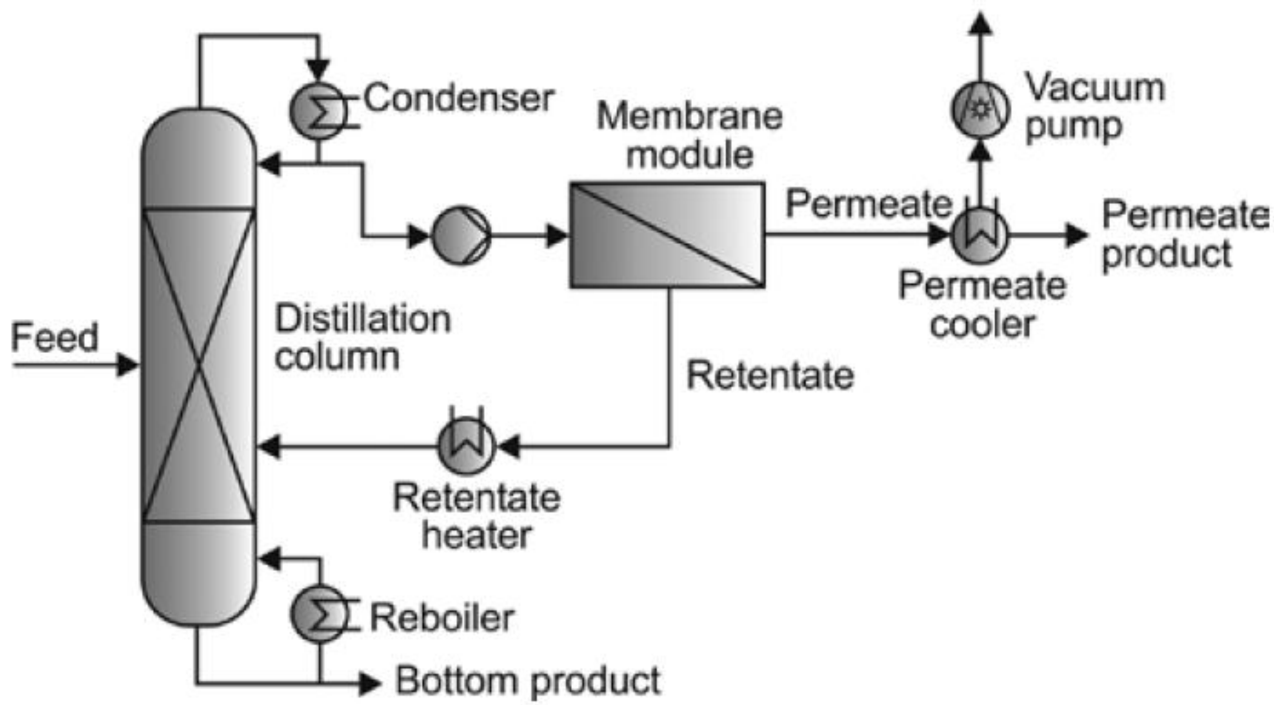



**Figure 25:** Hybrid process combining distillation with pervaporation for a binary system with a low-boiling azeotrope.

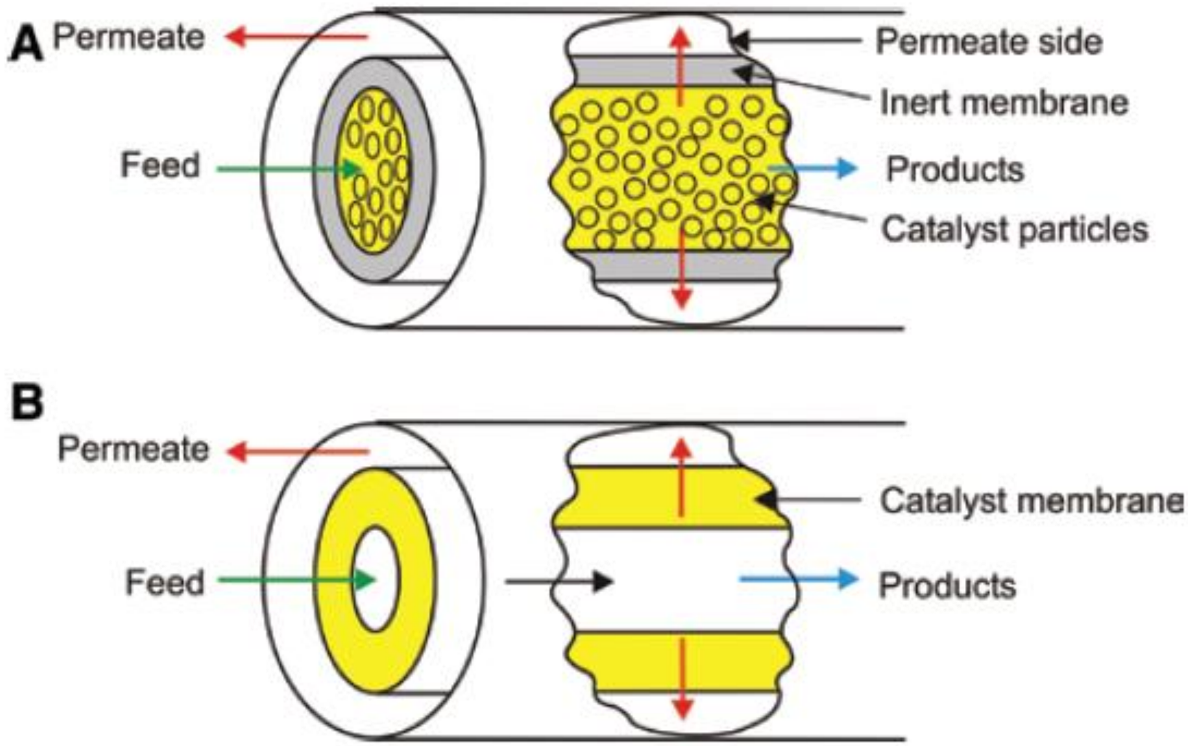



**Figure 27:** Catalytic membrane reactors: (A) catalytic packed bed, inert membrane; (B) catalyst membrane.

**Table 7. Main Industrial Applications of Membrane GS**

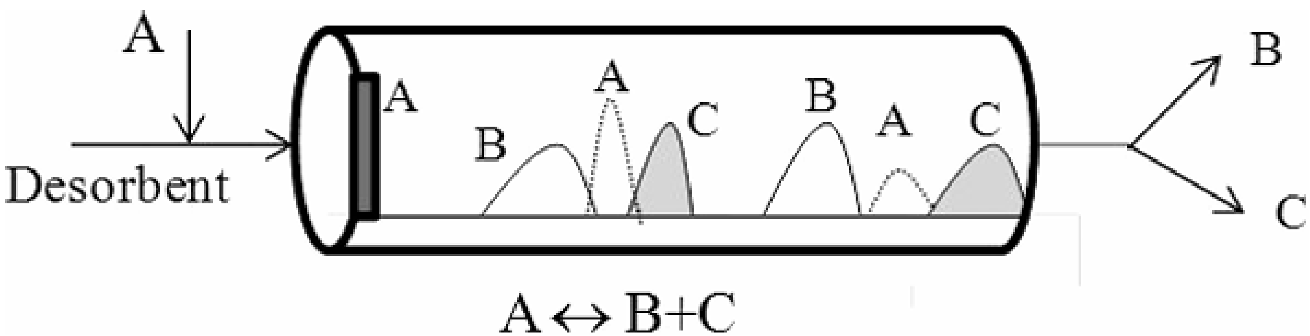
| separation | process |
|---|---|
| $H_2/N_2$ | ammonia purge gas |
| $H_2/CO$ | syngas ratio adjustment |
| $H_2$/hydrocarbons | hydrogen recovery in refineries |
| $O_2/N_2$ | nitrogen generation, oxygen-enriched air production |
| $CO_2$/hydrocarbons ($CH_4$) | natural gas sweetening, landfill gas upgrading |
| $H_2O$/hydrocarbons ($CH_4$) | natural gas dehydration |
| $H_2S$/hydrocarbons | sour gas treating |
| He/hydrocarbons | helium separation |
| He/$N_2$ | helium recovery |
| hydrocarbons/air | hydrocarbons recovery, pollution control |
| $H_2O$/air | air dehumidification |
| volatile organic species (e.g., ethylene or propylene)/ light gases (e.g., nitrogen) | polyolefin purge gas purification |

# ADSORPTION AND PROCESS INTENSIFICATION

# CHROMATOGRAPHIC BATCH REACTOR

problems

limited by equilibirum



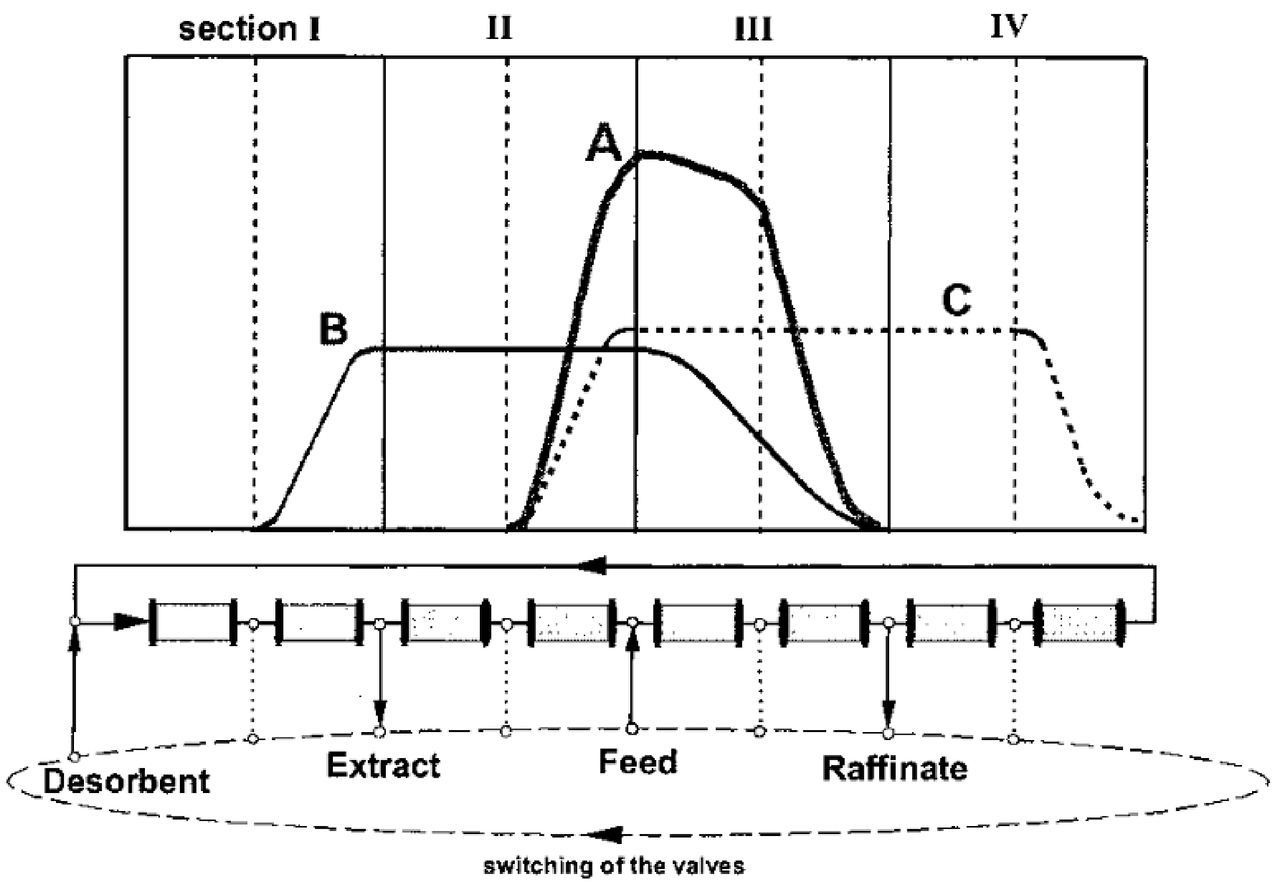
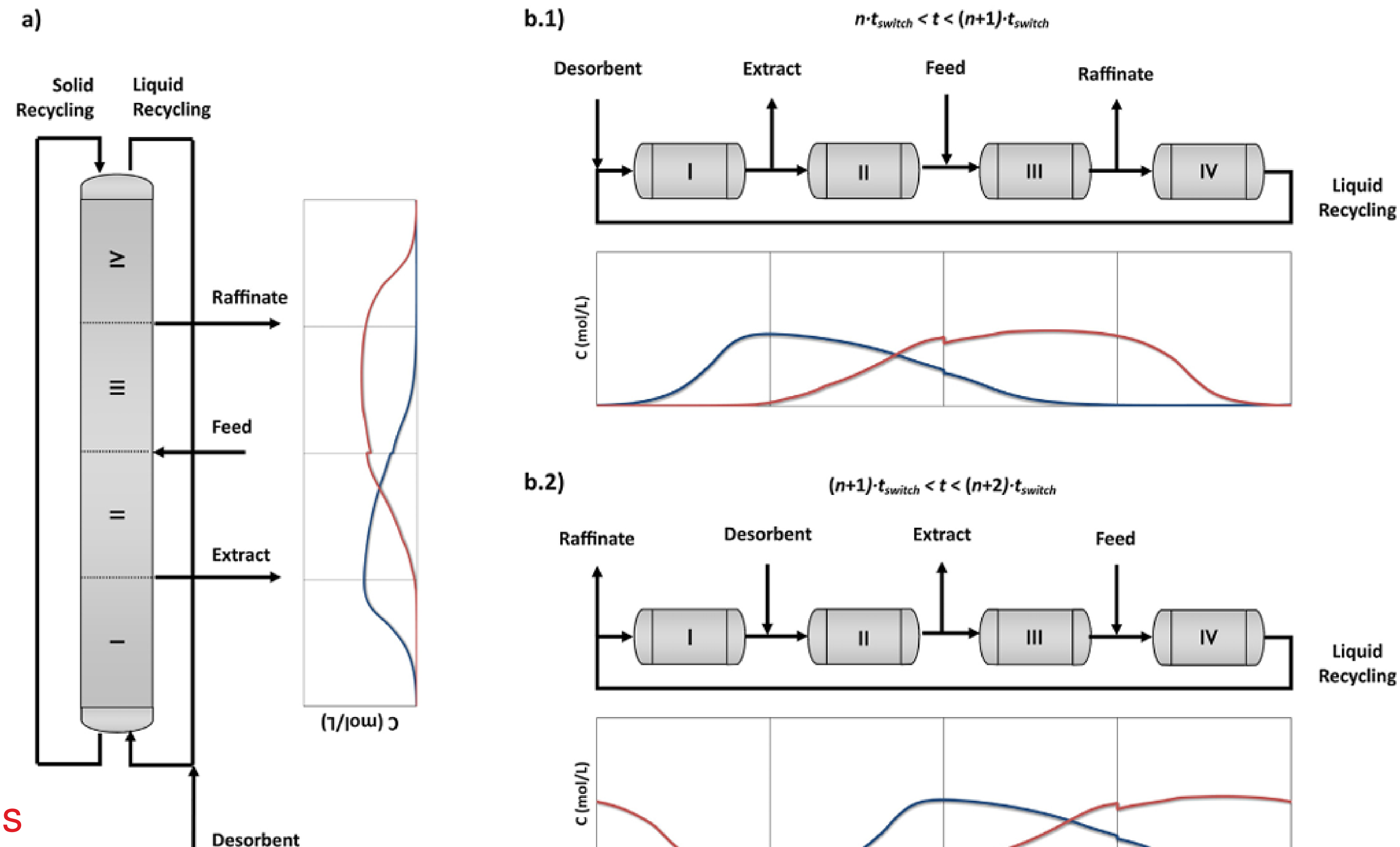



**Figure 1.** Schematic representation of a simulated moving bed reactor.




problems

difiiculty in transporting the solids

**Fig. 1.** Representation of a TMB (a) and a SMB (b) unit and the corresponding internal concentration profiles.

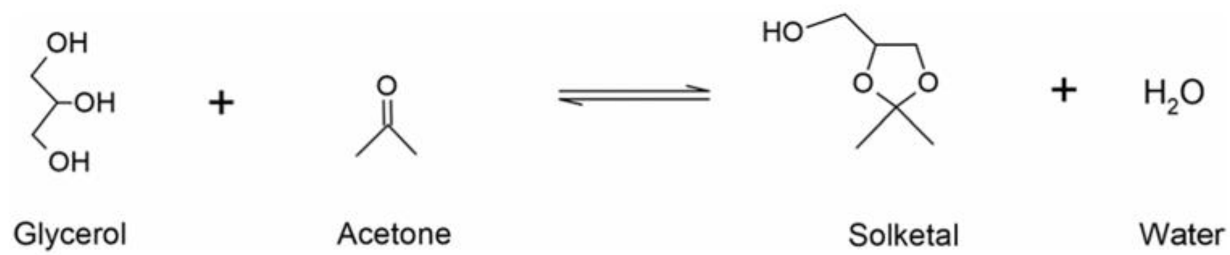



**Figure 1.** Glycerol ketalization with acetone to produce solketal and water.

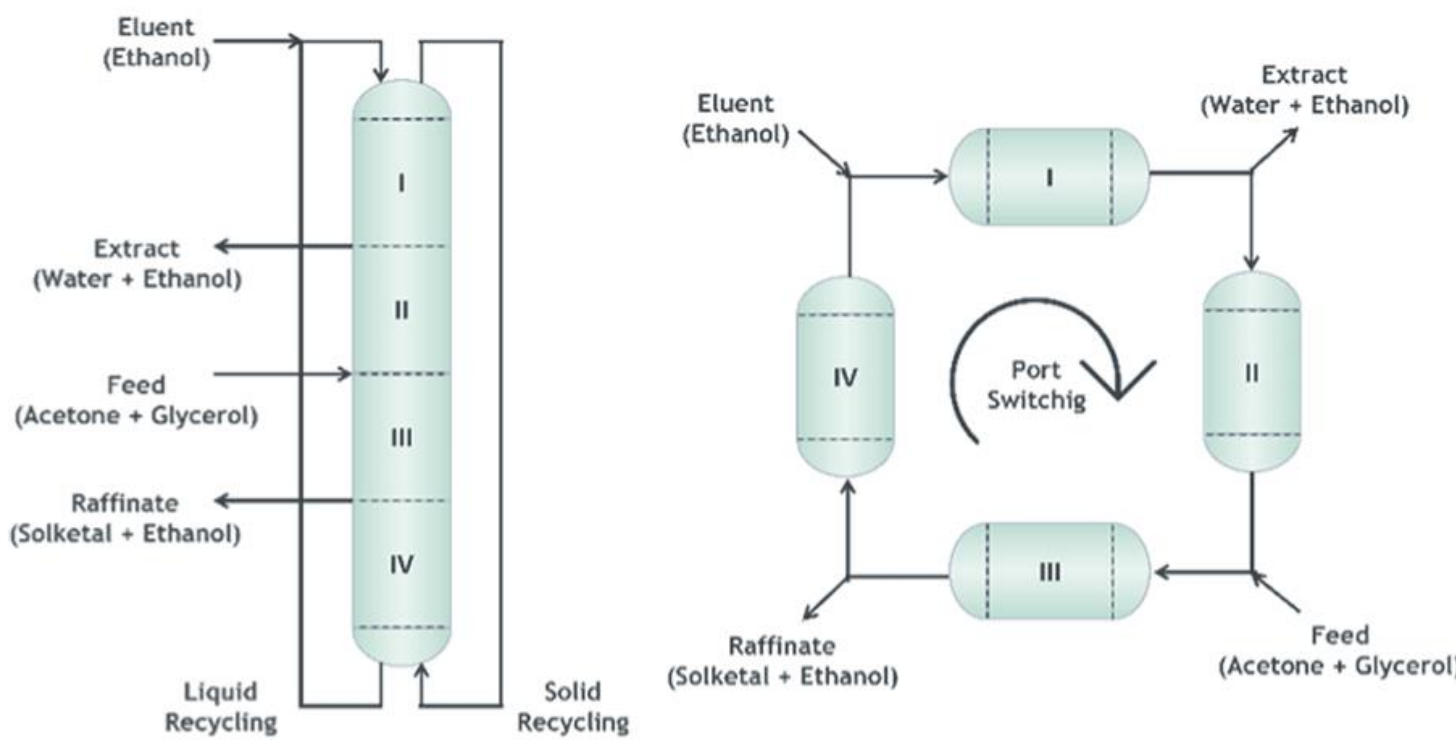



**Figure 2.** Scheme of (a) a true moving-bed reactor (TMBR) and (b) an SMBR.

# SIMULATED MOVING BED REACTOR

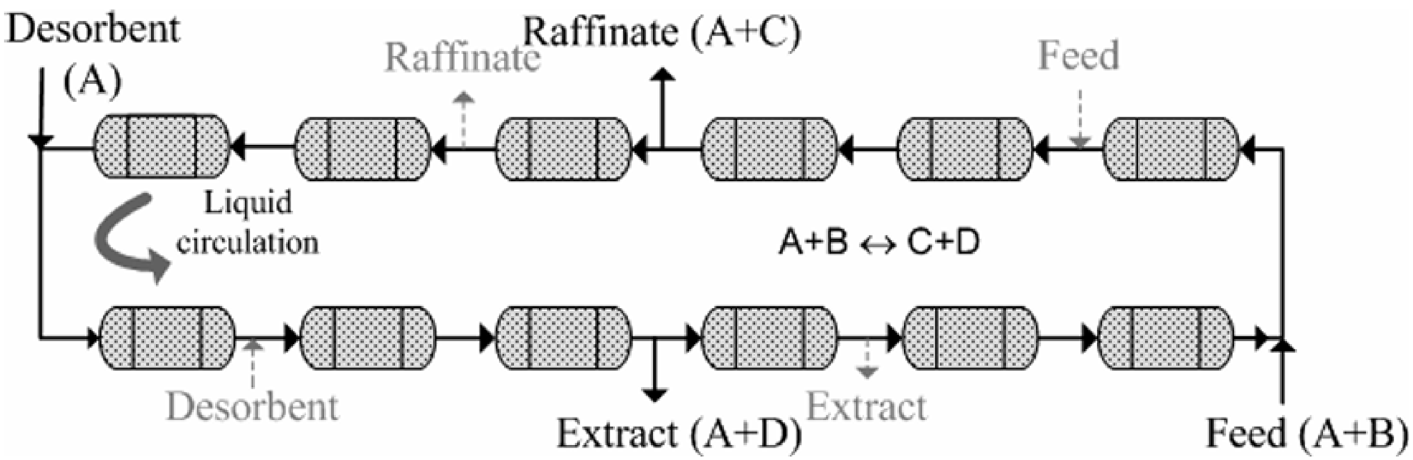



**Figure 5.** Schematic representation of a SMBR unit with four sections and three columns per section considering a reaction of type $A + B \rightarrow C + D$, where C is the less adsorbed product and D is the most adsorbed one.

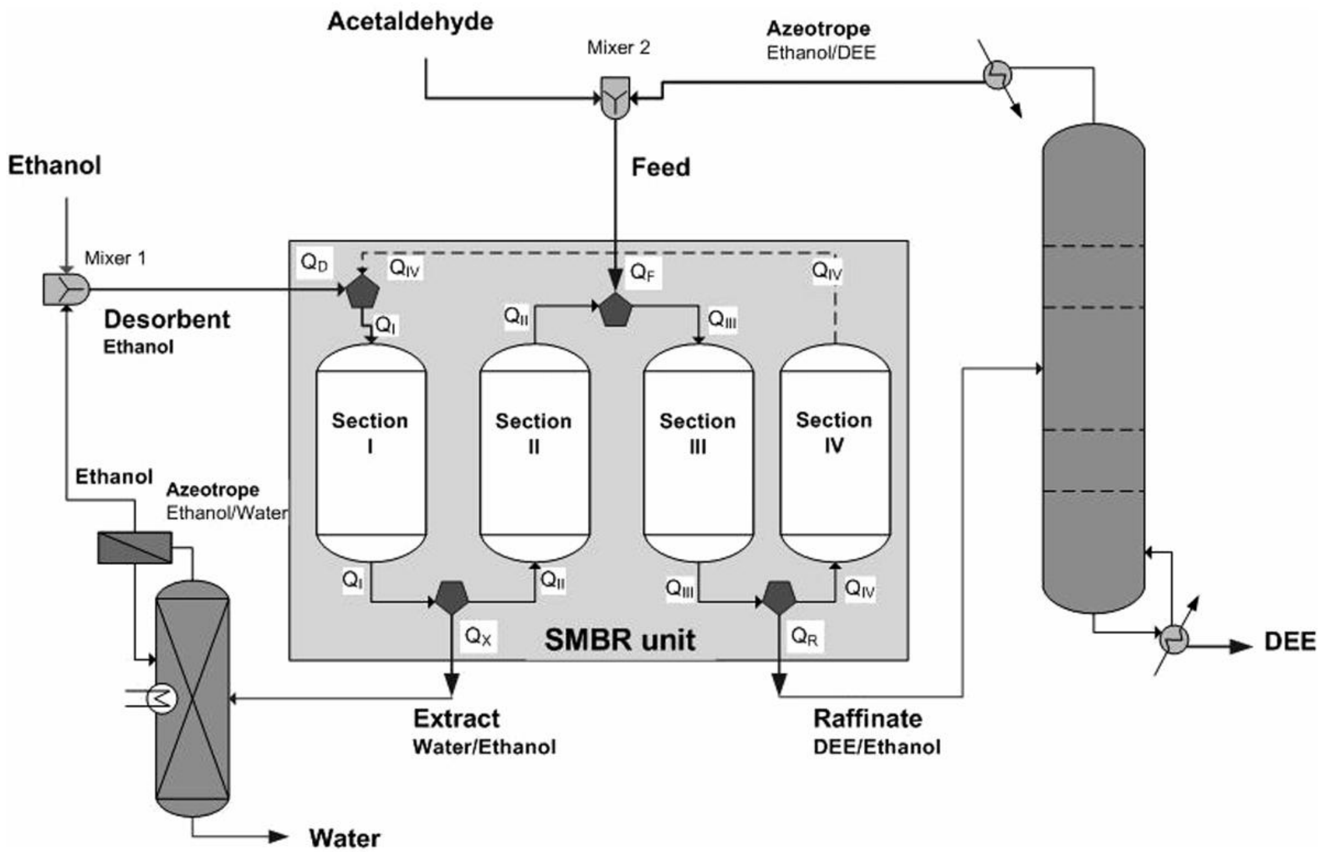
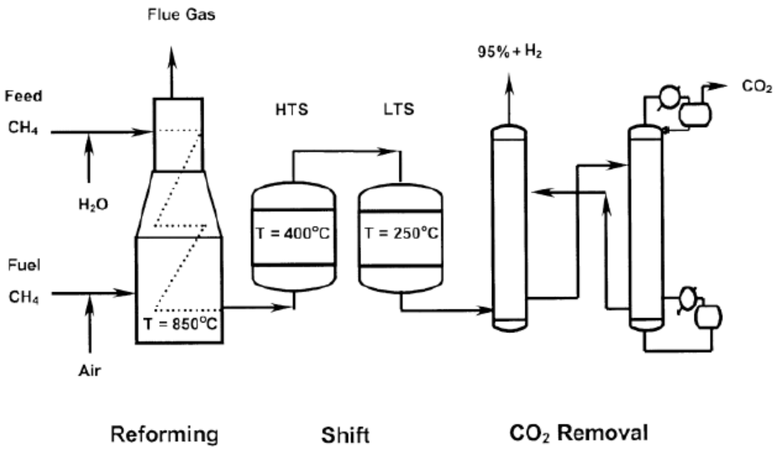


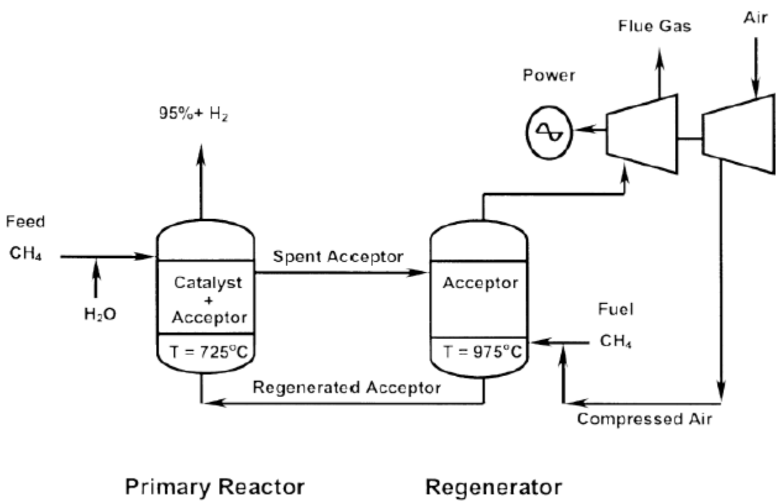

**Figure 6.** Process scheme for the production of DEE using an SMBR unit integrated with distillation and pervaporation units.

# SORPTION-ENHANCED REACTION PROCESSES

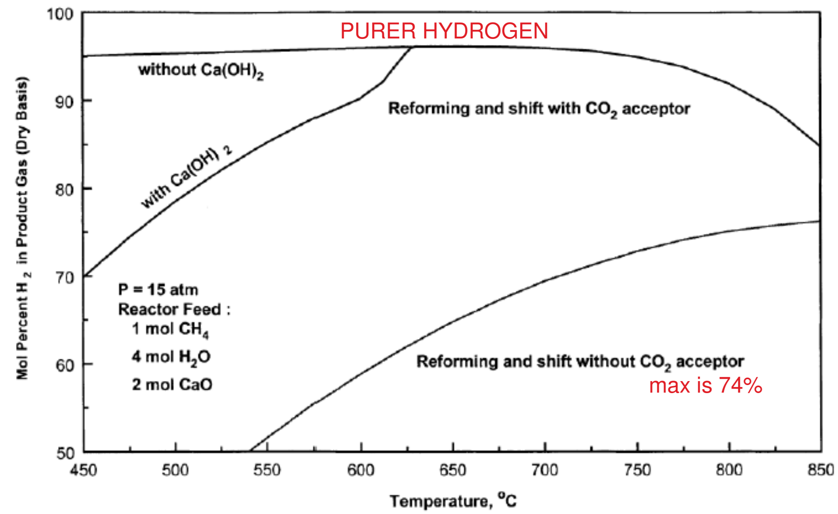
# Conventional and Sorption-Enhanced Steam Methane Reforming




Balasubramanian et al., Chem. Eng. Sci. 54 (1999) 3543




# Single-Step $CO_2$ Sorption-Enhanced Steam Methane Reforming




Thermodynamics for batch SMR reactor

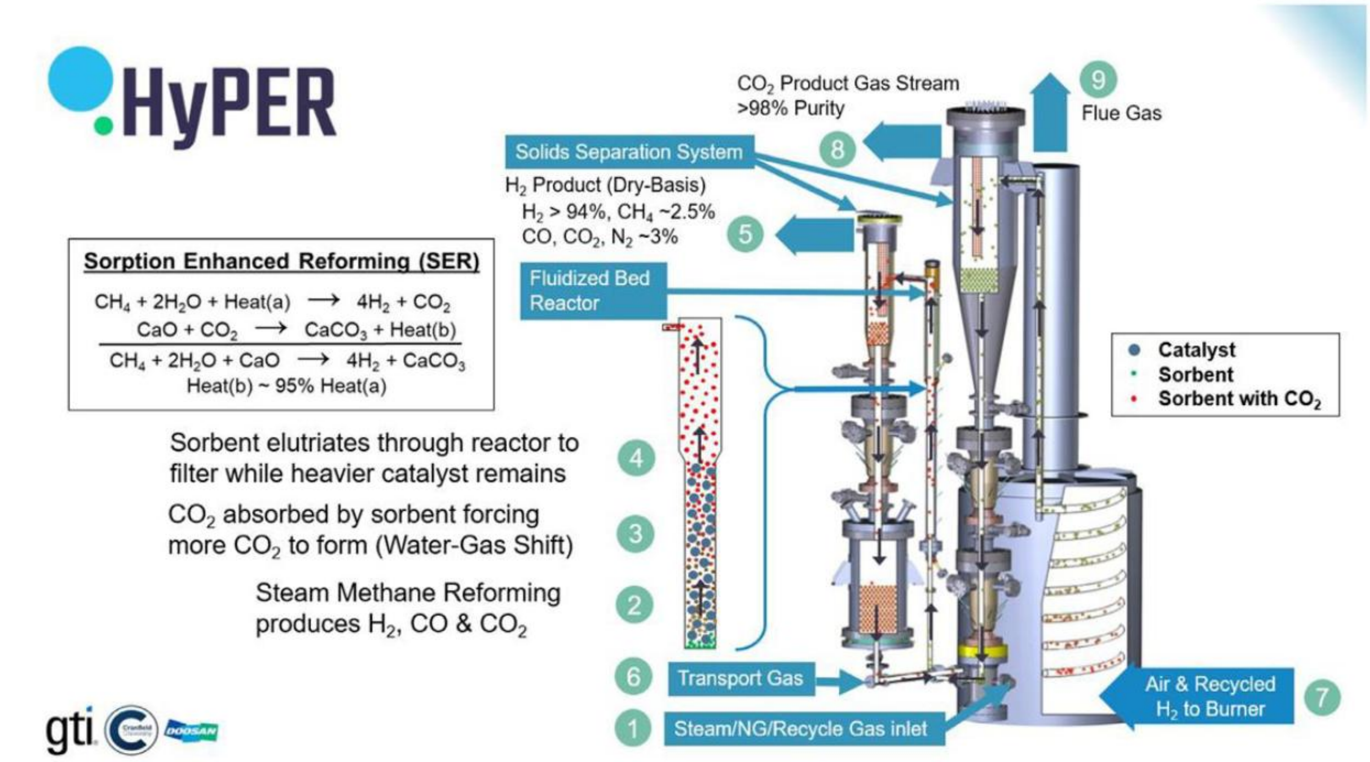
(T=450 $^0$C, P=1.5 atm, $\alpha$=6 mol $H_2O$/mol $CH_4$)

| fraction of $CO_2$ removed by chemisorption, $f$ (%) | equilibrium gas-phase mol % (dry basis) | | | | $CH_4$-to-$H_2$ conversion, $\chi$ (%) |
|---|---|---|---|---|---|
| | CO | $CO_2$ | $CH_4$ | $H_2$ | |
| 0.0 | 1.11 | 15.87 | 16.21 | 66.81 | 50.3 |
| 99.0 | 349 ppm | 0.24 | 3.31 | 96.42 | 87.9 |
| 99.5 | 189 ppm | 0.12 | 2.08 | 97.77 | 92.1 |
| 99.9 | 42 ppm | 248 ppm | 0.56 | 99.41 | 97.8 |



Hufton et al., AIChE J. 45 (1999) 248

# 1.5 MWth Pilot Plant System for SE-SMR Process




Farooqi et al., Carbon Capture Sci. Technol. 14 (2025) 100336

# INNOVATIONS IN CHEMICAL ENGINEERING AND TECHNOLOGY

| DAY | TOPIC | DETAILS |
|---|---|---|
| 1 | Getting started<br>CAT1 | Frequently asked questions on innovation |
| 2 | Innovations in chemical engineering<br>Business case for innovation<br>How to innovate during a crisis?<br>Innovation explained | Opportunities, examples, barriers, challenges<br>Phased gate process for innovation, SMEA<br>Innovation process similar to a chemical process<br>Types, fields, stages, strategies, measures, metrics, protection |
| 3 | Innovation in the chemical industry<br>Maturity of technology | Categories, product life cycle, levers, strategy, framework<br>TRL for the chemical industry, examples, TRL for entrepreneurs<br>Financial indicators for investment<br>Protection of innovation through IPR, patent filing<br>Ten lasting chemical engineering achievements |
| 4 | Examples of innovation in the chemical industry | Separations (distillation technology)<br>Process (ammonia, methanol)<br>Catalyst (FCC, alkylation<br>Business strategy (Refinery + petrochemicals)<br>Green chemistry, biorefinery, waste valorization<br>Process intensification and examples |
| 5 | Screening of innovative ideas | Examples at low and high TRLs |
| 6 | Newer concepts in innovation | Flow chemistry, batch vs. Continuous, Pilot plants, Scale-up, modularization in chemical industry, conceptual design of industrial plants and examples, assignment on process modifications |

| DAY | TOPIC | DETAILS |
|---|---|---|
| 7 | General concepts in innovation | Innovator’s dilemma<br>Strategies to enter a market<br>Disruptive technologies and examples<br>Innovative product design<br>Product life cycle<br>The innovation process<br>Stages of innovation<br>Classification in innovation<br>Risks of innovation<br>TRIZ methodology for innovation |
| 8 | Case studies in innovation | Fertilizer industry<br>Energy industry<br>Fine/specialty chemicals |
| 9 | | |
| 10 | | |

# The Innovator’s Dilemma

# The Innovator’s Dilemma

The innovator’s dilemma is very really problem that organizations face on a daily basis. They know that to survive into the future that they need to be producing the products of the future, and this means they need to be innovating. At the same time though, they also know that if they create a new product that they’ll be disrupting their existing, profitable products.

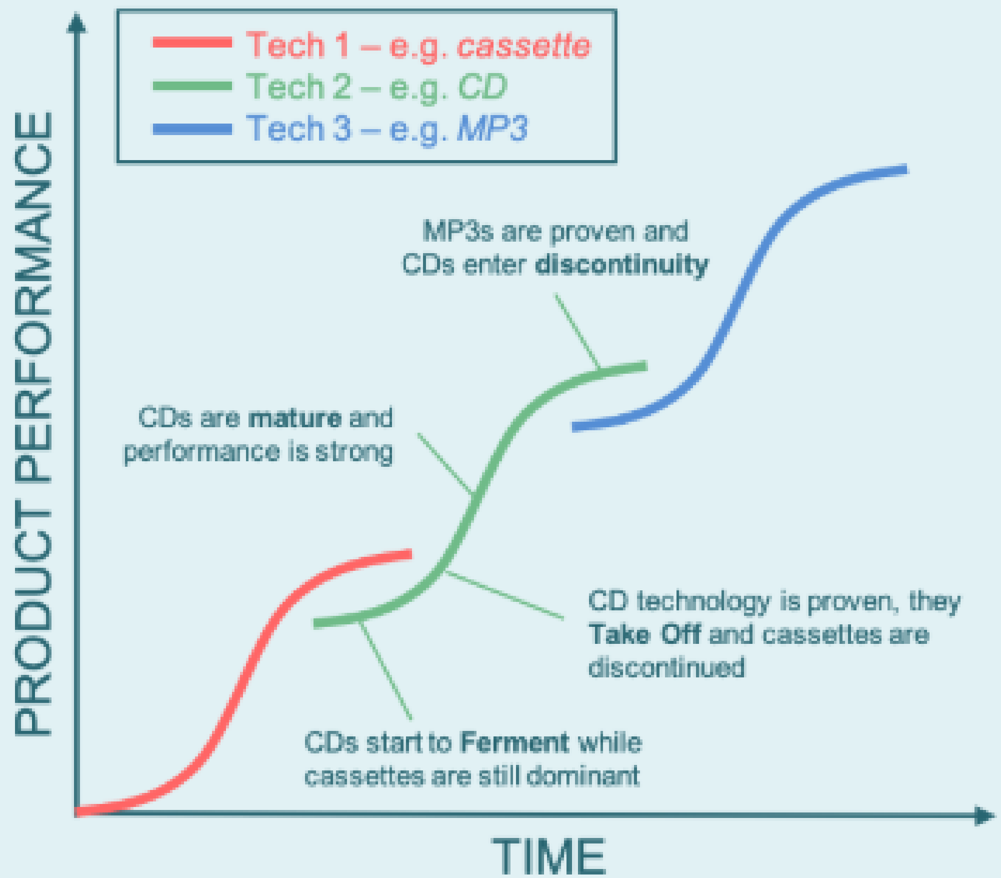
In our consideration of the innovator’s dilemma, we really focus on one aspect of this work, which is the concept of “S-curves” of innovation.

# Innovation S-Curves

So what is the innovation S-curve? Well, it is a model capturing a product or technology's life-cycle. The model effectively tracks the impact or popularity of a technology or product from the point at which it's first conceived of, to the point at which it is retired.

There are four stages in this model: ferment, take-off, maturity and discontinuity. The model plots performance (typically sales) vertically against time horizontally. The model says that the relationship between these dimensions for a product or technology adopts an "S" type shape. The gradient of this S curve shows how rapidly the new product or technology is being adopted (or dropped) by the market.

# THE INNOVATION S-CURVE




Technologies overlap. New ones initially grow slowly due to low investment while older technologies are mature and generating high profits for their companies.

As new technologies become proven, often by new market entrants, they take-off, displacing old technologies and becoming market leaders until they too are disrupted and replaced.

The pace of change and innovation and varies over time and between industries.

A particularly interesting feature of this model is how it effectively shows the transition between dominant technologies over a period of time. Each technology goes through one S in its lifetime. At the early stages of its life, it is competing with the technology that preceded it. At the later stages of its life it is competing against the technology that will ultimately replace it.

To help bring the S-curve to life, we'll consider these stages below, with a focus on CDs as a technology.

## At first, new technologies forment

And lo, before there were CDs there were cassettes. At the start of the technological life cycle of CDs, cassettes were the dominant technology in this domain. Their S-curve had a steep gradient. When CDs are introduced their curve has a shallow gradient. They don't make rapid inroads into the market, they simply ferment along as they attract early adopters and start to be accepted by the market.

At this stage they have some impact on the market dominance of cassettes, but not much. The cassette is king.

## Once established, new technologies take off

After a while though, the early adopters have tested the CDs and found them to be good. They promote them, marketers promote them, production volumes increase a bit, production costs fall a bit and the new technology finds a place in the world.

The curve steepens and the impact on the market is more pronounced. Mainstream consumers start to substitute their old cassette buying habits for new (and literally shiny) CDs.

At this stage new technologies really start to take off. Unfortunately though for the cassette, the rise of CDs signs the decline and discontinuity of the once mighty cassette tape.

# After take off, technologies mature

CDs are flying off the shelves now. Even late adopters know that they are the go-to technology for audio media. Everyone is buying them. Their increased sales volumes have hugely boosted production numbers. These scales of production, combined with incremental innovations have greatly reduced their costs as well.

The once new CD has fully arrived. It's loved by its consumers who consider it good value, and by its producers who generate great profits from it. The cassette is dead, long live the CD, the new mature technology in the market.

But unbeknownst to the CD, there are new stirrings in the market place. A few people have been asking if your even need CDs any more, or if you can just have hard-drives with audio files. These people are experimenting with MP3 players, and some early adopters are taking notice of this new technology that's beginning to ferment.

# After maturity comes discontinuity

The mighty CD now starts its inevitable decline. Its curve becomes more shallow. It's sales volumes fall. The incremental cost of production of each CD increases marginally as volumes decline. Organizations move their capital away from CD production towards new ventures to generate higher returns on equity. A few die-hard stragglers continue to buy CDs, but even they are starting to lose interest. Production splutters on for a while until it ultimately discontinues.

Of course, one technologies demise is another's rise, and it is now the time of the MP3 player.

However, as with all technologies, it too will be replaced. It's not long after the birth of the MP3 that people start asking why we need to own our music and why we can't just stream it.

## And so the cycle repeats

The S-curve tracks the rise and fall of technologies over time. Of course, not all make it through all the stages. Many ferment but never take off, but the dominant ones follow this trend. And it's this trend that causes some problems for innovators.

After all, if you were a dominant market player profiting well from a mature product, would you want to innovate a new replacement for it?

## The dilemma

So there is a clear dilemma. While market players who are not in dominant positions have a clear incentive to innovate, so that they can gain entry into a market, market leaders have no such incentive.

In fact, market leaders have a disincentive. All those producers who'd invested capital in CD production capability and were profiting well from their investments want nothing less than to see a new technology come to the market place.

At the same time though, they know that if they don't innovate themselves, someone else will and they may well lose their place in the market all together. Blockbuster, Kodak and other vanished entities are reminders of the perils of failing to move with the times.

At the same time though, they know that if they don’t innovate themselves, someone else will and they may well lose their place in the market all together. Blockbuster, Kodak and other vanished entities are reminders of the perils of failing to move with the times.

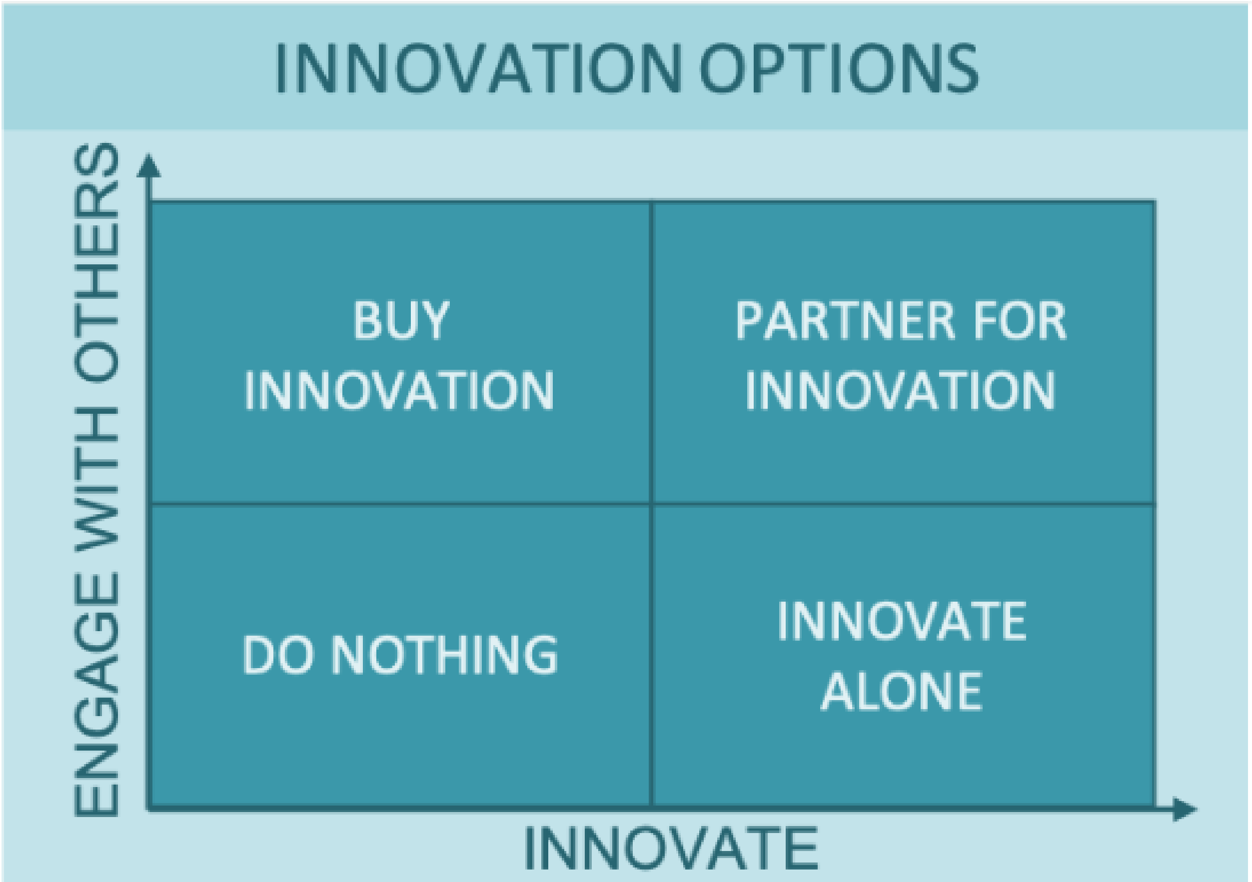
So what do they do? Should they invest in innovation and risk undermining their own market leading products in an effort to stay ahead of the curve? Or should they focus their resources solely on their market leading products? This is the question leaders must ask themselves every day when shaping their product portfolios and investment strategies.

Luckily for them, there are more options that just the two outlined above.

# And the responses

Organizations have a range of options that they can undertake in relation to innovation. We like to consider their options to be broadly speaking scalar between two dimensions: how much they engage with others, and how much they contribute to innovation themselves.

This structure leads to four broad strategies relating to innovation. Of course, organizations can adopt a mix of strategies across a mix of products or times. Despite this, we think this framework is a useful way to conceptualize some strategic decision making criteria.


INNOVATION OPTIONS
ENGAGE WITH OTHERS
BUY INNOVATION
PARTNER FOR INNOVATION
DO NOTHING
INNOVATE ALONE
INNOVATE

## Option 1: Don’t engage with others, and don’t innovate internally

This is the “do nothing” option. Here organizations choose not to invest in innovation internally. They also choose not to engage with others in relation to innovation.

In some industries, where product life-cycles are particularly long, or where product differentiation or marketing positions mean consumers are particularly loyal, this may be the right strategy. While Coke has innovated with new products and processes, it proudly states that its core product has retained its “original taste since 1886”.

# Option 2: Invest in innovation, but don’t engage others

This is the innovate alone option. Here organizations potentially become specialists in innovation, grow large research and development teams and become silos of creativity and product development.

This may be beneficial in industries where existing products have limited life-cycles and rewards are high, or where there are complexities associated with working with others. An example of where this strategy may work is in the pharmaceutical industry.

## Option 3: Don’t invest in innovation internally, but engage a lot with others

This is the buy innovation model. Here large organizations may know that their strengths lie in operational effectiveness and bringing products to market, but not product development. Instead of investing their capital in developing products, they instead trust themselves to pick winners, buy products, bring them to market and deliver them efficiently enough to generate suitable returns on capital.

Increasingly, some players in the tech industry can be seen shifting more towards acquisition of companies and innovations (as well as of innovative teams).

## Option 4: Invest in your own innovation, and engage with others

This is the partner for innovation model. Here organizations invest in and shape the innovative process, but know that they don't have the capability to innovate alone (or see other benefits in collaboration) so they work with other partner organizations on innovation.

This strategy has been popular recently in the FinTech space where traditional banks partner with newer, more agile and technologically capable partners for combined product innovations.

## So what’s the right answer?

Well, as ever, there isn’t one. Ever situation is a bit different, and every organization brings its own strengths, weaknesses, strategy, stakeholders and vision with it to the decision making process. While there is no one right answer, there are many options to choose from.

# Step-by-Step Guide to Creating an Effective Market Entry Strategy

## What is a Market Entry Strategy?

A market entry strategy is a plan businesses use to start selling their products or services in a new market. It helps you understand the new market, including customer needs, competitors, and potential challenges. This planning ensures that your business can effectively reach and attract new customers.

The strategy outlines the best way to enter the market, whether it's through exporting, partnering with local firms, or opening a new branch. By carefully planning and making informed decisions, you increase your chances of success and avoid common pitfalls.

## Key Components of a Solid Market Entry Strategy

Creating a successful market entry strategy requires careful planning and consideration of several important elements. Here are the key components that make up a solid market entry strategy:

1. **Market research** - Conduct thorough research to understand customer needs, market size, growth potential, and trends. Analyze the competition and potential entry barriers.
2. **Clear objectives** - Define clear business goals, such as increasing sales or expanding market share, to guide your strategy and measure success.
3. **Target audience** - Analyze your target audience's demographics, buying habits, and pain points to tailor your products or services to their needs.

4. **Competitive analysis** - Analyze competitors to identify their strengths and weaknesses. Use this information to differentiate your offerings and position your brand effectively.
5. **Entry strategy selection** - Choose the best market entry strategy, like exporting or joint ventures, based on your research, goals, and available resources.
6. **Financial plan** - Develop a budget that includes marketing, distribution, and staffing costs, ensuring it aligns with your objectives and risk tolerance.

7. **Marketing and sales strategy** - Create a marketing and sales plan that includes brand positioning, pricing, and promotion to effectively reach and resonate with your target audience.
8. **Risk management** - Identify potential risks like political or cultural challenges and develop strategies to mitigate them while staying flexible to adapt your plans.
9. **Performance metrics** - Set KPIs to track success, such as sales growth or market share, and regularly monitor progress to adjust your strategy as needed.

# Why are Market Entry Strategies Important

Market entry strategies are important because they help your business successfully start selling in a new market. Here's why you need to invest in market entry strategies.

## Better preparation

A market entry strategy helps you understand the new market in detail. This includes learning about local customer preferences, cultural differences, and what the market needs. With this knowledge, you can tailor your products or services to better meet local demands.

## Informed decisions

The strategy helps you decide the best way to enter the new market. Whether it's exporting products, partnering with local businesses, or opening a new branch, having a plan ensures you choose the method that fits your goals and market conditions.

## Avoid mistakes

By planning ahead, you can identify potential challenges before they arise. This helps you avoid costly mistakes, like failing to comply with local regulations or misunderstanding customer needs. A good strategy helps you address these issues in advance.

## Increased success

A well-thought-out strategy improves your chances of success in the new market. It helps you use your resources wisely, execute your plans effectively, and achieve your business goals, leading to better growth and profitability.

# How to Create a Market Entry Strategy

Expanding into a new market can boost your business, but it requires a solid plan. A market entry strategy helps you navigate this process by outlining how to introduce your products or services effectively. Here's how to create a successful market entry strategy in 7 steps.

## 1. Research the market

Start by learning about the new market. Find out who your potential customers are, what they need, and how big the market is. Look at your competitors and see what they are doing. This will help you understand if the market is a good fit for your business.

## 2. Define your goals

Decide what you want to achieve in the new market. Are you looking to increase sales, build brand awareness, or expand your customer base? Clear goals will guide your strategy and help you measure your success.

## 3. Choose your entry method

Decide how you will enter the market. Options include exporting your products, forming partnerships with local companies, franchising, or setting up a new branch. Each method has its pros and cons, so choose the one that best suits your goals and resources.

## 4. Develop a marketing plan

Create an action plan for how you will promote your products or services in the new market. This includes setting prices, deciding on advertising methods, and choosing how to distribute your products. Your plan should be tailored to the preferences and needs of the local customers.

## 5. Understand legal requirements

Learn about the local laws and regulations that affect your business. This includes rules on taxes, business registration, and product standards. Make sure you follow all legal requirements to avoid issues.

## 6. Build local connections

Find local partners, suppliers, or employees who can help you navigate the new market. Building relationships with local businesses and individuals can make it easier to enter and succeed in the market.

## 7. Monitor and adjust

Once you enter the market, keep track of how things are going. Measure your success against your goals and gather feedback from customers. Be ready to adjust your strategy if needed to improve your results.

## 15 Types of Market Entry Strategies

Choosing the right market entry strategy is essential for successful expansion. Each strategy offers different levels of risk, control, and investment. By understanding these options, you can select the one that best fits your business goals and resources. Here are 15 common market entry strategies to consider.

### 1. Exporting

Exporting means selling your products or services from your home country to customers in a new market. This is often the easiest and least expensive way to test a new market since you don't need to set up local operations. You just ship your products and handle sales from afar.

## 2. Licensing

Licensing allows a local company to use your brand name, technology, or product design in exchange for a fee or royalties. This means you can enter the new market with less risk and investment because you don't need to set up your own operations. The local company handles production and distribution.

## 3. Franchising

Franchising involves letting a local business use your brand and business model. The franchisee pays you fees and follows your system to operate their business. This helps you expand quickly because the franchisee brings local knowledge and manages day-to-day operations.

## 4. Joint ventures

A joint venture is a partnership with a local company to create a new, shared business. Both companies contribute resources, share risks, and split profits. This method combines the strengths of both partners and helps navigate local market conditions together.

## 5. Wholly owned subsidiaries

A wholly owned subsidiary is a company you fully own and control in the new market. You can set it up from scratch or buy an existing local company. This gives you full control over operations and business decisions, but it requires a significant investment and carries higher risks.

## 6. Strategic alliances

Strategic alliances are partnerships with local companies to work on specific projects or goals. Unlike joint ventures, these alliances are often less formal and can be focused on short-term objectives like joint marketing efforts or shared technology.

## 7. Piggybacking

Piggybacking means using the existing distribution network of a local company to sell your products. This allows you to enter the market quickly and cost-effectively by leveraging the local company's established relationships and infrastructure.

## 8. Direct investment

Direct investment involves putting money into building new facilities, offices, or stores in the new market. This method gives you full control over your operations and business practices, but it requires a large financial investment and involves higher risks.

## 9. E-commerce

E-commerce is selling your products online to customers in the new market. This approach allows you to reach a wide audience without needing a physical presence in the market. It’s a cost-effective way to test demand and build a customer base.

## 10. Contract manufacturing

Contract manufacturing means outsourcing the production of your products to a local manufacturer. This way, you can enter the market without investing in your own production facilities. The local manufacturer produces the goods according to your specifications.

## 11. Turnkey projects

Turnkey projects involve setting up a complete business operation in the new market and then handing it over to a local company. This means you handle everything from construction to set up and then transfer the fully operational business to the local partner.

## 12. Management contracts

Management contracts allow you to provide management expertise to a local company in exchange for fees. You oversee operations and help run the business, but the local company owns it. This method gives you operational control with less financial risk.

## 13. Acquisitions

Acquisitions involve buying an existing company in the new market. This gives you immediate access to the market, including its customer base, facilities, and staff. It’s a quick way to enter the market but requires significant investment and careful integration.

## 14. Greenfield investments

Greenfield investments mean building new facilities or operations from scratch in the new market. This approach offers complete control over the setup and operations but requires a substantial financial investment and a long time to become operational.

## 15. Cooperative agreements

Cooperative agreements involve working with local firms for specific activities like marketing or distribution. These agreements allow you to benefit from local expertise and networks without forming a formal joint venture or partnership. They can be flexible and focused on particular needs.

# Domestic vs International Market Entry

Expanding your business can mean entering either domestic or international markets, each with its own opportunities and challenges. Here's a simple breakdown of the differences:

## Domestic market entry

- **Familiarity**: Expanding within your home country means you already understand the culture, language, and customer preferences. This familiarity makes it easier to tailor your products or services to local needs.
- **Regulations**: You are already aware of local laws and regulations, which simplifies the process of compliance.
- **Logistics**: Managing logistics, such as shipping and distribution, is usually simpler and less costly within your own country.
- **Competition**: You may face more direct competition from established local businesses but can leverage your existing brand recognition.

## International market entry

- **New opportunities**: Expanding internationally opens up new customer bases and growth opportunities. You can reach markets with different demands and less competition for your products.
- **Cultural differences**: Understanding and adapting to different cultures, languages, and preferences is crucial for success.
- **Regulations and compliance**: Navigating foreign laws, trade regulations, and tariffs can be complex and may require expert assistance.
- **Logistics and costs**: International expansion often involves higher logistics costs, including shipping and potential import duties. It may also require setting up local operations or partnerships.

Choosing between domestic and international market entry depends on your business goals, resources, and risk tolerance. Domestic expansion is often simpler and less risky, while international expansion offers the potential for significant growth but comes with added complexity. Understanding these differences can help you make an informed decision on how best to expand your business.

# How to Select the Correct Market Entry Strategy

Choosing the right market entry method depends on your company's objectives, available resources, and the target market environment. Here are some steps to guide your decision:

## 1. Understand your goals

Clearly define what you want to achieve in the new market. Are you looking to increase sales, build brand awareness, or gain market share? Knowing your objectives will help you choose the most suitable strategy.

## 2. Analyze the market

Research the new market thoroughly. Understand customer preferences, cultural differences, and demand for your products or services. Identify potential competitors and analyze their strengths and weaknesses.

## 3. Assess your resources

Evaluate your financial and operational resources. Determine how much investment you can afford and whether you have the necessary expertise to enter the new market. This will help you decide if a low-cost strategy like exporting is better or if you can invest in a wholly owned subsidiary.

## 4. Evaluate risk

Consider the risks associated with each market entry strategy. Strategies like franchising and licensing involve less risk, while direct investment and acquisitions carry higher risks but offer more control.

## 5. Consider control

Decide how much control you want over your operations in the new market. Licensing and franchising give you less control, while joint ventures and wholly owned subsidiaries offer more.

## 6. Choose your entry strategy

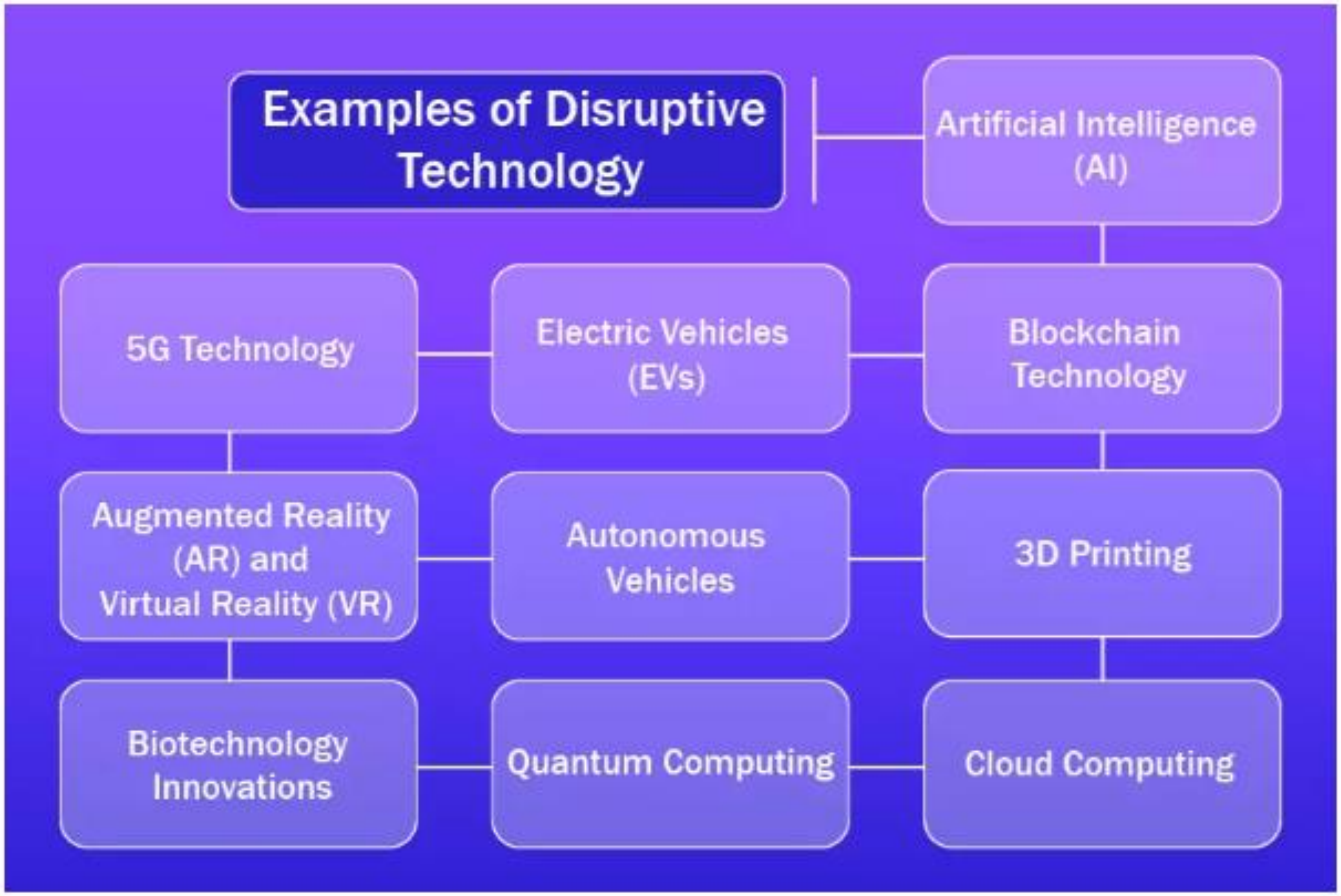
Based on your goals, market research, resources, risk tolerance, and desired level of control, choose the market entry strategy that best fits your needs. For example, if you want full control and have the resources, a wholly owned subsidiary might be ideal. If you want to minimize risk, exporting or licensing might be better.

## 7. Plan your execution

Once you have selected your strategy, create a detailed plan for execution. Outline the steps needed to enter the market, including timelines, budgets, and resources required. Be prepared to adapt your plan based on market feedback and performance.

# **Disruptive** Technology

## Examples of Disruptive Technology



## How is Disruptive Technology Different from Sustaining Technology?

## How to Implement Disruptive Technologies in a Company

## What Are The Challenges Regarding Disruptive Technology?

**Examples of Effective Disruptive Technology Management**

There are numerous examples of firms that have effectively managed disruptive technologies strategically.

- **Apple** is one of the most well-known, having continually been at the forefront of technical innovation with products such as the iPhone, iPad, and Apple Watch. Apple's success can be credited in part to its ability to predict and respond to evolving technology trends, as well as its dedication to making products that are both visually beautiful and user-friendly.

- **Amazon** is another example, having disrupted several industries with its innovative business models and use of new technology such as artificial intelligence and machine learning. Amazon's success can be attributed to its openness to trying new things, its emphasis on customer-centricity, and its ability to use technology to create new markets and opportunities.

- **Netflix** is a great illustration of how a firm may use disruptive technology to change its business model. Initially, the company operated as a DVD rental service. With the development of streaming technology, however, Netflix soon shifted to becoming a streaming service. The company invested substantially in building its streaming platform, and it is now one of the world's most popular streaming services. Netflix's success demonstrates the potential of disruptive technologies as well as the need to be able to quickly adjust to changing market conditions.

- **Tesla** is a game-changing technology business that has transformed the automotive industry. The company's emphasis on electric vehicles and renewable energy has upended the traditional automotive sector, which has long been dominated by gasoline-powered automobiles. Tesla has also made significant investments in autonomous driving technology, which has the potential to change the way we travel in the future.

- **Airbnb** is a technology startup that has revolutionized the lodging business. The organization runs a platform that allows homeowners to rent out their homes to tourists. This has generated a new market for travelers looking for unusual and economical lodging. Airbnb has successfully disrupted the hotel sector by harnessing technology to offer a more efficient and cost-effective method of connecting tourists with hosts.

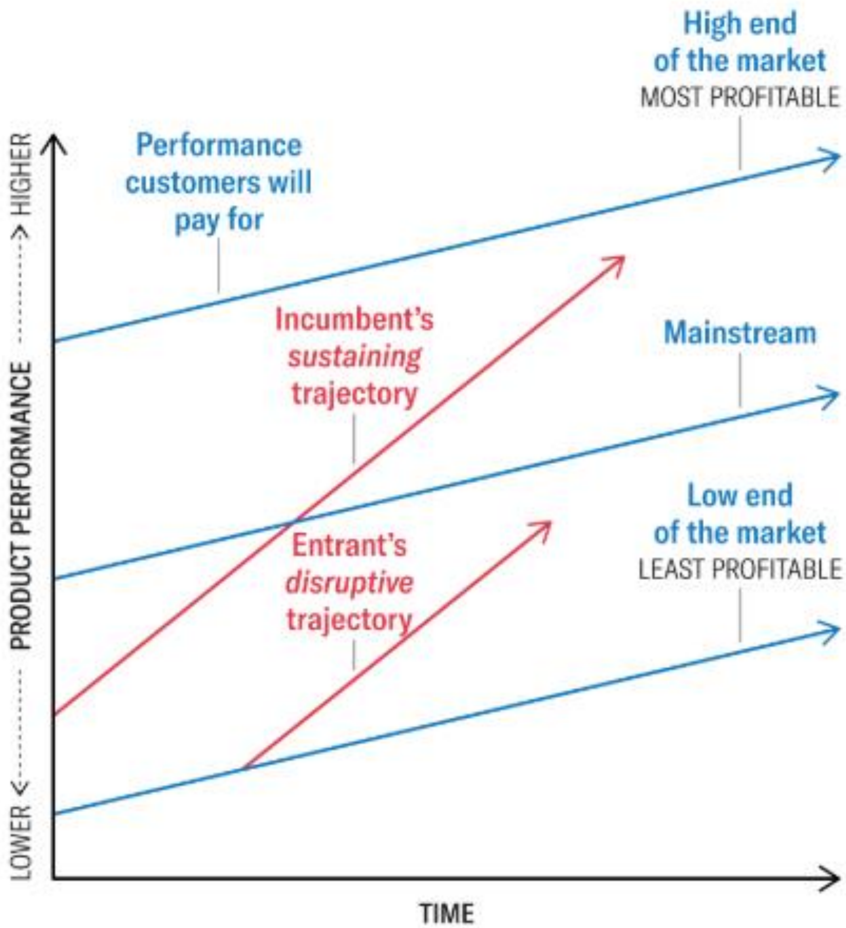
These examples show that disruptive technology management is about more than just embracing new technologies; it is also about being able to quickly respond to changing market conditions and consumer needs. Organizations must be nimble, innovative, and customer-focused to effectively manage disruptive technology. They must be willing to experiment and take chances while also managing the dangers connected with disruptive technologies. Strategic planning, organizational transformation, and effective implementation are all required for successful disruptive technology management.

# What is disruptive innovation?

The theory of disruptive innovation, popularized by Harvard Business School professor and frequent Harvard Business Review contributor Clayton Christensen, explains how new technologies, new products, or new services can start small but eventually surpass established offerings in the current market.

There are two key components to disruptive innovation:

- **New market disruption** occurs when a product or service reaches overlooked markets or customers not considered before. This typically involves targeting profitable customers and expanding into upmarket customer segments.
- **Low-end disruption** provides a more affordable or user-friendly alternative to existing products, appealing to mainstream customers or cost-conscious customers at the bottom of the market.




Source: Clayton M. Christensen, Michael E. Raynor, and Rory McDonald
From: "What Is Disruptive Innovation?" December 2015

HBR

# The Disruptive Innovation Model

This diagram contrasts *product performance trajectories* (the red lines showing how products or services improve over time) with *customer demand trajectories* (the blue lines showing customers' willingness to pay for performance). As incumbent companies introduce higher-quality products or services (upper red line) to satisfy the high end of the market (where profitability is highest), they overshoot the needs of low-end customers and many mainstream customers. This leaves an opening for entrants to find footholds in the less-profitable segments that incumbents are neglecting. Entrants on a disruptive trajectory (lower red line) improve the performance of their offerings and move upmarket (where profitability is highest for them, too) and challenge the dominance of the incumbents.

# 9 EXAMPLES OF INNOVATIVE PRODUCTS

## WHAT ARE THE THREE TYPES OF INNOVATION?

Innovation in business isn't limited to products. While there's no denying Apple's transformative products have made them an industry giant, it's important to remember that innovation has several applications. Here are three types of innovation your company can design and implement.

- **Product or service innovation**: Creating a new product or service or enhancing an existing one, such as the internet or the pivoting head of Gillette razor blades
- **Process innovation**: Implementing changes to make a process stronger or more efficient, such as assembly lines in manufacturing
- **Business model innovation**: Transforming business operations like ride-sharing platforms Uber and Lyft did by taking the taxi and car service companies' business model and altering it to a peer-to-peer, digitized one

# 3 Types of Innovation

1

**PRODUCT OR SERVICE INNOVATION**

Creating a new product or service or enhancing an existing one

2

**PROCESS INNOVATION**

Implementing changes to make a process stronger or more efficient

3

**BUSINESS MODEL INNOVATION**

Transforming business operations to increase your organization and customers' value

## WHAT MAKES A PRODUCT INNOVATIVE?

In business, innovation is an original idea that's useful to consumers. But what does that mean, and how can you ensure your idea has these two essential ingredients?

One way is by identifying and addressing the pain points your consumers are experiencing. There are two types of pain points innovation should address: explicit and latent.

- **Explicit pain points**: Customers are aware of and can easily define these pain points.
- **Latent pain points**: These are more difficult to define because most customers aren't aware of them.

Innovative ideas focused on users' challenges have a better chance of success and longevity. Understanding your innovation's viability can be harder than identifying pain points, but it's another crucial factor in this process.

Remember, innovations aren't inherently modern. Netflix's streaming service is a successful innovative product but grounded in the modern world. Even ancient innovations, such as the creation of language, are examples of how prospective innovators should approach the creative process. Consider this when looking for inspiration and guidance in your innovation process.

# 9 INNOVATIVE PRODUCT EXAMPLES

Recency bias—limiting your understanding of innovation to modern products and services—can be detrimental to the innovation process. Don't let a narrowed perspective of what successful innovation is negatively affect your creativity.

Here are nine incredibly successful innovations that have stood the test of time.

### 1. The Wheel

Invented around 4000 BCE, the wheel is one of the earliest recorded innovations. While it's often forgotten as an innovative product, it continues to have an impact. Its inventive design addressed a common pain point around moving multiple heavy objects at once. The result was a circular frame that allows users to transport many heavy items in a short time. Its significance is still felt today and has led to additional breakthroughs, such as carriages and today's more modern transportation methods.

### 2. The Printing Press

Your favorite book or magazine wouldn't exist without the printing press. This technological breakthrough was novel and useful in that it allowed for the mass production of written documents. It solved an explicit pain point in document production: time consumption and tediousness. Creating a product that eliminated the handwriting element transformed the publishing world by making the process easier and quicker.

### 3. The Lightbulb

Although there's some debate on who invented the lightbulb, no one denies its significance. It's a great example of an innovative product that solved explicit and latent pain points. Before lightbulbs, products like lanterns and oil lamps produced light but made houses more susceptible to fires. At the time, these accidents were accepted as a necessary risk until innovation showed people otherwise.

### 4. Automobiles

Tesla founder and CEO Elon Musk wouldn't be the business mogul he is today without the initial innovation of motorized automobiles. The automobile's invention in 1886 kickstarted a technological evolution by focusing on the transportation landscape's challenges, such as fatigue from walking or bicycling and caring for horses that pulled carriages. Today, horse-drawn carriages are nearly obsolete beyond tourist attractions and services.

### 5. Computers

Computers have completely changed everyday life. Since their humble beginnings of automating mathematical equations, computers have progressed and evolved according to users' ever-changing pain points. For example, computers were originally enormous, spanning nearly 50 feet long and weighing almost five tons. Over time, their size and portability have reduced from desktop computers to laptops and smartphones.

### 6. Cellular Phones

While cellular phones also evolved, they initially solved a specific problem for phone users: landlines weren't portable. People were tethered to house phones, beepers, and phone booths if they wanted to receive a call. Cellular phones allowed users to take calls from anywhere. As more consumers bought cellular phones, this product solved latent pain points about safety outside the house and emergency contacting.

### 7. The Internet

The internet is such a widely used product that it's hard to imagine a world without it. In this way, it may be the most successful modern innovation. It was originally based on the expression "information at your fingertips." Although limited information was accessible to those with a library card, basic cable, and a newspaper subscription, waiting for information was still inconvenient. The internet solved this latent pain point by becoming a vast hub of instantaneous knowledge and information.

### 8. Bagless Vacuum Cleaner

Bagless vacuums may seem like an odd addition to this list, but it's a great example of how simple updates to a product can impact an industry. James Dyson, an industrial designer, was frustrated with the process of emptying his vacuum cleaner bags. They sometimes caused clogs and buildup that affected the vacuum's performance. With these pain points in mind, he built the first bagless vacuum cleaner. Since then, Dyson has revolutionized cleaning technology and continues to innovate based on its users' key pain points.

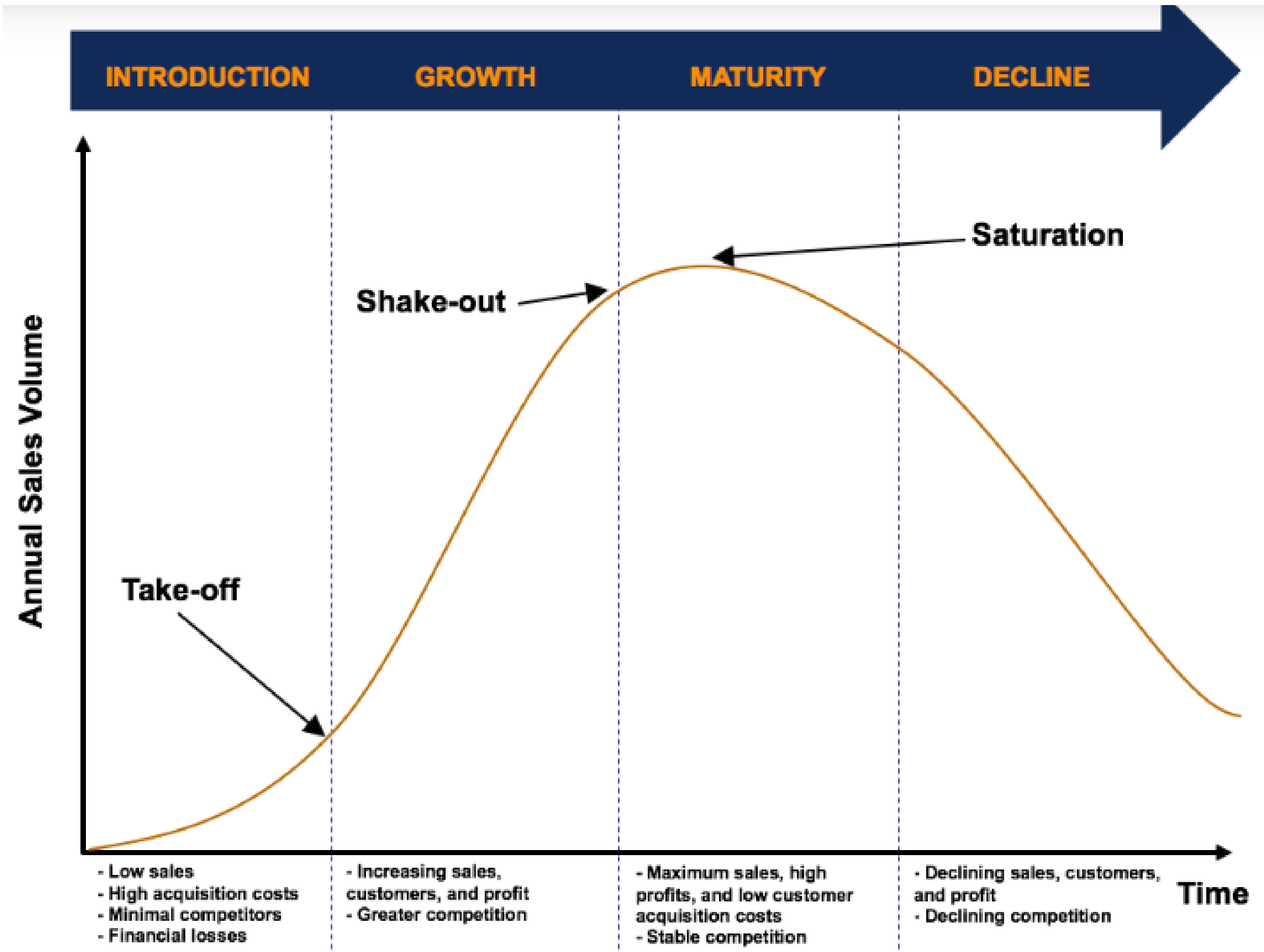
### 9. iPhones

It's no surprise that Apple products are almost always mentioned in any discussion about innovation. The iPhone is a modern innovation that revolutionized cellular phone technology. While computers and cell phones were constantly evolving, Steve Jobs understood that consumers' latent need for portability and speed couldn't be solved with a computer or phone alone. This is what led to the iPhone.

# PRODUCT LIFE CYCLE

## What is the Product Life Cycle?

The Product Life Cycle (PLC) defines the stages that a product moves through in the marketplace as it enters, becomes established, and exits the marketplace. In other words, the product life cycle describes the stages that a product is likely to experience. It is a useful tool for managers to help them analyze and develop strategies for their products as they enter and exit each stage.



INTRODUCTION
GROWTH
MATURITY
DECLINE
Annual Sales Volume
Saturation
Shake-out
Take-off
- Low sales
- High acquisition costs
- Minimal competitors
- Financial losses
- Increasing sales, customers, and profit
- Greater competition
- Maximum sales, high profits, and low customer acquisition costs
- Stable competition
- Declining sales, customers, and profit
- Declining competition
Time

### 1. Introduction Stage

When a product first launches, sales will typically be low and grow slowly. In this stage, company profit is small (if any) as the product is new and untested. The introduction stage requires significant marketing efforts, as customers may be unwilling or unlikely to test the product. There are no benefits from economies of scale, as production capacity is not maximized.

The underlying goal in the introduction stage is to gain widespread product recognition and stimulate trials of the product by consumers. Marketing efforts should be focused on the customer base of innovators – those most likely to buy a new product. There are two price-setting strategies in the introduction stage:

- **Price skimming**: Charging an initially high price and gradually reducing ("skimming") the price as the market grows.
- **Price penetration**: Establishing a low price to quickly enter the marketplace and capture market share, before increasing prices relative to market growth.

### 2. Growth Stage

If the product continues to thrive and meet market needs, the product will enter the growth stage. In the growth stage, sales revenue usually grows exponentially from the take-off point. Economies of scale are realized as sales revenues increase faster than costs and production reaches capacity.

Competition in the growth stage is often fierce, as competitors introduce similar products. In the growth stage, the market grows, competition intensifies, sales rise, and the number of customers increases. Price undercutting in the growth stage tends to be rare, as companies in this stage can increase their sales by attracting new customers to their product offerings.

### 3. Maturity Stage

Eventually, the market grows to capacity, and sales growth of the product declines. In this stage, price undercutting and increased promotional efforts are common as companies try to capture customers from competitors. Due to fierce competition, weaker competitors will eventually exit the marketplace – the shake-out. The strongest players in the market remain to saturate and dominate the stable market.

The biggest challenge in the maturity stage is trying to maintain profitability and prevent sales from declining. Retaining customer brand loyalty is key in the maturity stage. In addition, to re-innovate itself, companies typically employ strategies such as market development, product development, or marketing innovation to ensure that the product remains successful and stays in the maturity stage.

### 4. Decline Stage

In the decline stage, sales of the product start to fall and profitability decreases. This is primarily due to the market entry of other innovative or substitute products that satisfy customer needs better than the current product. There are several strategies that can be employed in the decline stage, for example:

- Reduce marketing efforts and attempt to maximize the life of the product for as long as possible (called milking or harvesting).
- Slowly reducing distribution channels and pulling the product from underperforming geographic areas. Such a strategy allows the company to pull the product out and attempt to introduce a replacement product.
- Selling the product to a niche operator or subcontractor. This allows the company to dispose of a low-profit product while retaining loyal customers.

# Benefits and Drawbacks of Using the Product Life Cycle

### Benefits

- Clarify portfolio of offerings
- Better allocation of resources
- Positive impact on economic growth
- Promotes innovation

### Drawbacks

- Not appropriate for every industry or product
- Legal or trademark restrictions
- Planned obsolescence
- Product or resource waste

## Prolonging the Mature Stage

Many of the most successful products on earth are suspended in the mature stage for as long as possible, undergoing minor updates and redesigns to keep them differentiated. Examples include:

- Apple computers and iPhones
- Ford trucks
- Starbucks' coffee

All of these products undergo minor changes accompanied by major marketing efforts, which are designed to keep them feeling unique and special in the eyes of consumers.

# EXAMPLE OF THE PRODUCT LIFE CYCLE

## Coca-Cola

On April 23, 1985, Coca-Cola announced a new formula for its popular beverage, referred to as "new Coke." Coca-Cola's market-share lead had been decreasing over the past 15 years, and the company decided to launch a new recipe in hopes of reinvigorating product interest. After its launch, Coca-Cola's phone line began receiving 1,500 calls per day, many of which were to complain about the change. Protest groups recruited 100,000 individuals to support their cause of bringing "old" Coke back. [6]

A stunning 79 days after its launch, "new Coke's" full product life cycle was complete. Though the product didn't experience much growth or maturity, its introduction to the market was met with heavy protest. Less than three months after it announced its new recipe, Coca-Cola announced it would revert its product back to the original recipe.

## What Are Product Life Cycle Strategies?

Depending on the stage a product is in, a company may adopt different strategies along the product life cycle. For example, a company is more likely to incur heavy marketing and R&D costs in the introduction stage. As the product becomes more mature, companies may then turn to improving product quality, entering new segments, or increasing distribution channels. Companies also strategically approach divesting from product lines including the sale of divisions or discontinuation of goods.

## What Is Product Life Cycle Management?

Product life cycle management is the act of overseeing a product's performance over the course of its life. Throughout the different stages of product life cycle, a company enacts strategies and changes based on how the market is receiving a good.

## Why Is Product Life Cycle Important?

Product life cycle is important because it informs management of how its product is performing and what strategic approaches it may take. By being informed of which stage its product(s) are in, a company can change how it spends resources, which products to push, how to allocate staff time, and what innovations they want to research next.

## Which Factors Impact a Product's Life Cycle?

Many factors can affect how a product performs and where it lies within the product life cycle. In general, the product life cycle is heavily impacted by market adoption, ease of competitive entry, rate of industry innovation, and changes to consumer preferences. If it is easier for competitors to enter markets, consumers can change their minds frequently about the goods they consume, or the market may quickly become saturated. Then, products are more likely to have shorter lives throughout a product life cycle.

## The Bottom Line

Broadly speaking, almost every product sold undergoes the product life cycle. This cycle of market introduction, growth, maturity, and decline may vary from product to product, as well as from industry to industry.

This cycle can help a company make decisions about resource allocation, track the outlook of products, and strategically plan for bringing new products to market.

# What is Innovation Strategy? Stages, types & examples

# What is innovation strategy?

**An innovation strategy** is a detailed roadmap with a series of strips that helps your organization reach its future goals. This roadmap isn't just a guide for business success — it's a guide that helps you keep up in your industry by thinking of new and innovative ways to tackle problems.

These innovations can be creating new products to solve future problems, optimizing business processes to be more efficient, or thinking of entirely new ways to disrupt your industry.

Ultimately, your innovation strategy will help you align your business with your customer's needs and create products and services to give your company a competitive advantage.

# What are the stages of innovation strategy?

The innovation process doesn't happen in a vacuum — it requires a clear process that takes a new idea and brings it to market.

## 1. Idea generation and ideation

Idea generation is the foundation of an innovation system. It's the process that helps companies develop ideas to compete in the market and explore new opportunities without taking too much risk.

Many companies innovate by taking advantage of new technologies for insights — learning from data to see what problems customers have that you can solve. Innovators can use this information to explore new ideas and brainstorm ways to bring those ideas to life.

The vital part of this process is customer feedback. A great idea needs a customer base available to see success — and not every idea will have people around willing to pay money to access it. New ideas need customer validation to determine if they are viable.

## 2. Idea evaluation and selection

Idea evaluation and selection is assessing new ideas to see if they are feasible. During this process, you'll look at costs, time to market, potential profit, and other business metrics to see if it's worth pursuing an idea.

If it looks like a product you're considering has a chance to succeed, then the next step is to compare it with what's currently on the market. Is your new product something new that doesn't exist yet, or does it improve on existing products? Will this help your company stand out?Use this information to find the unique value proposition and determine if the idea is worth pursuing.

## 3. Implementation and execution

The implementation and execution phase is where you take your new idea and turn it into a concrete plan to make it a reality. But this stage is more than building a product — it's also about creating a successful business strategy and finding help to create your idea.

The first step is product development. Which vendors can supply the materials for your product or help you manufacture it? Which members of your team will be the best fit for development?

Once you find the right people, implementation becomes a matter of executing a plan. It involves project management, resource management, process management, and continuous improvement.

## 4. Monitoring and evaluation

You're breaking new ground when innovating in an industry, so you may not get everything right. Monitoring and evaluation will help you handle those problems and adapt your innovation efforts.

Your goal when monitoring a new project is to assess your performance. Is what you're doing helping you achieve your goals and keeping you on track for release?

A key part of doing this is metrics. These numbers will tell you how well you are doing. Track your time to release, product progress, defects, and eventual return on investment. Reassess your strategy and make changes if your metrics show you aren't meeting your goals.

# What are the types of innovation strategy?

There isn't one type of innovation in business. Let's look at four of the main ways you can create new and innovative ideas.

- **Product innovation**: this is the process of building new products or upgrading existing ones to meet customer needs in a new way. It helps companies offer better products and stay competitive in evolving markets.
  This process happens when you pay attention to product markets. You see the problems customers have and their issues with the current offerings. Use that information to develop novel ways to solve those problems and offer a better customer experience.

- **Process innovation:** this type of innovation is changing how you do things inside an organization. You aim to apply new initiatives and technology to optimize efficiency and your ability to compete in the market.
  Take the inclusion of a new IT system, for instance. New technological innovation can improve your employees' ability to collaborate and get more done — allowing them to serve customers better in your core business.
- **Business model innovation:** it is about making big changes to how a business runs. It isn't just a few changes in business processes or a new product. Your goal is to create new business models to capture value in the market.
  One of the common ways this happens is finding new sources of revenue. An excellent example of this is service businesses adding subscriptions to their revenue.

Take an HVAC company. Instead of relying on incoming calls to get business, HVAC companies offer customers a maintenance plan — allowing companies to get a new source of predictable revenue.

- Disruptive innovation: this one is about shaking things up. You aren't launching products that are a minor upgrade of what exists — you're making game-changing innovation initiatives that disrupt existing business models.
  Radical innovation often comes from startups, such as rideshare companies taking on the taxi industry. You're rewriting some of the rules to take on the established industries to provide a better service and give people more options.

# What are some examples of successful innovation strategies?

Now that you've seen what types of innovations there are, let's look at a few innovation strategy examples.

## Amazon

Amazon was one of the first companies to use the internet to sell products. It started as an online bookstore, but as time passed, it changed its focus to become a one-stop marketplace for every product you can imagine.

Since then, Amazon has continued to use technology to innovate in the online market. It now offers subscription services to offer more customer value, video streaming, audiobooks, eBooks, and countless other value-added services for its customers.

## Apple

Apple's innovation projects are a blend of customer-centric design and quality. Apple doesn't often create brand-new product categories. Instead, it waits to see how technology trends play out and creates innovative approaches to those products.

One of the best examples of this is the iPhone. Apple didn't create the first smartphone. Blackberry was one of the first to market and was popular for a while.

But smartphones didn't start making their way into most homes until Apple unveiled the iPhone — their easy-to-use smartphone offering that every consumer can use.

## Startups and entrepreneurs

Innovation is at the heart of the startup and entrepreneurship ecosystem. Let's look at how a few companies changed the game with their new innovations.

- **Uber:** Took on the taxi industry by allowing regular people to use their cars to offer rideshare services to customers using mobile apps.
- **Airbnb:** Competed against the established hotel industry to give people more lodging options and allow homeowners to rent parts of their homes.
- **Netflix:** Used a subscription model and on-demand streaming approach instead of a monthly fee for live TV.
- **Stripe:** Changed the payment industry by allowing companies to integrate payment processing into their websites easily.

# What is Innovation? Definition, Types, Examples and Process

### Key components that make innovation possible in an organization

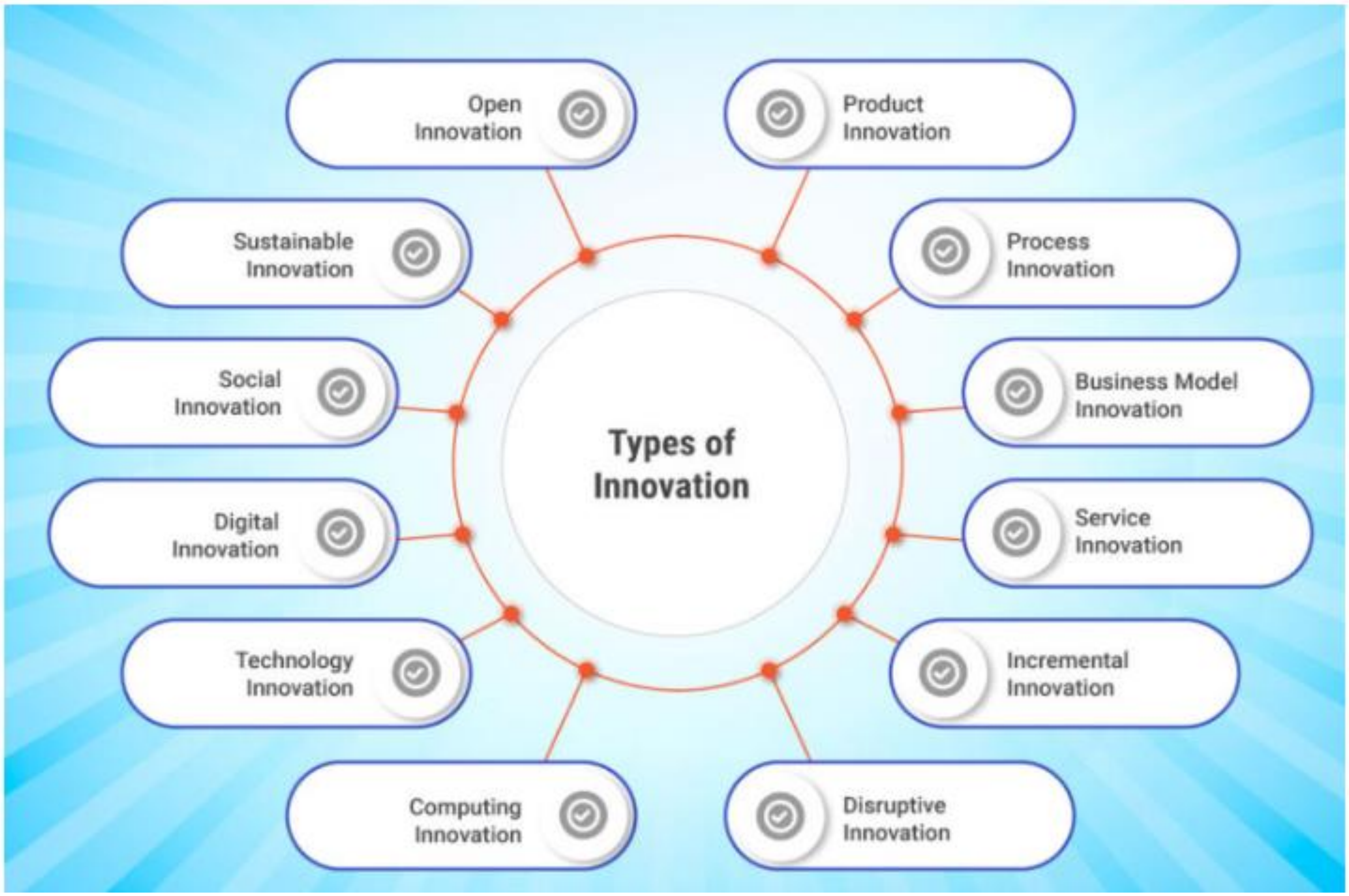
Several key components contribute to making innovation possible in an organization. These components provide the foundation and enable organizations and individuals to drive innovation effectively. Here are the key essential components:

- **Creativity and Ideas:** Creativity is the fuel for innovation. It involves generating new and original ideas, challenging assumptions, and thinking beyond conventional boundaries. It is the ability to connect disparate concepts and envision novel possibilities. The generation of diverse ideas, both incremental and disruptive, serves as the starting point for innovation.
- **Culture of Innovation:** An organizational culture that fosters and supports innovation is crucial. It includes values, attitudes, and behaviors that encourage curiosity, risk-taking, collaboration, and experimentation. A culture of innovation promotes an open and inclusive environment where individuals feel empowered to contribute their ideas and embrace change.

- **Resources and Infrastructure:** Adequate resources, both financial and non-financial, are essential for innovation. This includes dedicated funding, skilled human capital, technology infrastructure, research and development capabilities, and access to relevant information and data. Organizations need to allocate resources strategically to support innovation initiatives.
- **Leadership and Vision:** Effective leadership plays a vital role in driving and supporting innovation. Leaders set the vision, create a sense of purpose, and provide guidance and resources for innovation initiatives. They foster an environment that encourages risk-taking, empowers employees, and leads by example. Leadership commitment and support are crucial in nurturing a culture of innovation and driving innovation efforts.

- **Feedback and Adaptation:** Innovation requires a feedback loop that allows for continuous innovation. Feedback can come from customers, users, stakeholders, and market trends. Organizations need mechanisms to gather and analyze feedback, learn from successes and failures, and iterate on their innovation initiatives. The ability to adapt and pivot based on feedback is essential to refine and enhance innovative solutions.
- **Effective Risk Management:** Innovation involves inherent risks and uncertainties, which makes effective risk management crucial to mitigate potential challenges and ensure successful outcomes. Organizations need processes to identify, assess, and manage risks associated with innovation initiatives. This includes evaluating the feasibility, viability, and potential impact of innovative ideas and implementing risk mitigation strategies.

# Types of Innovation



- **Product Innovation:** Product innovation involves developing novel products or enhancing existing ones to meet evolving market demands and customer expectations. It encompasses advancements in functionality, design, performance, and user experience.
- **Process Innovation:** Process innovation revolves around optimizing internal workflows, systems, and technologies to streamline operations, boost efficiency, and reduce costs. It enhances organizational agility and competitiveness.
- **Business Model Innovation:** Business model innovation redefines how businesses create, deliver, and capture value. It explores new revenue streams, cost structures, partnerships, and customer engagement strategies to drive market disruption and differentiation.
- **Service Innovation:** Service innovation focuses on transforming customer experiences through new service delivery methods, customization options, and enhanced accessibility. It aims to exceed customer expectations and strengthen brand loyalty.

- **Incremental Innovation:** Incremental innovation entails continuous enhancements and refinements to existing products, services, or processes. It fosters gradual improvements in efficiency, quality, and user satisfaction, ensuring sustained market relevance.
- **Disruptive Innovation:** Disruptive innovation introduces groundbreaking solutions that challenge existing market norms and create new value propositions. It starts in niche markets and progressively displaces incumbents by addressing unmet customer needs.
- **Open Innovation:** Open innovation involves collaborating with external stakeholders—such as customers, suppliers, and research institutions—to co-create innovative solutions. It leverages external expertise and resources to accelerate innovation cycles and foster industry leadership.
- **Sustainable Innovation:** Sustainable innovation focuses on developing environmentally friendly products, services, and business models that minimize ecological footprints and promote social responsibility. It addresses global challenges like climate change and resource conservation.

- **Social Innovation:** Social innovation tackles societal challenges through novel products, services, or approaches that enhance community well-being, promote inclusivity, and drive positive social change. It fosters sustainable development and ethical business practices.
- **Digital Innovation:** Digital innovation harnesses digital technologies—such as AI, IoT, and blockchain—to revolutionize products, services, and operational processes. It optimizes efficiency, enhances data-driven decision-making, and drives digital transformation.
- **Technology Innovation:** Technology innovation drives advancements in scientific and technological fields—ranging from biotechnology to renewable energy—to address complex global issues and improve quality of life.
- **Computing Innovation:** Computing innovation pioneers breakthroughs in computing technologies and IT systems—enabling new capabilities, enhancing cybersecurity, and driving digital evolution across industries.

Furthermore, any of the above types of innovation can be approached through continuous innovation or discontinuous innovation:

- **Continuous Innovation:** Continuous innovation is the iterative improvements that refine existing offerings and sustain competitive advantage through ongoing adaptation.
- **Discontinuous Innovation:** Discontinuous innovation is radical transformations that disrupt markets, redefine industry standards, and create new growth opportunities through visionary leaps.

Each type of innovation offers distinct advantages and strategic opportunities for organizations aiming to innovate effectively in today’s dynamic market landscape. Implementing a tailored innovation strategy that integrates these types can propel your business towards sustained growth and market leadership.

# Innovation Examples: 10 Best Across Industries

Here are the ten best innovation examples from various industries:

**1. Airbnb:** Airbnb disrupted the traditional hospitality industry by creating an online marketplace that allows individuals to rent out their homes or spare rooms to travelers. This peer-to-peer accommodation model provided an alternative to traditional hotels and revolutionized the way people travel and experience accommodations.

**2. Electric Vehicles:** Electric vehicles by brands like Tesla, have revolutionized the automotive industry by pioneering electric vehicles (EVs) that offer long-range capabilities, high-performance features, and sustainable energy solutions. Their technology innovation in battery and charging infrastructure has played a significant role in accelerating the adoption of EVs worldwide.

**3. SpaceX's Reusable Rockets:** SpaceX, founded by Elon Musk, developed reusable rockets that can be landed and reused, significantly reducing the cost of space exploration and making space travel more accessible. This innovation has opened up new possibilities in the aerospace industry.

**4. 3D Printing:** 3D printing, also known as additive manufacturing, has transformed various industries by enabling the production of complex and customized objects with precision. It has revolutionized manufacturing, healthcare (e.g., prosthetics), and design prototyping, among other sectors.

**5. CRISPR Gene Editing:** CRISPR-Cas9 is a revolutionary gene-editing technology innovation that enables precise and efficient modification of DNA sequences. This innovation has the potential to revolutionize healthcare, agriculture, and biotechnology by offering new approaches to treating genetic diseases, enhancing crop resilience, and developing new therapies.

**6. Netflix's Streaming Service:** Netflix disrupted the traditional video rental and television industry by introducing a streaming service that allows users to watch movies and TV shows on demand. This innovation led to a shift in how content is consumed, paving the way for other streaming platforms.

**7. Mobile Payment Solutions:** Companies like Apple Pay, Google Pay, and PayPal have transformed the way payments are made by enabling secure and convenient mobile transactions. This innovation has simplified payment processes and enhanced financial inclusivity.

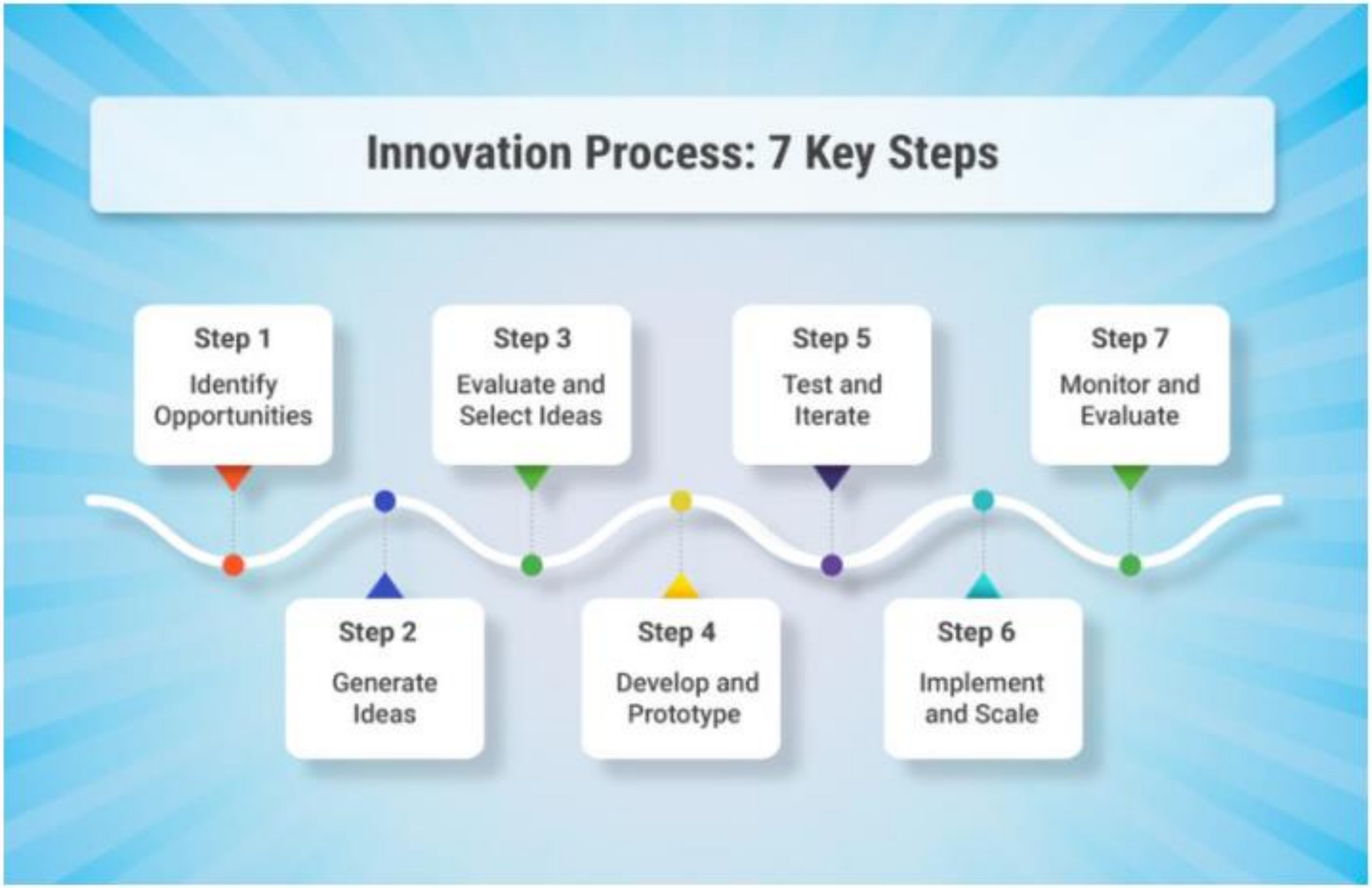
**8. Amazon's Alexa Voice Assistant:** Amazon's voice-controlled assistant, Alexa, introduced a new way of interacting with technology leveraging digital innovation through natural language processing. It has revolutionized the smart home industry and paved the way for voice-controlled devices and services.

**9. Electric Scooters and Bike-Sharing Services:** Electric scooters and bike-sharing services, such as Lime and Citi Bike, have provided eco-friendly alternatives for urban transportation. These innovations have facilitated short-distance travel, reduced congestion, and promoted sustainable mobility options.

**10. Solar Energy Technologies:** Advancements in solar energy technologies, including more efficient photovoltaic cells and cost reductions, have made solar power increasingly accessible and economically viable. This innovation has driven the growth of renewable energy sources and contributed to the transition towards a sustainable energy future.

These examples showcase the transformative power of innovation across various sectors, highlighting how ideation and advancements can reshape industries, improve lives, and drive positive change.

# Innovation Process: 7 Key Steps



The innovation process typically involves a series of key steps that organizations follow to foster and implement innovation. While specific approaches may vary, here are the common steps involved in the innovation process:

**Step 1. Identify Opportunities**

The first step is to identify opportunities for innovation. This can be done through market research, customer insights, trend analysis, or internal assessments. The goal is to uncover unmet needs, emerging trends, or areas for improvement that can be addressed through innovation.

**Step 2. Generate Ideas**

Once opportunities are identified, the next step is to generate ideas. This can be done through brainstorming sessions, idea competitions, customer feedback, or cross-functional collaboration. The aim is to generate a wide range of creative and innovative ideas that have the potential to address the identified opportunities.

### Step 3. Evaluate and Select Ideas

After ideation, the next step is to evaluate and select the most promising ones. This involves assessing the feasibility, viability, and desirability of each idea. Consider factors such as market potential, technical feasibility, resource requirements, alignment with strategic goals, and potential impact. The goal is to identify the ideas that are worth pursuing further.

### Step 4. Develop and Prototype

Once ideas are selected, they can be further developed and prototyped. This involves translating the selected ideas into tangible prototypes, mock-ups, or minimum viable products (MVPs). The aim is to test and validate the concepts, gather feedback, and refine the ideas based on customer insights and technical feasibility.

### Step 5. Test and Iterate

In this step, the prototypes or MVPs are tested with users or in real-world scenarios. Customer feedback is collected, and the concepts are iterated and refined based on the insights gained. This iterative process helps to validate assumptions, uncover potential issues, and improve the innovation before moving to the next stage.

### Step 6. Implement and Scale

Once the innovation has been tested and refined, it can be implemented and scaled up. This involves developing a detailed implementation plan, allocating resources, and executing the necessary actions to bring the innovation to market or implement it within the organization. The goal is to ensure a smooth transition from the development phase to full-scale implementation.

### Step 7. Monitor and Evaluate

After implementation, it is important to monitor and evaluate the performance and impact of the innovation. This involves tracking key metrics and performance indicators to assess the success of the innovation. Regular evaluation helps identify areas for improvement, make necessary adjustments, and capture learnings for future innovation initiatives.

Innovation is an ongoing process, and organizations should foster a culture of innovation. This involves capturing feedback, promoting learning from both successes and failures and continuously seeking new opportunities for innovation. Regularly revisiting and refining the innovation process itself is also essential to optimize the organization's ability to innovate effectively.

# WHAT ARE THE BENEFITS AND RISKS OF INNOVATION?

## BENEFITS OF INNOVATION

### Improved productivity & reduced costs

A lot of process innovation is about reducing unit costs. This might be achieved by improving the production capacity and/or flexibility of the business – to enable it to exploit economies of scale

### Better quality

By definition, better quality products and services are more likely to meet customer needs. Assuming that they are effectively marketed, that should result in higher sales and profits

### Building a product range

A business with a single product or limited product range would almost certainly benefit from innovation. A broader product range provides an opportunity for higher sales and profits and also reduces the risk for shareholders

**To handle legal and environmental issues**

Innovation might enable the business to reduce it carbon emissions, produce less waste or perhaps comply with changing product legislation. Changes in laws often force business to innovate when they might not otherwise do so

**More added value**

Effective innovation is a great way to establish a unique selling proposition ("USP") for a product – something which the customer is prepared to pay more for and which helps a business differentiate itself from competitors

**Improved staff retention, motivation and easier recruitment**

Not an obvious benefit, but often significant. Potential good quality recruits are often drawn to a business with a reputation for innovation. Innovative businesses have a reputation for being inspiring places in which to work.

Innovation is the act of creating and implementing new ideas, products, or processes that can bring value to an organization or an individual. However, innovation is not without risk. Risk is the potential for loss or uncertainty that arises when one ventures into the unknown.

**Risk can come from various sources, such as:**

- **Technical Uncertainty:** Related to the technical ability to implement the innovative idea, such as problems in developing the technology or lack of technical knowledge required.
- **Market Uncertainty:** Related to how the market will react to the new product or service, such as questions about market demand, competition, and consumer trends.
- **Financial Risk:** Innovation often requires large investments, and there is financial risk associated with developing and launching new products or ideas.

- **Psychological Risk:** This involves the fear or resistance of individuals or teams to the change required for innovation, which can be a significant barrier.
- **Leadership and Political Risk:** Innovation sometimes shakes up the organizational hierarchy or status quo, so there is risk related to how the leadership and organizational politics respond to the change.
- **Security and Legal Risk:** Risk related to data security, patents, or regulations that must be followed in certain innovations.

These risks are not mutually exclusive, and they can interact and amplify each other. For example, technical uncertainty can increase market uncertainty, which can increase financial risk, which can increase psychological risk, and so on. Therefore, innovation is not a simple or linear process, but a complex and dynamic one that involves multiple actors and factors.

# What is innovation risk?

**Innovation risk deals with likeliness and repercussions. It refers to the probability of any untoward happening that may affect the innovation process at a given rate or time.**

It also involves any unfavorable circumstance that may affect the success of a specific innovation measure or project. For instance, what are the odds of a certain concept not working out the way you expected it to?

Suppose the chances of this happening are high. In that case, the innovation team may experience several setbacks that could force them to go another direction or shelf out the project altogether.

For the business, this means wasted resources, effort, and time.

Although this scenario has not happened yet, the perception of failure keeps this idea from being implemented. At some point, the company may devise a specific set of contingency measures to address this and proceed with its vision.

Yet, for most businesses, particularly those who hold onto their natural inclination of risk avoidance, such an action may not be worth spending a lot of time on, so they decide to go for another idea instead.

In other words, despite their negative connotation, innovation risks are vital for organizations to evaluate and take note of to prevent further damage. It is crucial for innovation teams to uncover and identify them early on to prepare for them.

They need to be discussed in collaboration spaces and meetings so investors, stakeholders, and authorized executives can create the necessary decisions that may alter an idea for the better.

## What causes innovation risks?

Technically, risks are inevitable. However, some actions may cause unnecessary troubles.

A few of these are:

- **Experimenting too late.** Organizations need to test ideas to refine them out. Once you have them organized, conduct rapid experimentation right away to incorporate findings in the early stages of product development.
- **Not spending sufficient time in finding the problem**. Most innovations fail because they didn't go deep enough in the first stages (problem finding, solution finding). Usually, this results in big changes later on.
- **Proceeding without really addressing an issue.** In innovation, it is vital to find a problem or a situation to improve. Without it, an idea will only become a product without value – so if you want your innovation to be successful, find an issue and solve it.
- **Creating products that only increase people's curiosity** instead of responding to their needs. Make sure that your product or service stays relevant. Match your innovation with your customers' needs and add value to their lives by helping them out.

- **Focusing too much on technology.** Innovation is more than just coming up with the next Apple or Spotify. Innovation is the process of realizing new products, processes, propositions, or business models to create added value for customers and employees.
- **Deciding with an inadequate number of ideas.** One idea is not enough – you need at least 300 ideas for success. The more ideas that you have, the more solutions you can generate. Get varied insights and collaborate with other departments for diversity.
- **Not involving sponsors from the start.** Involving sponsors will not only help you with funding. But with them, you can also understand your value proposition better and tailor your business models before officially launching them for public use.
- **Refusing to include experts and stakeholders.** Collaborating with experts and stakeholders gives you access to a pool of useful information that can speed up projects forward and uncover risk areas to increase the project's success.
- **Building the business case too soon.** Rushing the business case may result in an unclear and unstructured innovation. This will prohibit you from creating the best products, concepts, projects, and changes that could have made your innovation better.

## What are the types of innovation risks?

Innovation comes with a number of risks. Some of these are:

### Operational Risks

**Operational risks involve challenges that are met by organizations in accomplishing innovation, particularly with materials, budget, and other necessary elements included in bringing ideas to life.**

This type of risk occurs when decisions related to priorities, production, and management fail to meet what is imperative to ensure innovation success. Operational risks are also referred to as human risks as a way of attributing these repercussions to human error.

Some areas that involve operational risks are systems, equipment, third-party services, and other business components that are utilized by the organization in its daily business activities.

Examples of operational risks are failing to meet product quality standards and cost requirements.

## Commercial Risks

**Commercial risks deal with possible losses that an organization may encounter from its business partners or from its market. These are problems that arise from the company's assets, liabilities, and cash flows.**

Technically, this type of risk happens when buyer-related problems result in nonpayment. In innovation, such risks take place when an idea does not clearly establish its target audience and when demands are not met because of problems with suppliers and other related parties.

Other instances, such as not meeting the requirements and preferences of the product's target audience, can also result in commercial risks. A concrete example of this is when a product fails to attract sufficient customers, leading the business to incur some losses.

**Financial risks are issues that may arise from losing money in innovation (negative return on innovation).**

Depending on the nature of your industry, financial risks may mean failing to control monetary policies, inability to solve debt issues, and undertaking projects that place a financial burden on the organization.

It may also come as a result of flawed financial reasoning. This type of risk may also arise from failures in financial transactions and capital structure.

Other circumstances that may result in financial risks are changes in market interest rates that may push companies into lower-debt paying securities and negative returns.

Financial risks come in various forms.

Being aware of possible financial risks can help you prepare for it and in time, mitigate its effects to avoid negative outcomes. A simple example of financial risk is when a company invests in an innovation project that ends up being unsuccessful.

## Why innovation risk may be different from other types of risks?

**Innovation leads companies to implement certain changes that may revolutionize various processes and improve the way businesses and customers handle specific activities.**

Organizations that like to innovate are able to enhance or formulate their own solutions, designs, products, technologies, capabilities, and services that add value to their employees or customers.

As innovation involves going beyond traditional boundaries, this complex process is made up of different phases that involve various experimentations.

Provided that these activities result in trial-and-error situations, innovation risk is different from other types of risks as organizations regularly expect failures to determine what works and what doesn't.

## What are the ways to manage innovation risks?

In innovation, making it right the first time just doesn't seem to happen sometimes. Ideas get modified from time to time to fit certain requirements, and the risks that come with them have to be mitigated.

It's important to ensure that any temporary setback won't keep the concept from becoming a product that adds value to both the customers and the business.

Here are some methods that you can utilize in managing innovation risks:

## Evaluate risk levels in the early stage of product development

**Identify risks as early as possible. Collaborate with other departments to determine all types of risks that may occur in the entire innovation process.**

Gather insights from both inside and outside of your project team to obtain diverse information, knowledge, and expertise. It is also imperative for businesses to evaluate the risk acceptance level of their stakeholders.

Some innovation teams like to set goals that might seem impossible for others, believing that taking enormous risks would lead to monumental success.

While this may be ideal for some, others would like to take a more conservative approach.

Whatever the aspirations of your innovation team are, make sure that it matches the risk tolerance of your stakeholders so you can take appropriate steps in implementing your ideas.

## Address the bigger risks first, then smaller risks second

Bigger risks have a larger tendency to push projects off the trail.

Dealing with them early on avoids wasting time and resources while increasing the project's success. Once done, smaller risks should be addressed right after as they can pile up and become derailing at some point.

Smaller risks also have a cumulative effect. Failing to respond to them immediately may worsen the negative effects of other risks, particularly if these risks are related to one another.

## Take every risk as a chance to improve

Innovation risks may reduce a project's success – but successfully mitigating them can be advantageous for companies. Addressing them boosts an organization's creativity and problem-solving capabilities.

This leads innovation teams to come up with solutions to existing problems. Concepts are transformed into workable strategies, and successful breakthroughs are gradually attained over time.

In the end, managing risks is endless:

Progress and risks sometimes come as a package, and being proactive in looking out for them and addressing them will enable you to predict negative outcomes and avoid innovation roadblocks.

To make risk management easy, you can use software to manage innovation portfolios and establish a framework that converts roughly produced ideas into real investment opportunities.

# Introduction to TRIZ: The Innovation and Problem-Solving Methodology

TRIZ, the Russian acronym for "Theory of Inventive Problem Solving," is a powerful, systematic methodology for innovation and problem-solving. Developed by Genrich Altshuller and his colleagues in the former Soviet Union, TRIZ is based on the idea that there are universal principles underlying the process of innovation, which can be identified, studied, and applied to enhance creative thinking and solve complex problems. In this post, we will provide an overview of TRIZ, discuss its key principles and tools, and explore how it can be used to foster innovation and enhance problem-solving capabilities in various fields.

## Key Principles of TRIZ:

TRIZ is built on several foundational principles that guide the problem-solving process:

1. **Patterns of innovation:** TRIZ posits that technological systems evolve following predictable patterns, which can be identified and leveraged to generate innovative solutions.

2. **Contradictions:** TRIZ recognizes that problems often arise from inherent contradictions within a system, and solving these contradictions can lead to breakthrough solutions.

3. **Ideality:** TRIZ emphasizes the pursuit of ideality, which is achieved when a system delivers maximum value with minimal resources and complexity.

4. **Resources:** TRIZ encourages the efficient use of available resources, including materials, energy, and knowledge, to create innovative solutions.

## Key Tools and Techniques in TRIZ:

TRIZ offers a range of tools and techniques designed to facilitate the problem-solving process, including:

1. **40 Inventive Principles:** A collection of 40 general strategies that can be applied to resolve contradictions and generate innovative solutions.

2. **Separation Principles:** Techniques for resolving contradictions by separating conflicting requirements in time, space, or condition.

3. **Contradiction Matrix:** A tool that helps identify the most relevant inventive principles to apply based on the specific contradiction at hand.

4. **Function Analysis:** A method for understanding and optimizing the interactions between components within a system.

5. **ARIZ (Algorithm for Inventive Problem Solving):** A step-by-step algorithmic approach that guides users through the TRIZ problem-solving process.

## Applications of TRIZ:

TRIZ has been successfully applied across various industries, including automotive, aerospace, electronics, manufacturing, and healthcare, among others. Its versatility allows it to be adapted to a wide range of problems, from incremental improvements to radical innovations. Some potential applications of TRIZ include:

**1. Product development:** TRIZ can be used to identify new product features, improve existing designs, and optimize performance by addressing contradictions and leveraging patterns of innovation.

**2. Process improvement:** TRIZ can help identify and resolve bottlenecks, inefficiencies, and contradictions within a process, leading to enhanced productivity and reduced costs.

**3. Organizational problem solving:** TRIZ can be employed to address complex organizational challenges, such as optimizing resource allocation, improving communication, and managing conflicts.

## Conclusion:

TRIZ is a powerful and systematic approach to innovation and problem-solving that offers a range of tools and techniques to help users identify and resolve contradictions, leverage patterns of innovation, and pursue ideality. By incorporating TRIZ into their problem-solving toolkit, individuals and organizations can enhance their creative thinking capabilities, overcome complex challenges, and drive breakthrough innovations in a wide variety of fields.

## Unlocking Innovative Solutions with TRIZ: A Powerful Problem-Solving Methodology

Businesses constantly search for innovative fixes to stay ahead and handle complex puzzles. But creation paths often hit snags, contradictions, and walls that seem impossible to climb. This is where TRIZ, the Theory of Inventive Problem Solving, shows its worth with a powerful method fueling breakthrough solutions and a constantly upgrading culture.

Developed decades ago by Soviet brainiac Genrich Altshuller after analyzing millions of patents across industries, TRIZ identifies the patterns behind successful ideas.

By furnishing a methodical framework for problem-solving and imagination, TRIZ empowers organizations to expertly handle technical and business complications.

TRIZ recognizes most problems stem from built-in clashes in a system. Where improving one aspect often damages another.

## What is TRIZ?

TRIZ, an acronym for the Russian phrase **"Teoriya Resheniya Izobretatelskikh Zadatch"**, stands for the Theory of Inventive Problem Solving.

This powerful methodology, developed by Genrich Altshuller and his colleagues in the Soviet Union, represents a systematic approach to innovation and problem-solving rooted in analyzing patents and inventions across various fields.

TRIZ operates on the premise that the evolution of technical systems follows distinct patterns and principles.

By studying these patterns and the underlying mechanisms that drive successful innovations, TRIZ provides a structured framework for understanding and resolving contradictions within a system, unlocking creative solutions that transcend traditional trade-offs and compromises.

## The Essence of TRIZ

The essence of TRIZ lies in its ability to identify and resolve contradictions, which are often at the heart of many complex problems.

These contradictions can manifest as technical trade-offs, where improving one aspect of a system leads to the deterioration of another.

For instance, increasing the strength of a material may result in an increase in its weight, which is undesirable in certain applications.

TRIZ recognizes that true innovation occurs when these contradictions are addressed and resolved systematically.

By generalizing problems and solutions across industries, TRIZ leverages the collective knowledge and ingenuity of inventors and problem-solvers throughout history, providing a vast reservoir of proven techniques and principles that can be adapted and applied to specific challenges.

Moreover, TRIZ emphasizes the importance of understanding the patterns of technical evolution, which describe the natural progression of systems and technologies over time.

By anticipating and aligning with these patterns, problem-solvers can proactively identify opportunities for innovation and develop solutions that are effective and aligned with the system's evolutionary trajectory.

Through its structured approach and extensive knowledge base, TRIZ empowers organizations to navigate the complexities of problem-solving and innovation, fostering a culture of systematic thinking and creative problem-solving that can yield breakthrough solutions and drive continuous improvement across various domains.

## Key Principles and Tools of TRIZ

TRIZ is a powerful arsenal of principles and tools that provide a structured approach to problem-solving and innovation.

These principles and tools have been meticulously derived from the analysis of countless patents and inventions, distilling the collective wisdom and ingenuity of innovators throughout history.

By mastering these key principles and tools, problem-solvers can unlock a world of creative solutions and drive continuous improvement within their organizations.

### The Contradiction Matrix

One of the fundamental tools in the TRIZ methodology is the Contradiction Matrix. This matrix serves as a powerful framework for identifying and resolving contradictions, which often lie at the root of many complex problems.

## The Contradiction Matrix

One of the fundamental tools in the TRIZ methodology is the Contradiction Matrix. This matrix serves as a powerful framework for identifying and resolving contradictions, which often lie at the root of many complex problems.

TRIZ recognizes two distinct types of contradictions:

**Technical contradictions and trade-offs**

These contradictions arise when improving one aspect of a system leads to the deterioration of another.

For instance, increasing the strength of a material may result in an increase in weight, which is undesirable in certain applications.

**Physical contradictions and inherent conflicts**

These contradictions occur when an object or system is subject to conflicting requirements or characteristics.

For example, a product may need to be both durable and lightweight, presenting an inherent contradiction.

The Contradiction Matrix provides a systematic approach to resolving these contradictions by mapping them to a set of 40 inventive principles.

These principles, derived from the analysis of countless patents, offer proven strategies and techniques for overcoming specific contradictions, enabling problem-solvers to generate innovative solutions that transcend traditional trade-offs.

## Functional Analysis and Trimming

Effective problem-solving often requires a deep understanding of a system's functions, interactions, and components.

TRIZ offers powerful tools for functional analysis and trimming, which enable problem-solvers to deconstruct and analyze complex systems in a structured manner.

Analyzing system functions and interactions: TRIZ employs techniques such as functional modeling and substance-field analysis to identify the relationships and interactions between various components within a system.

By understanding these interactions, problem-solvers can pinpoint potential areas for improvement or optimization.

## Ideality and Patterns of Evolution

TRIZ is not merely a collection of tools but also a guiding philosophy that emphasizes the pursuit of ideality and the recognition of patterns in technical evolution.

**Striving for the Ideal Final Result (IFR):** The concept of ideality in TRIZ encourages problem-solvers to envision an ideal state where the desired functions are achieved without any adverse effects or limitations.

This aspirational mindset drives the search for innovative solutions that continually approach the ideal, pushing the boundaries of what is possible.

**Understanding technical evolution patterns (S-curves):** TRIZ recognizes that technical systems and technologies follow predictable patterns of evolution, often represented by S-curves.

By studying these patterns, problem-solvers can anticipate future developments, identify opportunities for innovation, and align their solutions with the natural trajectory of technical evolution.

## Applying TRIZ: A Step-by-Step Approach

While the principles and tools of TRIZ may seem complex, the true power of this methodology lies in its structured and systematic approach to problem-solving.

By following a well-defined step-by-step process, organizations can harness the full potential of TRIZ to tackle even the most daunting challenges and drive innovation within their domains.

This approach not only ensures a thorough understanding of the problem but also provides a roadmap for leveraging the various TRIZ tools and techniques to generate and evaluate innovative solutions.

## Defining the Problem and Identifying Contradictions

The first step in the TRIZ approach is to clearly define the problem at hand and identify the desired outcomes. This process involves a thorough analysis of the current situation, including the limitations, constraints, and requirements that must be addressed.

By establishing a clear understanding of the problem, problem-solvers can better recognize the inherent contradictions that lie at its core.

Recognizing technical and physical contradictions is a crucial aspect of this stage.

Technical contradictions, as we've discussed, arise when improving one aspect of a system leads to the deterioration of another. Physical contradictions, on the other hand, stem from conflicting requirements or characteristics within an object or system.

TRIZ provides specific tools and techniques for identifying and analyzing these contradictions, laying the foundation for the subsequent steps in the problem-solving process.

## Utilizing TRIZ Tools and Techniques

Once the problem and its contradictions have been clearly defined, the next step is to leverage the various TRIZ tools and techniques to generate potential solutions.

This stage involves a systematic application of the principles and concepts we've discussed earlier, including:

**Leveraging the Contradiction Matrix and Inventive Principles**

By mapping the identified contradictions to the Contradiction Matrix, problem-solvers can access a wealth of inventive principles that offer proven strategies for resolving those contradictions.

These principles, derived from the analysis of countless patents, serve as a powerful source of inspiration and guidance for generating innovative solutions.

**Conducting functional analysis and trimming**

TRIZ's functional analysis and trimming tools enable problem-solvers to deconstruct and analyze complex systems, identifying unnecessary components or functions that can be streamlined or eliminated.

**Exploring patterns of evolution and ideality**

TRIZ encourages problem-solvers to explore the patterns of technical evolution and strive for the Ideal Final Result (IFR).

By understanding these patterns and envisioning an ideal state, organizations can align their solutions with the natural trajectory of technical evolution and continually push the boundaries of what is possible.

## Generating and Evaluating Solutions

With a deep understanding of the problem and its contradictions, and armed with the TRIZ tools and techniques, problem-solvers can then embark on the process of generating potential solutions.

This stage often involves ideation techniques based on TRIZ principles, such as the use of analogies, functional modeling, and the application of scientific effects.

However, generating solutions is only half the battle. TRIZ also provides a framework for evaluating potential solutions based on their feasibility, impact, and alignment with the desired outcomes.

This evaluation process involves assessing factors such as resource availability, technical feasibility, and potential risks or unintended consequences.

## Engineering and Manufacturing Applications

TRIZ has found widespread adoption in the engineering and manufacturing sectors, where its principles have been instrumental in driving product design and development, as well as optimizing processes and reducing waste.

**Product design and development**

TRIZ has played a pivotal role in the creation of innovative products that push the boundaries of what is possible.

By resolving contradictions inherent in design requirements and leveraging the principles of technical evolution, TRIZ has enabled engineers to develop cutting-edge solutions that meet customer needs while overcoming technical limitations.

Examples include the development of more fuel-efficient aircraft designs, improved engine performance, and the creation of advanced materials with unique properties.

**Process optimization and waste reduction**

In the realm of manufacturing, TRIZ has been a driving force behind the optimization of production processes and the reduction of waste.

By applying functional analysis and trimming techniques, organizations have been able to streamline their operations, eliminate unnecessary steps, and enhance efficiency.

This has not only resulted in cost savings but has also contributed to sustainability efforts by minimizing resource consumption and environmental impact.

## Business and Management Applications

While TRIZ originated in the realm of engineering and technology, its principles have proven equally valuable in the business and management spheres, where strategic planning, decision-making, and innovation management are paramount.

## Cross-Industry Applications

The versatility and universality of TRIZ principles have enabled its application across a diverse range of industries, transcending the boundaries of engineering and business.

Pharmaceutical and biomedical innovations: TRIZ has played a crucial role in advancing the fields of pharmaceuticals and biomedicine.

By resolving contradictions inherent in drug formulation, delivery mechanisms, and medical device design, TRIZ has facilitated the development of innovative treatments, drug delivery systems, and cutting-edge medical technologies that improve patient outcomes and enhance the quality of healthcare.

Software and IT problem-solving: In the rapidly evolving world of software and information technology, TRIZ has proven to be a valuable asset for tackling complex technological challenges.

These success stories and applications serve as a testament to the power and versatility of TRIZ.

TRIZ has enabled organizations across various industries to achieve remarkable breakthroughs, drive continuous improvement, and maintain a competitive edge in an ever-changing business landscape by providing a structured and systematic approach to problem-solving and innovation.

# Innovations in Chemical Engineering & Technology

P D Vaidya

## Getting started

Q1. Discovery, invention & innovation – are they different? If so, how?
Q2. What is the true meaning of innovation? Discuss with examples.
Q3. Is the chemical process industry (CPI) viewed as a very innovative field? Explain.
Q4. Is innovation related to the customer? How do companies identify business case for innovation?
Q5. Discuss ten innovative chemical engineering solutions with suitable examples.
Q6. Does a chemical engineer have opportunities to innovate in the CPI? If so, where and how?
Q7. What are the mistakes and misconceptions about innovations?
Q8. What are the challenges and barriers to innovations?
Q9. How to learn from challenges and failures during innovation?
Q10. How to move towards innovative solutions in the CPI?
Q11. What are the fields in innovation?
Q12. What are the four broader types of innovation?
Q13. Give examples of innovation types (based on applications).
Q14. What are the stages of corporate innovation?
Q15. How to encourage innovation in business?
Q16. What are the strategies of innovation?
Q17. How to measure innovation? How to protect innovation?
Q18. What are the four phases in the life cycle of a product?
Q19. Patent, trademark, copyright and trade secret – how are they different?
Q20. What is technology readiness level?
Q21. How to take investment decisions in innovation projects?
Q22. What are the characteristics of today's and tomorrow's chemical sectors?

Analyze the given flow-sheet and suggest modifications using the knowledge of Green Chemistry and Engineering. Comment on each and every equipment and what changes you can make to make the process greener and safer. Discuss the nature of pollution and its atom economy. Name the principles used or which could be used in the new flow-sheet you want to propose. The process description is given. Read it carefully (Marks 10)

Ethylene oxide manufacture is described here.

The reaction is:

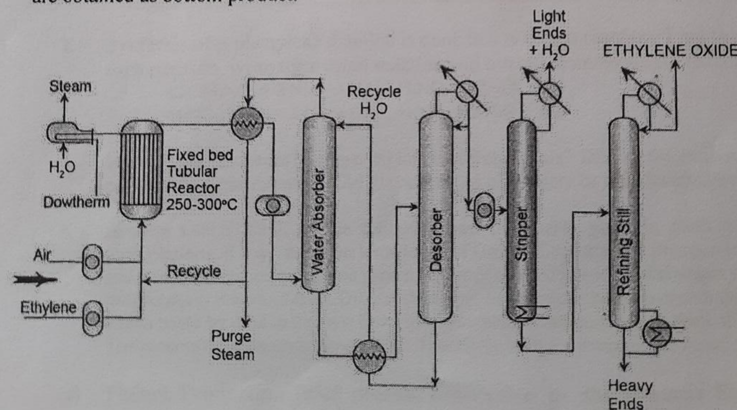
$C_2H_4 + 0.5\ O_2 \rightarrow CH_2\ O.CH_2O$

- Ethylene to air ratio: 3 – 10 %
- Side reaction products: $CO_2$ , $H_2O$
- Catalyst: Silver oxide on alumina
- Operating temperature and pressure: 250 – 300°C and 120 – 300 psi
- Supressing agent for side reactions: Ethylene dichloride
- Reaction is exothermic

P.T.O.

**Process Description**

- Air and ethylene are separate compressed and along with recycle stream are sent to the shell and tube reactor.
- The reactor is fed on the shell side with Dowtherm fluid that serves to maintain the reaction temperature. A dowtherm fluid is a heat transfer fluid, which is a mixture of two very stable compounds, biphenyl and diphenyl oxide. The fluid is dyed clear to light yellow to aid in leak detection.
- The hot dowtherm fluid from the reactor is sent to a waste heat recovery boiler to generate steam.
- The vapour stream is cooled using a integrated heat exchanger using the unreacted vapour stream generated from an absorber.
- The vapour stream is then sent to the heat integrated exchanger and is then sent back to the reactor and a fraction of that is purged to eliminate the accumulation of inerts such as Nitrogen and Argon.
- The product vapours are compressed and sent to a water absorber which absorbs ethylene oxide from the feed vapours. Eventually, the ethylene oxide rich water stream is sent to a stripper which desorbs the ethylene oxide + water as vapour and generates the regenerated water as bottom product. The regenerated water reaches the absorber through a heat integrated exchanger.
  The ethylene oxide + water vapour mixture is compressed (to about 4 - 5 atms) and then sent to a stripper to generate light ends + $H_2O$ as a top product and the bottom product is then sent to another fractionators to produce ethylene oxide as top product. The heavy ends are obtained as bottom product.



# TRL for Entrepreneurs

### TRL 1: BASIC RESEARCH

At TRL 1, the technology is in its earliest stages, with only basic principles and scientific research available. Startups at this stage typically focus on exploring potential applications for their technology and conducting preliminary tests.

Examples:

1. Graphenea - A company researching the potential applications of graphene, a one-atom-thick layer of carbon with unique properties.
2. Emrod - A startup working on wireless electricity transmission through electromagnetic waves.

### TRL 2: APPLIED RESEARCH

At TRL 2, the technology concept is more defined, and applied research is conducted to investigate its feasibility. Startups at this stage work on refining their ideas and determining whether the technology can be developed further.

Examples:

1. DeepBranch - A biotech startup working on converting carbon dioxide into proteins for animal feed using microbial gas fermentation.
2. Solexel - A company developing high-efficiency silicon solar cells using thin-film technology.

### TRL 3: PROOF OF CONCEPT

TRL 3 marks the stage where a startup has a working proof of concept for their technology. This involves demonstrating its functionality in a laboratory setting or through simulations.

Examples:

1. CarbonCure - A startup that injects carbon dioxide into concrete to reduce its carbon footprint.
2. Zap&Go - A company developing ultra-fast charging energy storage devices based on carbon-ion technology.

### TRL 4: TECHNOLOGY VALIDATION

At TRL 4, the technology has been validated in a lab setting, and startups are now focused on refining it further and preparing for real-world testing.

Examples:

1. Bioelektra Group - A company working on a waste treatment technology that can convert mixed waste into reusable materials.
2. HelixNano - A biotech startup developing mRNA-based cancer vaccines using gene editing technology.

### TRL 5: TECHNOLOGY DEMONSTRATION

TRL 5 involves demonstrating the technology in a relevant environment, such as a pilot project or a small-scale test site. Startups at this stage work on optimizing their technology and gathering data to support future development.

Examples:

1. Echogen Power Systems - A company developing a waste heat recovery system that converts industrial waste heat into electricity.
2. Pavegen - A startup that has developed energy-harvesting floor tiles that convert footsteps into electricity.

### TRL 6: PROTOTYPE SYSTEM

At TRL 6, a startup has developed a prototype system that is fully functional and ready for testing in a relevant environment. This stage often involves scaling up the technology and refining its design.

Examples:

1. Aquaporin - A company that has developed a biomimetic membrane technology for water purification.
2. Ecovative Design - A startup producing sustainable materials, such as packaging and insulation, from mycelium, the root structure of mushrooms.

### TRL 7: SYSTEM DEMONSTRATION

TRL 7 represents the stage where the technology has been successfully demonstrated in an operational environment. Startups at this level are focused on refining their systems and preparing for large-scale implementation.

Examples:

1. Solidia Technologies - A company that has developed a sustainable cement and concrete production process that reduces CO2 emissions.
2. CarbonCure - A startup that has successfully deployed its CO2 injection technology in multiple real-world concrete production facilities.

### TRL 8: SYSTEM INTEGRATION

At TRL 8, the technology has been integrated into a complete system, and startups are focused on optimizing performance, reliability, and manufacturability.

Examples:

1. Lilium - A company that has built a fully-functioning prototype of its electric vertical takeoff and landing (eVTOL) aircraft.
2. Xeros Technology Group - A startup that has integrated its water-saving laundry technology into commercial washing machines.

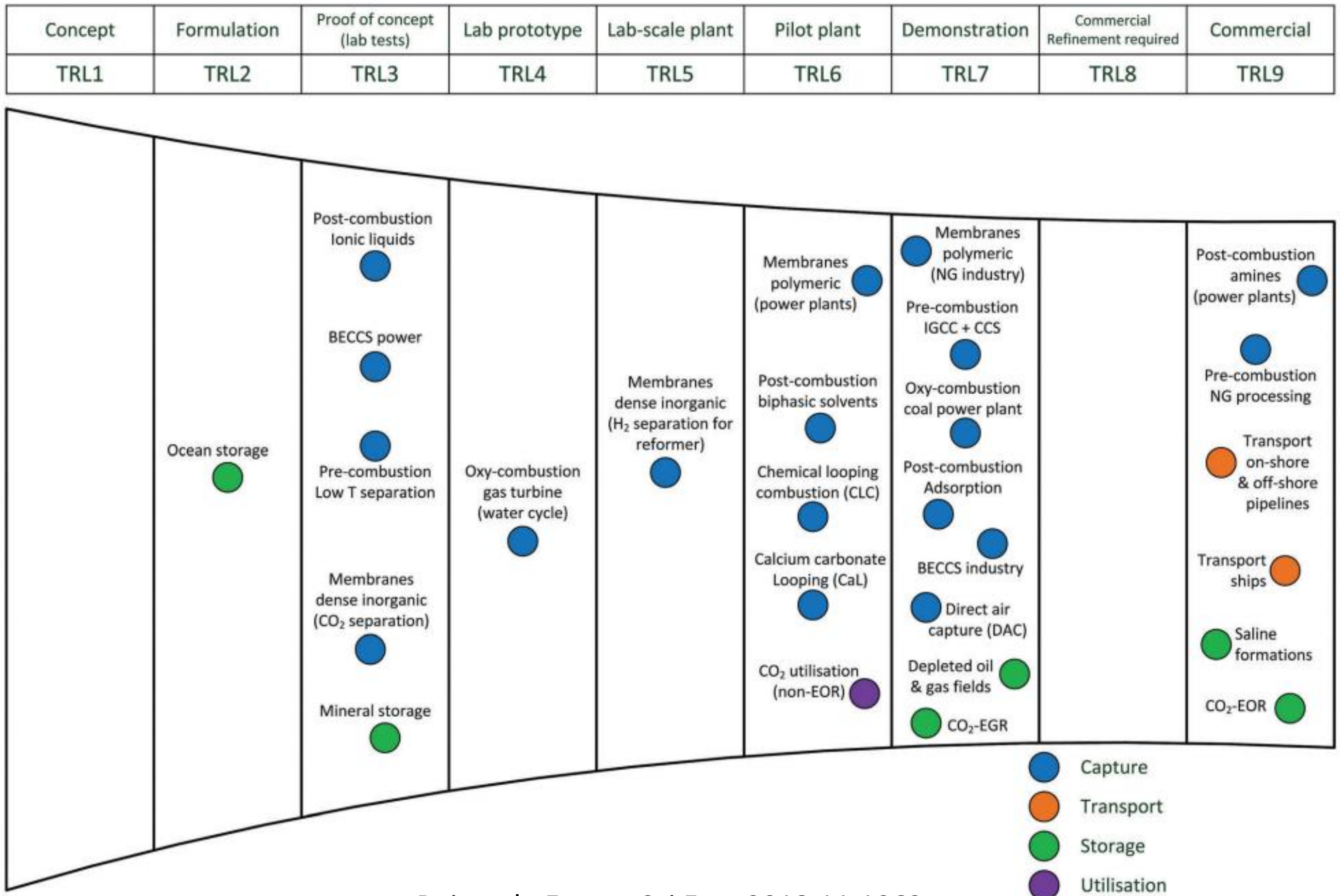
### TRL 9: FULLY DEPLOYED TECHNOLOGY

TRL 9 marks the stage where the technology is fully mature and has been deployed in its intended operational environment. Startups at this level have a proven product that has been successfully commercialized.

Examples:

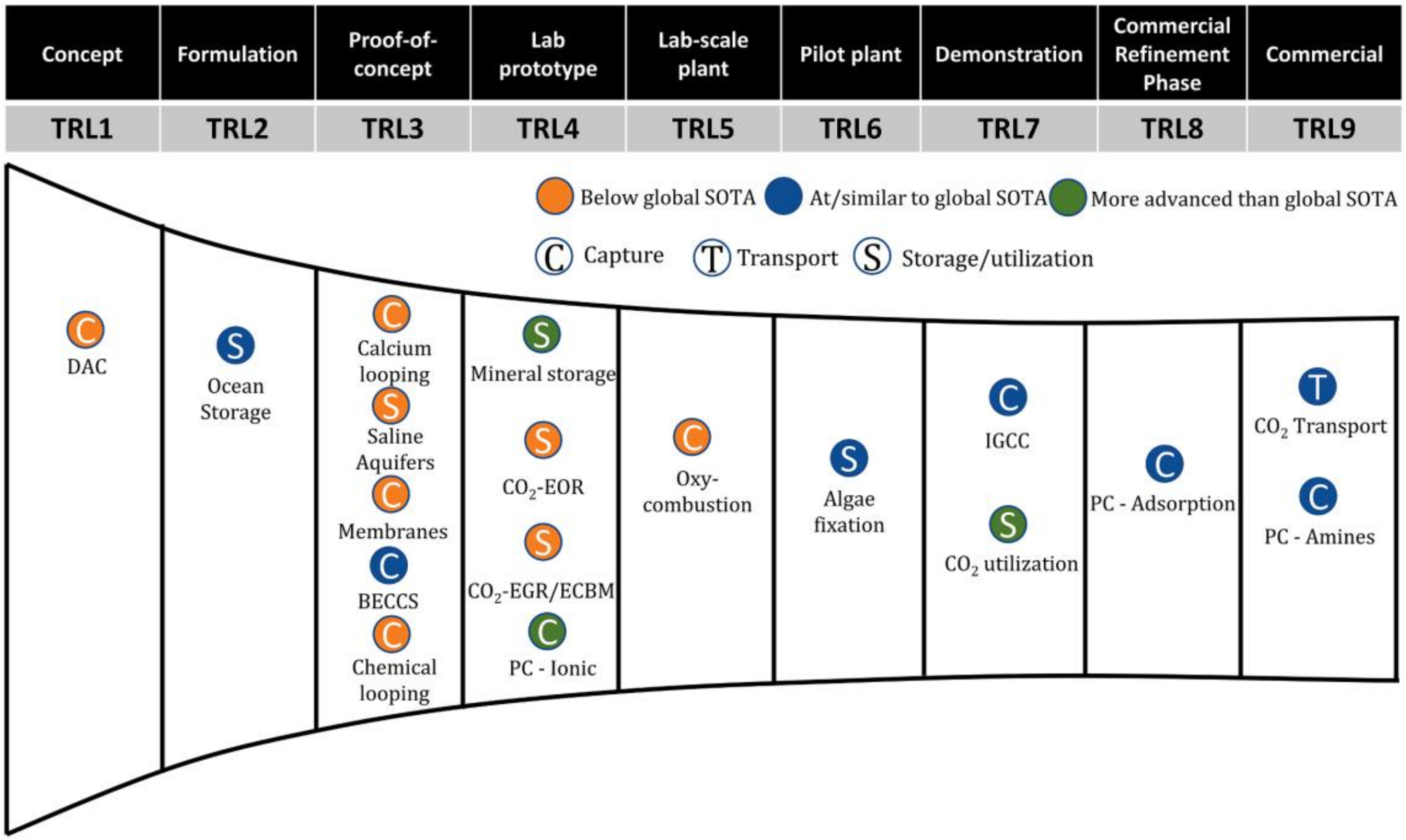
1. Tesla - A well-known electric vehicle manufacturer that has successfully deployed its technology on a large scale.
2. Beyond Meat - A company that has commercialized its plant-based meat alternatives in supermarkets and restaurants worldwide.

# TRL of CCUS Technologies Worldwide



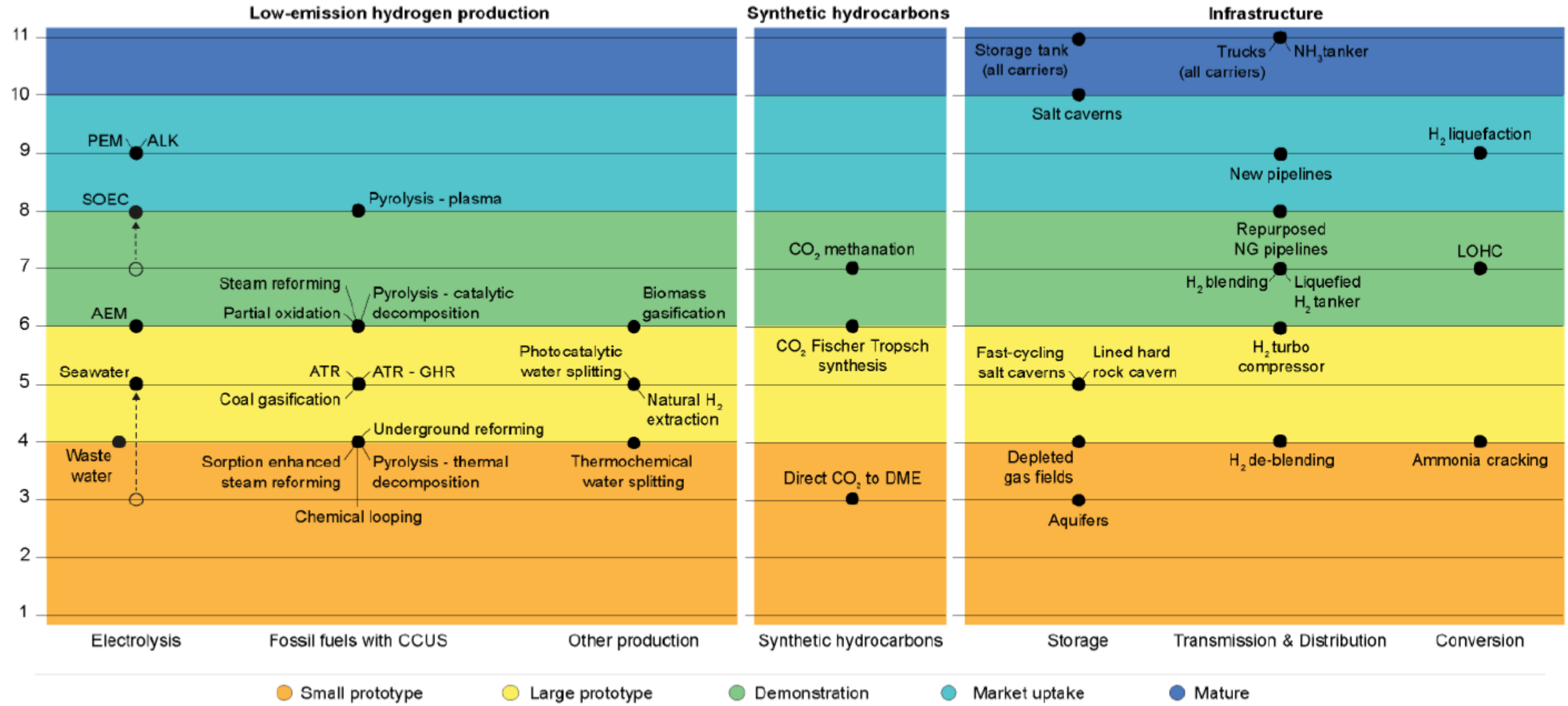

Bui et al., Energy Sci Eng. 2018;11:1062

# TRL of CCUS Technologies Available in India




Colour schemes compare technology stages with the global state-of-the-art (SOTA).

Vishal et al., Resources, Conservation & Recycling 2021;175:105829






Notes: AEM = anion exchange membrane; ALK = alkaline; ATR = autothermal reformer; CCUS = carbon capture, utilisation and storage; $CH_4$ = methane; DME = dimethyl ether; GHR = gas-heated reformer; LOHC = liquid organic hydrogen carrier; $NH_3$ = ammonia; PEM = proton exchange membrane; SOEC = solid oxide electrolyser cell. Biomass refers to both biomass and waste. Arrows show changes in technology readiness level as a consequence of progress in the past year. For technologies in the CCUS category, the technology readiness level refers to the overall concept of coupling production technologies with CCUS and high $CO_2$ capture rates. Pipelines refer to onshore transmission pipelines. Storage in depleted gas fields and aquifers refers to pure hydrogen and not to blends. LOHC refers to hydrogenation and dehydrogenation of liquid organic hydrogen carriers. Ammonia cracking refers to low-temperature ammonia cracking. Technology readiness level classification based on Clean Energy Innovation (2020).

Sources: IEA Clean Tech Guide (2023); IEA Hydrogen Technology Collaboration Programme.

# 9 Latest Technology Trends Shaping the Chemical Sector

## 1. Digital Transformation and Industry 4.0

The chemical industry undergoes a significant transformation with digital transformation, offering new avenues for efficiency, innovation, and competitiveness. In Industry 4.0, digitalization of processes links advanced technologies like IoT, big data analytics, and cloud computing with traditional processes in chemical manufacturing. He processes work optimally. This also implies benefits such as better decision-making and sustainability on top of mere operational efficiency.

**Main Components of Digital Transformation**

- **IoT Sensors and Cloud Computing**
- **IoT Integration:** IoT sensors capture the real-time data from the equipment and processes to enable monitoring and control accurately. The sensors may be measuring temperature, pressure, and flow rates amongst other critical parameters that might ensure the processes run within optimal conditions.
- **Cloud Computing:** All the data of devices and machines get stored and get processed within the cloud environment in order to develop insights which now become readily available, making cross-border operation much easier.
- **Advantages**: It enhances the efficiency of the process; energy saving during production; also reduces the costs of production.

- **Predictive Maintenance**
- **AI-Based Systems:** The use of AI and machine learning algorithms in the analysis of history and real-time data indicates what the patterns of probable equipment failures could be.
- **Cost Savings:** Companies avoid costly repairs and unplanned downtime through predictive maintenance by addressing issues before they escalate.
- **Improved Reliability**: Improvements ensure reliable and uniform product quality; prevent chain breakdown.
- **Digital Twins**
- **Virtual Simulations:** Digital twins are virtual replicas of a physical asset, which might be as simple as the manufacturing plant or the specific equipment in use.Such virtual simulations let the processes go through a form of simulation for testing out scenarios, optimal operation, and even prediction without affecting the original production.
- **Better Decision Making:** Real-time visualization of change enables companies to make data-driven decisions to enhance efficiency and minimize risk.

## 2. Artificial Intelligence (AI) and Machine Learning (ML)

Artificial Intelligence and Machine Learning have significantly transformed the chemical industry. Through these chemical industry trends and technologies, research and development processes can be improved along with manufacturing. Powerful tools help the companies to simulate chemical reactions and predict the material properties along with optimizing formulations in order to develop faster innovations along with higher efficiency operations.

## Applications of AI/ML in Chemical Industry

- **Simulation and Prediction**
- AI algorithms can mimic complicated chemical reactions without the necessity for many physical experiments. This property accelerates the discovery of new material and formulations.
- ML models predict material properties and their behavior based on conditions. It thereby allows the scientists to design novel solutions that fit their application requirements.
- **Process Optimization**
- AI analytics identify inefficiencies in the production process. It allows companies to enhance their yield, waste reduction, and energy usage.
- Real-time ML algorithms learn the actual production conditions, ensuring constant quality while minimizing operational risks.

## 3. Green Chemistry

Sustainable Practices Sustainability has really been the thrust in innovation regarding chemical industry issues.

- Green chemistry emerged as that change, changing everything, aiming at designing such products and their respective processes with diminished or eliminated hazard for humans and society while conserving or improving human health and human safety. Leveraging renewable feedstocks and ecofriendly technologies is sought in companies to limit the impact and be responsive in filling the burgeoning needs for sustainable options.

### Key Principles of Green Chemistry

- **Use of Renewable Feedstocks**
- Moving from petroleum-based raw materials to bio-based alternatives, such as plant-derived compounds, reduces dependence on finite fossil fuels.
- These renewable feedstocks are not only abundant but also often biodegradable, which minimizes long-term environmental damage.
- **Energy Efficiency**
- This green chemistry develops processes that expend less energy or even those run at ambient conditions, like conditions of temperature and pressure. Consequently, it minimizes resource loss and emissions from greenhouse gases.
- **Waste Minimization**
- Closed-loop systems and cleaner production ensure no by-product is generated, and minimal waste in companies.

## 4. Circular Economy and Waste Management

They turn waste streams into valuable resources based on recycling and reuse material. The conservation of resources happens while the issues of waste challenges are being tackled.

- **Key Principles**
- Focus on the reuse and recycling of materials to reduce waste.
- Convert streams of waste into useful resources while conserving finite natural resources.
- **Innovative Example – Eastman Chemical's Molecular Recycling**
- Cracking plastic waste down into molecular constituents for high-quality raw materials
- Produces new products without virgin feedstocks
- **Chemical Recycling**
- It is the only sustainable solution for the global plastic waste crisis.
- Non-recyclable plastics are processed while eradicating the traditional limitations of mechanical recycling.
- Environmental pollution is reduced because material usage will now be a loop.
- **Effects on Waste Management**
- Waste management will not rely on a linear "take-make-dispose" model; it will use a circular "reuse-recycle-regenerate" model.
- Promote sustainability: Reduce resource use and fossil fuel reliance.
- Promotes a global initiative for a clean, greener future.

## 5. Advanced Process Automation and Robotics

Integrating advanced process automation and robotics into chemical manufacturing is revolutionizing processes through enhanced efficiency, precision, and safety. Robotics with the application of artificial intelligence is changing every aspect of production-from material handling to laboratory testing. Automation systems now reduce human error while boosting throughput and allowing for far more precise control of all production processes.

- **Key Innovations**

Covestro has been a material science leader, embracing automation into its manufacturing. Automated material handling systems blend materials with high accuracy, leading to more consistent quality in the finished product and at lower operational cost. The other advantage is it reduces safety risk by eliminating humans from hazardous work environments.

- **Cobots**

Among the many trends happening within this realm is the one with collaborative robots or cobots. Unlike conventional industrial robots, cobots interact directly with humans while they perform activities in an operational facility. In a chemical plant, for example, cobots will help to load or unload materials and keep monitoring equipment; then operations in such plants would smoothly run as well as ensure safe running.

## 6. Blockchain for Supply Chain Transparency

Supply chains in chemistry are increasingly involving blockchain technology toward more transparency and security and enhancing accountability. Their ability to give an immutable record that is kept decentralized makes these systems powerful aids in enhancing the traceability associated with supply chains of products, towards the assurance to be ethical as well as compliance with environmental concerns and regulatory statutes.

- **Key features and benefits**

Every transaction gets recorded on a digital ledger, which all members of the supply chain have access to. Since once recorded, no changes are allowed in transactions, this guarantees all information-from where raw materials came from to their shipment status and handling-are tamper-proof. It establishes trust and openness among the supply chain participants-from the manufacturer to the consumer.

It improves transparency and can significantly reduce fraud and the risk of counterfeit goods entering the supply chain. It enables faster and more accurate reconciliation of payments and deliveries, thus improving efficiency.

- **Use Case: BASF's Sustainable Palm Oil Tracking**

The most important example is that of the application of blockchain by BASF. It has launched a blockchain for recording the sustainable use of palm oil. With its implementation of blockchain technology, BASF can be ensured to make use of only sustainably and ethically sourced palm oil in its products. In this way, blockchain ensures sustainability-related compliance at all the stages of the supply chain, right from plantations to final products.

## 7. Additive Manufacturing and 3D Printing in the Chemical Sector

Additive manufacturing, commonly known as 3D printing, is a transforming technology, proving to amongst the greatest chemical industry trends. This practice gives way to complex chemical products and components to be made by the deposition of layer upon layer and has many benefits regarding customization, material efficiency, and rapid prototyping. It's rapidly transforming how chemicals and materials are designed, tested, and produced. With 3D printing, the impossible becomes possible.

Major Application of 3D Printing in Chemical Industry

- **Rapid Prototyping and Customization**
- The chemical manufacturer can develop a prototype within days through 3D printing. It lets them test and iterate product designs far faster than in any conventional manufacturing process.
- Being able to exactly print custom chemical formulations would mean a potentially more customized product, especially for pharmaceuticals, automotive, and electronics industries.
- **Material Innovation and Optimization**
- It is possible to embed different materials such as polymers, metals, and ceramics in 3D printing to achieve new material properties. It means lightweight, high-demand industrial products.
- For example, within the automobile world, 3D-printed polymers result in the very lightweight yet effective materials. Also, the car becomes lighter but at the same time enhances its fuel efficiency.

- **Reduce Material Waste**
- The traditional manufacturing process will involve the removal of material from a larger block, which will create waste. 3D printing, on the other hand, uses only the material required for the item under production. Therefore, waste gets reduced and results in lowering down the environmental implication of manufacturing.
- This reduction in waste is particularly helpful for the chemical industry, which may have raw materials that are expensive and resource-intensive.

- **Rapid Development Cycles**
- 3D printing can prototype, test, and refine products rapidly for chemical companies. It accelerates the product development cycles, which can reduce time-to-market for new chemical innovations and solutions.
- Customized catalysts are also produced through 3D printing technologies. Such catalysts often play a very significant role in many chemical reactions, and it makes the R&D process of new formulations much faster.
- **Supply Chain Optimization**
- With 3D printing, companies are able to print on demand the special chemicals or components, without keeping inventories of extensive proportions. This also leads to less delay and cost in logistics, making it really agile and responsive supply chains.

## 8. Renewable Energy Integration

The importance of integrating renewable energy sources in chemical manufacturing operations is increasing since companies are facing the challenge of reducing their carbon footprints and trying to meet the global decarbonization targets. Chemical companies will not only make a cleaner environment but also ensure long-term cost savings and improvement in operational efficiency by adopting renewable energy, which includes solar, wind, and bioenergy.

### Renewable Energy Key Drivers in Chemicals

- **Environmental Responsibility**
- Chemical companies are beginning to realize that their activities harm the environment. The use of renewable energy, as a whole, is going to reduce emissions and help achieve the sustainability requirements set by regulators for companies.
- Transition to cleaner sources of energy supports international climate agreements and local governmental policies to reduce carbon emissions.

- **Cost reduction**
- Renewable energy can save more in the long run since these technologies such as solar and wind power continue to become cheaper and cheaper.
- Others invest in on-site source renewable energy generation through chemical companies that have ‘solar panels or wind turbines’, making such a business less reliant on grid electricity supply and having much control over their energy costs.
- **Energy Security and Reliability**
- The integration of renewable energy helps the chemical companies diversify their sources of energy and thus decreases their vulnerability to volatility in fossil fuel prices and interruptions in supply chains.
- Renewable energy will form a secure supply of power due to the current instability of these energy markets for such companies in operational areas.

## 9. Advanced Materials and Nanotechnology

Advanced materials with more superior properties bring about major break-throughs with nanotechnology-driven changes in nearly every industry in the world today. Advanced strength and conductivity property materials are engineering catalysts toward making everything-everything ranging from electronics, healthcare, everything much more effective.

**Key Developments in Advanced Materials**

- **Nanocomposites:** Nanotechnology aids in the production of nanomaterial-based materials through conventional materials combined together. Therefore, nanocomposites possess enhanced properties like mechanical strength, lightweight, and friction as well as abrasion resistance. Nanocomposites are becoming significant in the applications of automobiles, aerospace, and construction sectors with improved strength and durability.
- **Catalysts:** Nanomaterials are increasingly showing their very effective role in catalysis and accelerating chemical reaction rates, besides enhancing efficiency in chemical processes. This is critically important in chemicals and energy companies, where high-performance catalysts can result in more sustainable process manufacturing and the production of clean energy.

**Highly Emphasized Innovation-Graphene**

- The other exciting development is graphene, famous for its superior properties, with electrical conductivity unlike any other known, tensile strength that can almost rival steel and is incredibly lightweight. Graphene is being considered for the manufacture of next-generation energy storage, including ultra-efficient batteries and capacitors, thus significantly improving electronic devices.
- Medical Applications: Given its biocompatibility and potential for very large surface area improvements, graphene could be integrated with medical devices and drug delivery systems for effective drug absorption as well as effective monitoring.

## Future Predictions

The chemical industry will witness unimaginable innovation, embracing this transformatory trend. Companies would continue to keep driving AI-powered R&D and blockchain transparency through advanced manufacturing in the future.

High implementation costs and strict regulation remain a challenge. Companies who put a premium on sustainability, along with technological adoption, will lead the way and create an efficient greener industry.

## Conclusion

The chemical industry is, in fact on the threshold of its revolutionary era, be it about digitalization, AI, and green chemistry plus circular economy in practice. Thereby, advances in technology alter the production pattern and consumption behaviors of chemicals significantly.

This would provide chemical companies a chance to sustain themselves, develop efficiency in running operations, and be responsive to changing market demands. With year 2024 at the threshold, the industry is ready to transform- innovating, staying resilient, and stewarding its environment.

# CALCULATE FINANCIAL INDICATORS TO GUIDE INVESTMENTS

| Table 1. Consider these key R&D inputs when preparing financial analyses of proposed projects. | |
|---|---|
| **Input to Financial Calculations** | **Inputs from R&D Required to Ensure Valid Financial Analysis** |
| Revenue/Market | Is the market new or established? How large is the market?<br>What advantage will the new product/process offer and what is its value?<br>Based on the answers to the two previous questions, what share of the market can be obtained?<br>Are prices for the products well known?<br>Are byproduct credits established?<br>Are byproduct credits vulnerable to changes in demand (such that they could become negative)?<br>Does product demand already exist or will it increase over time? How much time?<br>Will revenues from existing products be affected (portfolio extensions, cannibalization, etc.)?<br>Has the market been assessed in all parts of the world?<br>Are product registrations and/or other regulatory clearances required? |
| Raw Material Prices | Have suppliers been identified for each of the key raw materials?<br>Are any of the key raw materials available from only one source?<br>Have price quotes/estimates been validated? |

| | |
|---|---|
| Raw Material Usage Rates | Are reaction yields known?<br>How many data points support the assumptions?<br>Is the ability to recycle solvents, catalysts, etc. proven?<br>At what scale have these usage rates been demonstrated?<br>Must operating parameters be tightly controlled to achieve the assumed results?<br>What are the consequences of operating outside the ordinary control limits? |
| Capital and Conversion Costs | What volume is required by the market?<br>Will the plant be operated at this volume? Will it be operated year-round?<br>Has the process been run in its entirety so that all operations (*e.g.*, waste treatment) are accounted for?<br>Have all unit operations been identified?<br>Is equipment for any unit operation highly specialized or unique?<br>Are the materials of construction requirements well understood?<br>Are all utility requirements known?<br>Are labor rates known?<br>Are labor productivities well understood?<br>Are equipment import restrictions and duties known? |

| | |
|---|---|
| Nonmanufacturing Costs | Will investment be required long before the product is available (*e.g.*, a new plant needs to be built)? How much earlier?<br>What rate of return on loans and/or investments is expected?<br>Is project timing contingent on resource availability? How will timing be affected?<br>Are local tax rates, import duties, etc. that apply to products well known?<br>How long will the product be in demand (*i.e.*, are replacement and/or competitive products being developed)?<br>Does all required infractructure for product transport exist?<br>Is the level of ongoing sales, R&D, and technical support required by the product application(s) well understood? |

# CALCULATION OF BASIC FINANCIAL INDICATORS

| Table 2. Example 1: Financial analysis for a proposed new product launch. | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | A | B | C | D | E | F | G | H | I | J | K |
| 1 | Year | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
| 2 | Product Volume, MM lb | — | — | 10 | 15 | 70 | 90 | 100 | 100 | 100 | 100 |
| 3 | Sales Price, $/lb | — | — | 3.00 | 3.00 | 3.00 | 3.00 | 3.00 | 3.00 | 3.00 | 3.00 |
| 4 | Total Revenue, $MM<br>= Product Volume × Sales Price | — | — | 30 | 45 | 210 | 270 | 300 | 300 | 300 | 300 |
| 5 | Raw Material Price, $/lb of product | — | — | 2.00 | 2.00 | 2.00 | 2.00 | 2.00 | 2.00 | 2.00 | 2.00 |
| 6 | Raw Material Cost, $MM<br>= Product Volume × Raw Material Price | — | — | 20 | 30 | 140 | 180 | 200 | 200 | 200 | 200 |
| 7 | Other Costs: Labor, Maintenance, etc., $MM | — | — | 10 | 11 | 17 | 20 | 21 | 21 | 21 | 21 |
| 8 | Total Cash Cost, $MM<br>= Raw Material Cost + Other Costs | — | — | 30 | 41 | 157 | 200 | 221 | 221 | 221 | 221 |
| 9 | Depreciation, $MM | — | — | 20 | 20 | 20 | 20 | 20 | 0 | 0 | 0 |
| 10 | Tax at 30%, $MM<br>= 0.3 × (Revenue – Cash Costs – Depreciation) | — | — | –6 | –5 | 10 | 15 | 18 | 24 | 24 | 24 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 11 | Net Income, \$MM<br>= Revenue – Cash Costs – Depreciation – Taxes | — | — | –14 | –11 | 23 | 35 | 41 | 55 | 55 | 55 |
| 12 | Capital, \$MM | 20 | 80 | — | — | — | — | — | — | — | — |
| 13 | Cash Flow, \$MM<br>= Revenue – Cash Costs – Taxes – Capital | –20 | –80 | 6 | 9 | 43 | 55 | 61 | 55 | 55 | 55 |
| 14 | Discount Factor at *r* = 10% | 0.952 | 0.861 | 0.779 | 0.705 | 0.638 | 0.577 | 0.522 | 0.473 | 0.428 | 0.387 |
| 15 | Discounted Cash Flow, \$MM<br>= Discount Factor × Cash Flow | –19 | –69 | 5 | 6 | 27 | 32 | 32 | 26 | 24 | 21 |
| 16 | $NPV_{10}$ | \$85MM | | | | | | | | | |
| 17 | IRR | 23.3% | | | | | | | | | |
| 18 | Steady-State ROI | 55% | | | | | | | | | |
| 19 | Payback | ~6 years | | | | | | | | | |

| Table 3. Example 2: Financial analysis for a hypothetical plant improvement project. | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | A | B | C | D | E | F | G | H | I | J | K |
| 1 | Year | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
| 2 | Product Volume, MM lb | — | — | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 |
| 3 | Cost Reduction, $/lb | — | — | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 |
| 4 | Total Revenue, $MM<br>= Product Volume × Cost Reduction | — | — | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 | 5.0 |
| 5 | Raw Material Price, $/lb of product | — | — | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 6 | Raw Material Cost, $MM<br>= Product Volume × Raw Material Price | — | — | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 7 | Other Costs: Labor, Maintenance, etc., $MM | — | — | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 |
| 8 | Total Cash Cost, $MM<br>= Raw Material Cost + Other Costs | — | — | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 |
| 9 | Depreciation, $MM | — | — | 1.2 | 1.2 | 1.2 | 1.2 | 1.2 | 0 | 0 | 0 |
| 10 | Tax at 30%, $MM<br>= 0.3 × (Revenue – Cash Costs – Depreciation) | — | — | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.4 | 1.4 | 1.4 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 11 | Net Income, \$MM<br>= Revenue – Cash Costs – Depreciation – Taxes | — | — | 2.4 | 2.4 | 2.4 | 2.4 | 2.4 | 3.2 | 3.2 | 3.2 |
| 12 | Capital, \$MM | 1.0 | 5.0 | — | — | — | — | — | — | — | — |
| 13 | Cash Flow, \$MM<br>= Revenue – Cash Costs – Taxes – Capital | –1.0 | –5.0 | 3.6 | 3.6 | 3.6 | 3.6 | 3.6 | 3.2 | 3.2 | 3.2 |
| 14 | Discount Factor at $r$ = 10% | 0.952 | 0.861 | 0.779 | 0.705 | 0.638 | 0.577 | 0.522 | 0.473 | 0.428 | 0.387 |
| 15 | Discounted Cash Flow, \$MM<br>= Discount Factor × Cash Flow | –0.95 | –4.31 | 2.79 | 2.52 | 2.28 | 2.07 | 1.87 | 1.52 | 1.38 | 1.25 |
| 16 | $NPV_{10}$ | \$10.4MM | | | | | | | | | |
| 17 | IRR | 42% | | | | | | | | | |
| 18 | Steady-State ROI | 54% | | | | | | | | | |
| 19 | Payback | ~4 years | | | | | | | | | |

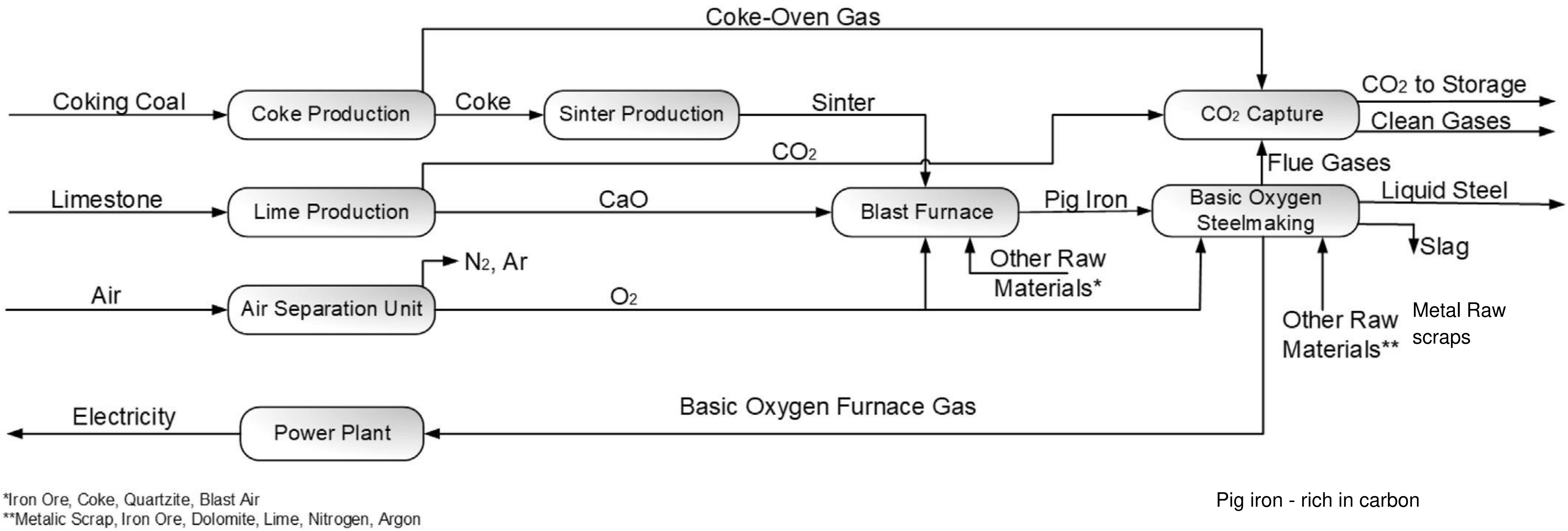
**Table 4. This comparison of Examples 1 and 2 shows that although Example 1 generates a higher $NPV_{10}$, Example 2 requires far less capital expenditures and may be a less-risky project.**

| Example | Description | $NPV_{10}$, \$MM | Capital Investment, \$MM | $NPV_{10}$/ Capital Investment |
|---|---|---|---|---|
| 1 | New Product | \$85 | \$100 | 0.85 |
| 2 | Plant Improvement | \$10 | \$6 | 1.67 |

# Steel industry

Problems in steel production
$CO_2$
managing slag
energy efficiency

# General scheme of steelmaking

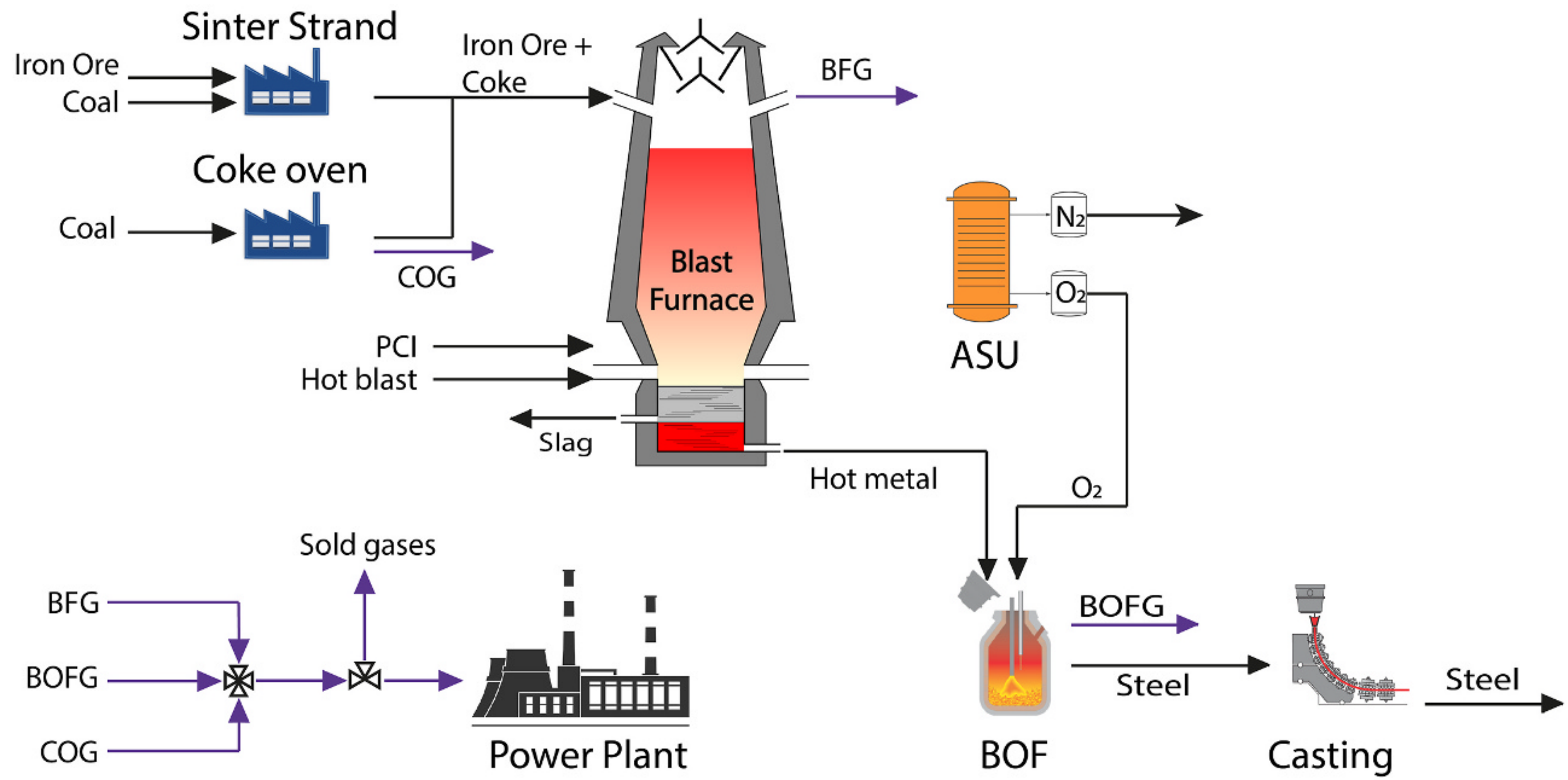



Metal Raw scraps

Pig iron - rich in carbon

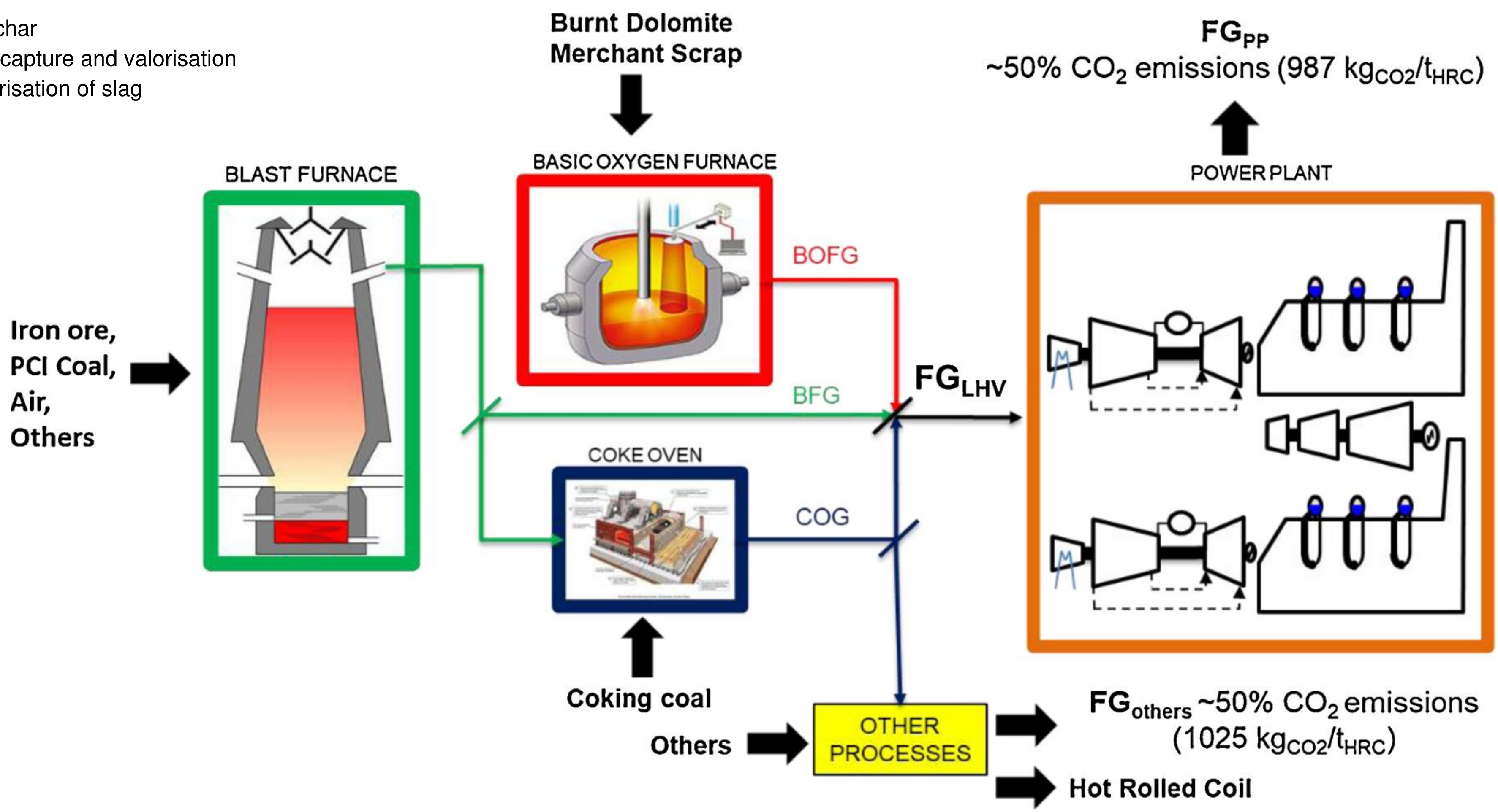
DRIEAF - current method

# Conventional iron and steel plant



# Schematic of integrated steelmaking

bio char
co2 capture and valorisation
valorisation of slag



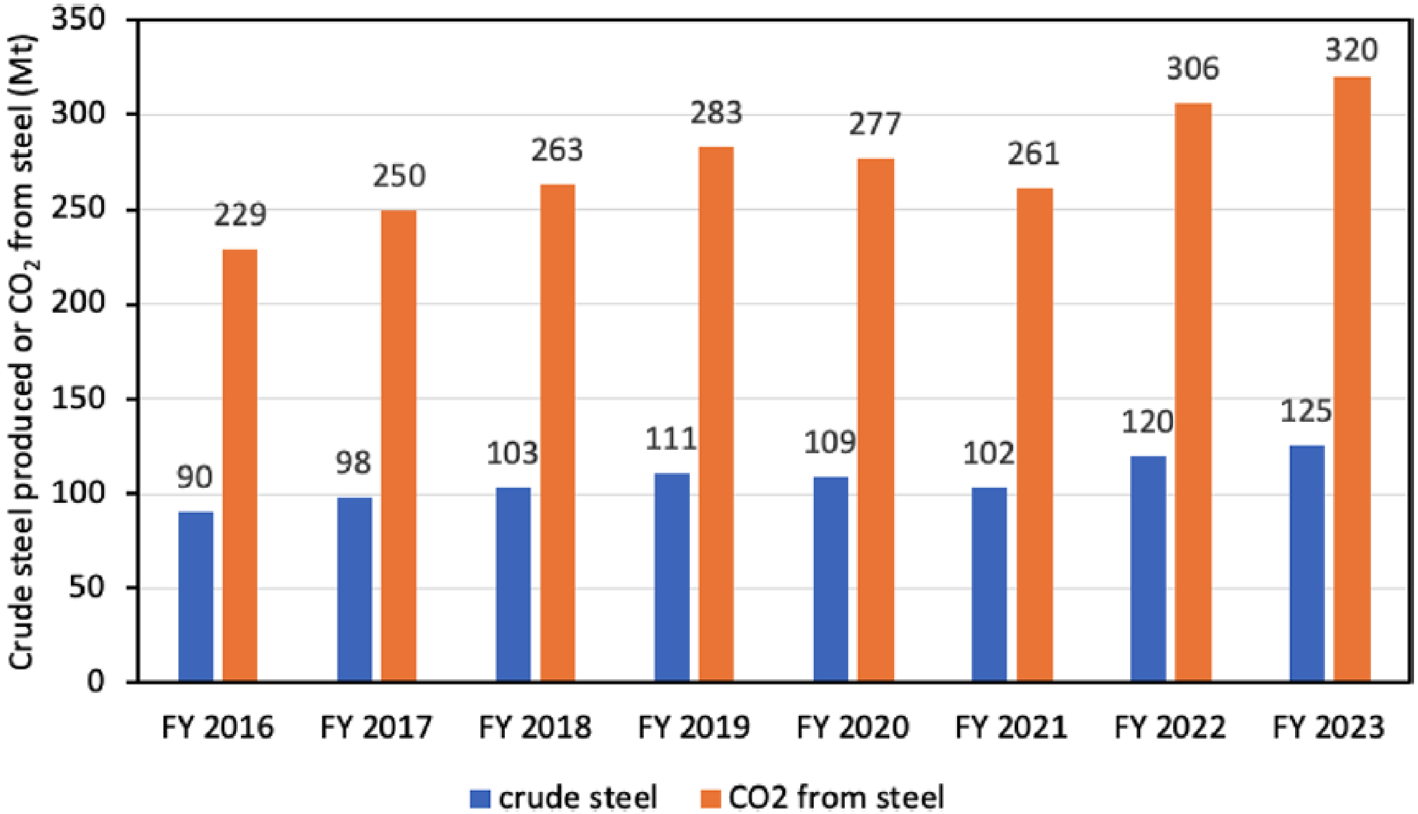
why high co2/ton of coal?
1. coal not natural gas
2. quality of iron ore is not too great (requires more energy for reduction)
3. unavailability of scrap iron

# Steelmaking in India

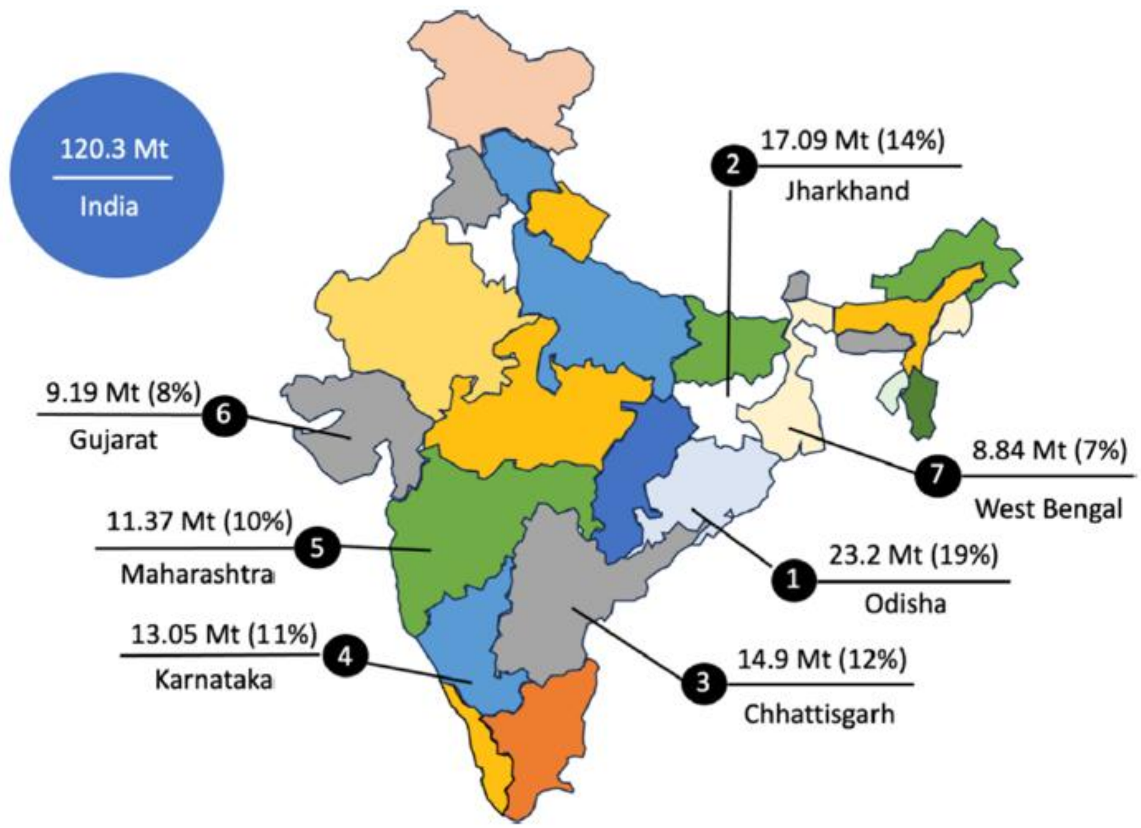
steel production to 500 by 2050

2.55 co2/steel has to reduce to ~2.2

net zero CO2 by 2070



# Crude steel production in India in FY2022



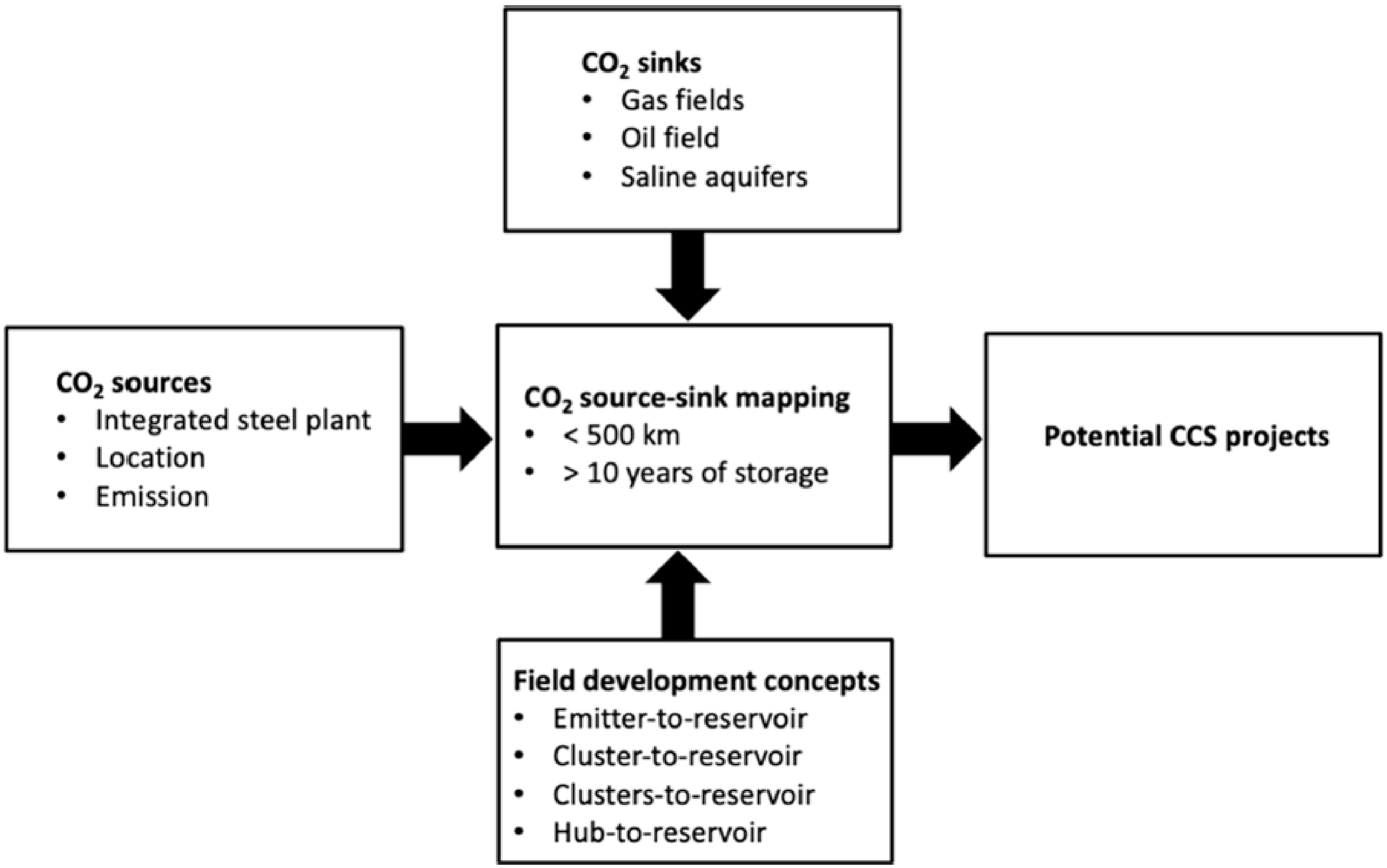
**Problems in decarbonization of steelmaking**

1. Availability of natural gas
2. Quality of iron ore
3. Availability of scrap iron

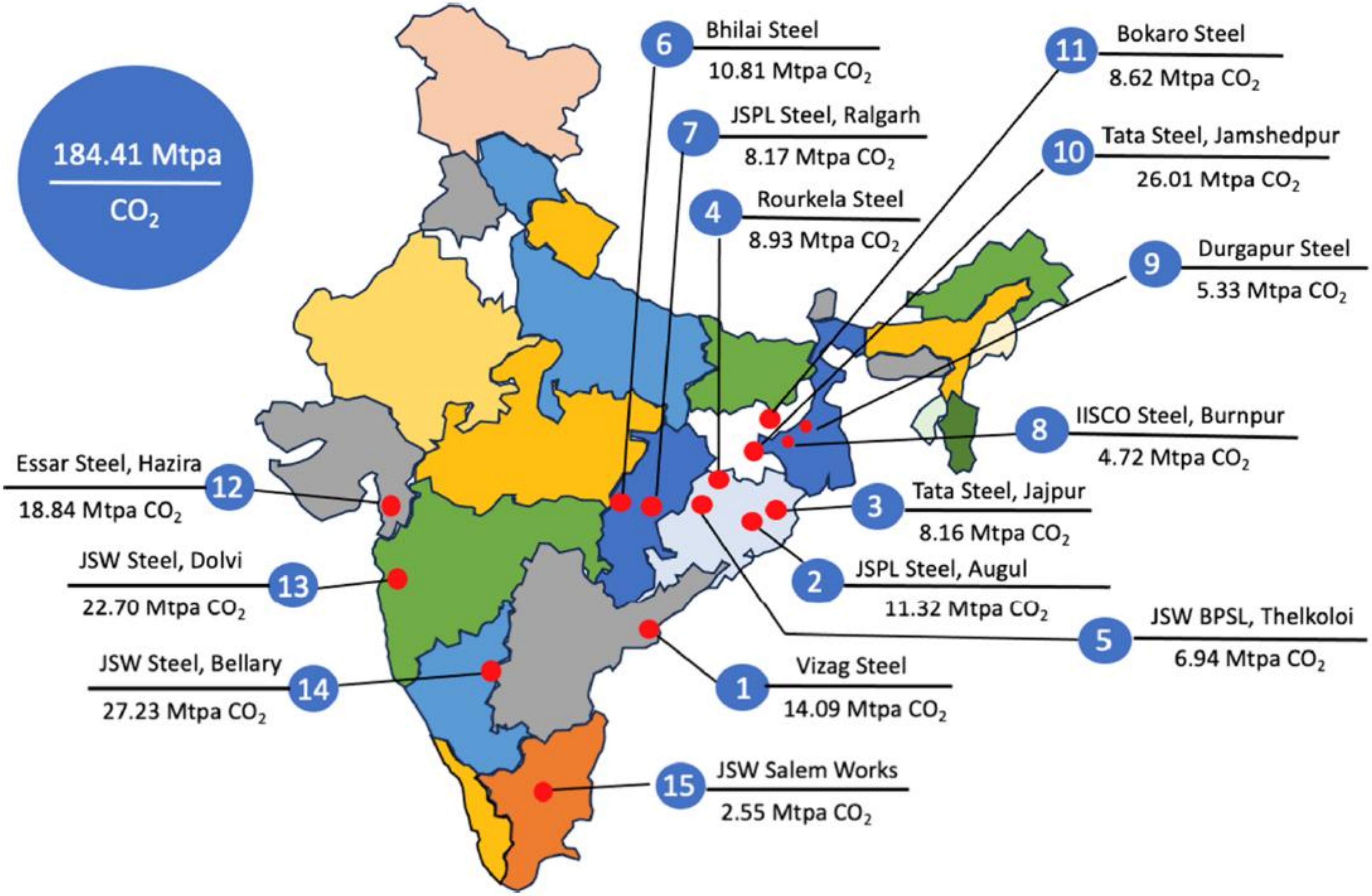
**Options for decarbonization of steelmaking**

1. Replace coal with NG
2. Use better technology (smelting reduction)
3. CCUS

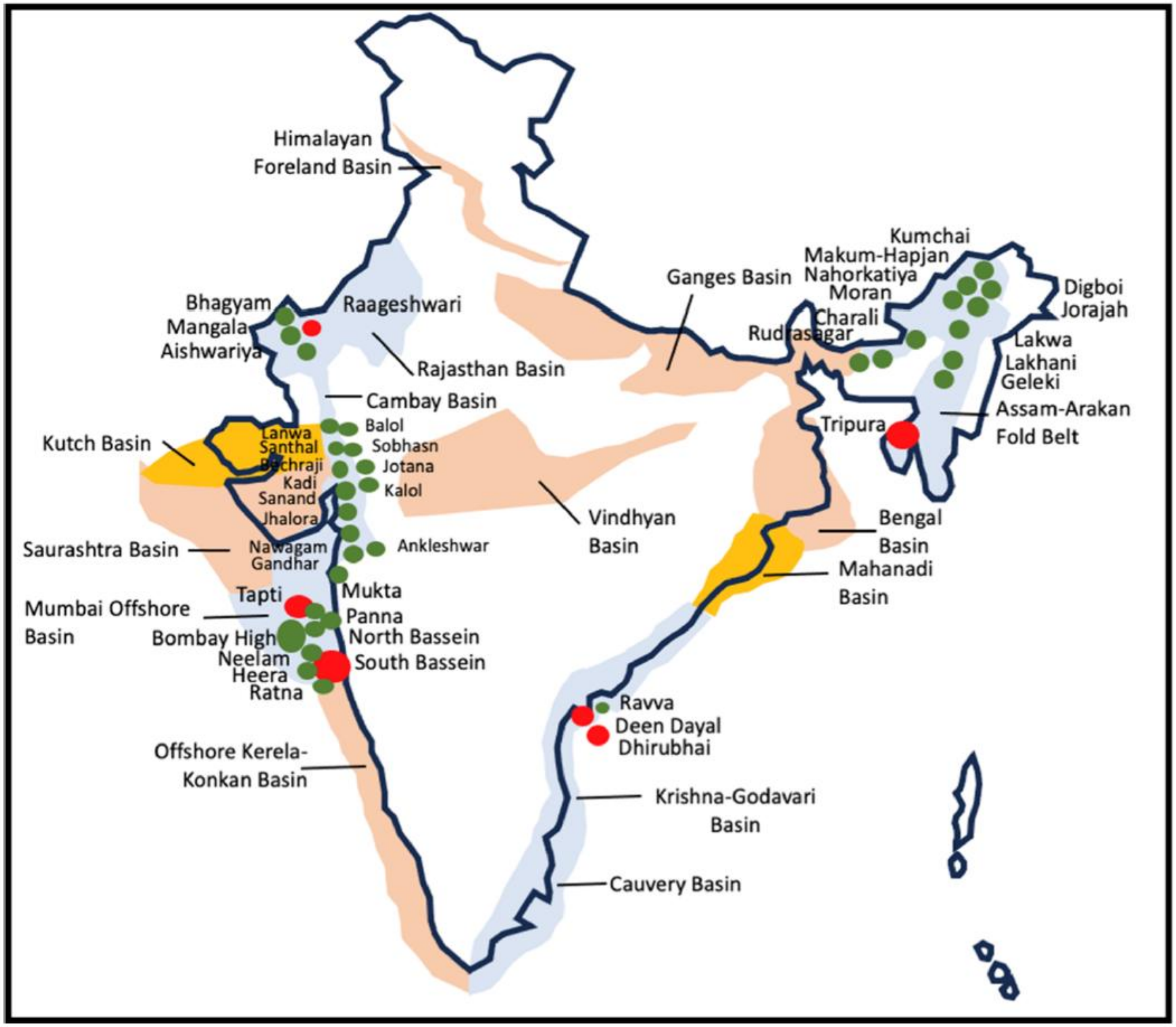
# CCS in steelmaking

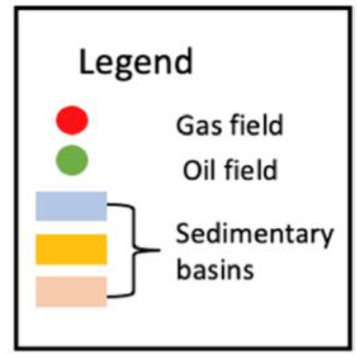


**Integrated steel plants in India**



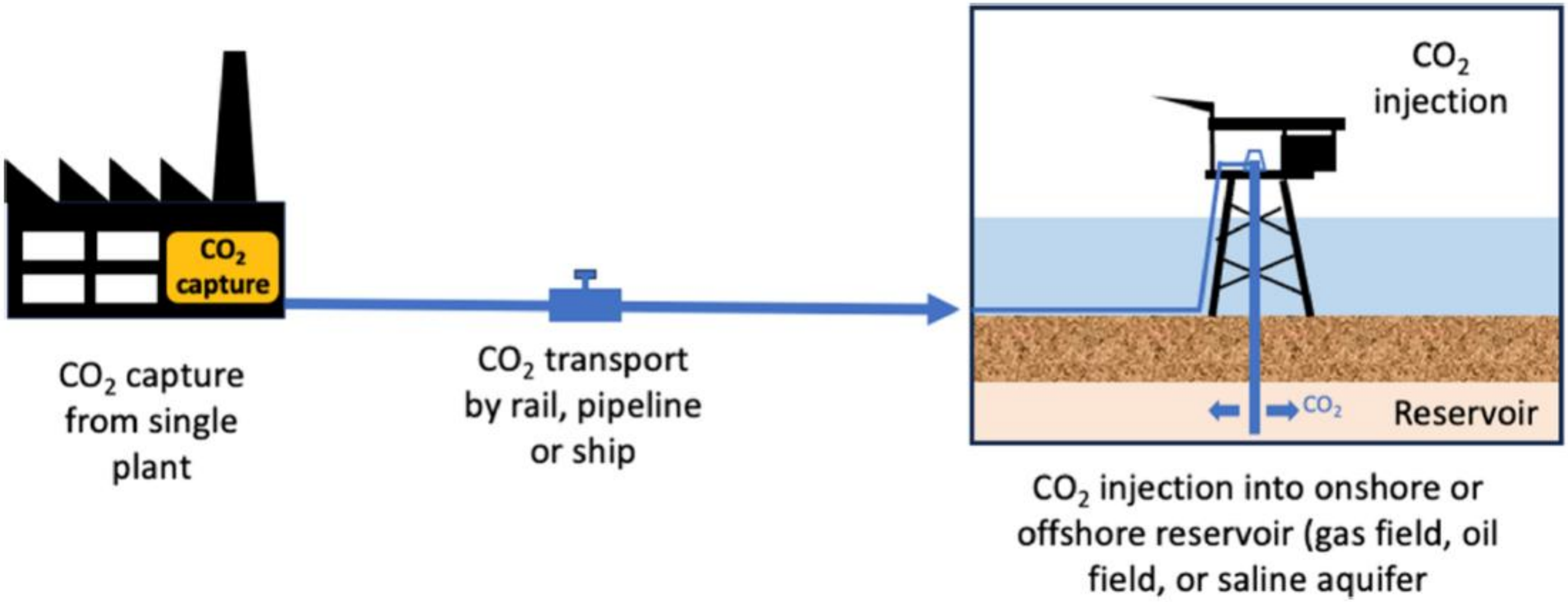
# Sedimentary basins, oil and gas fields in India



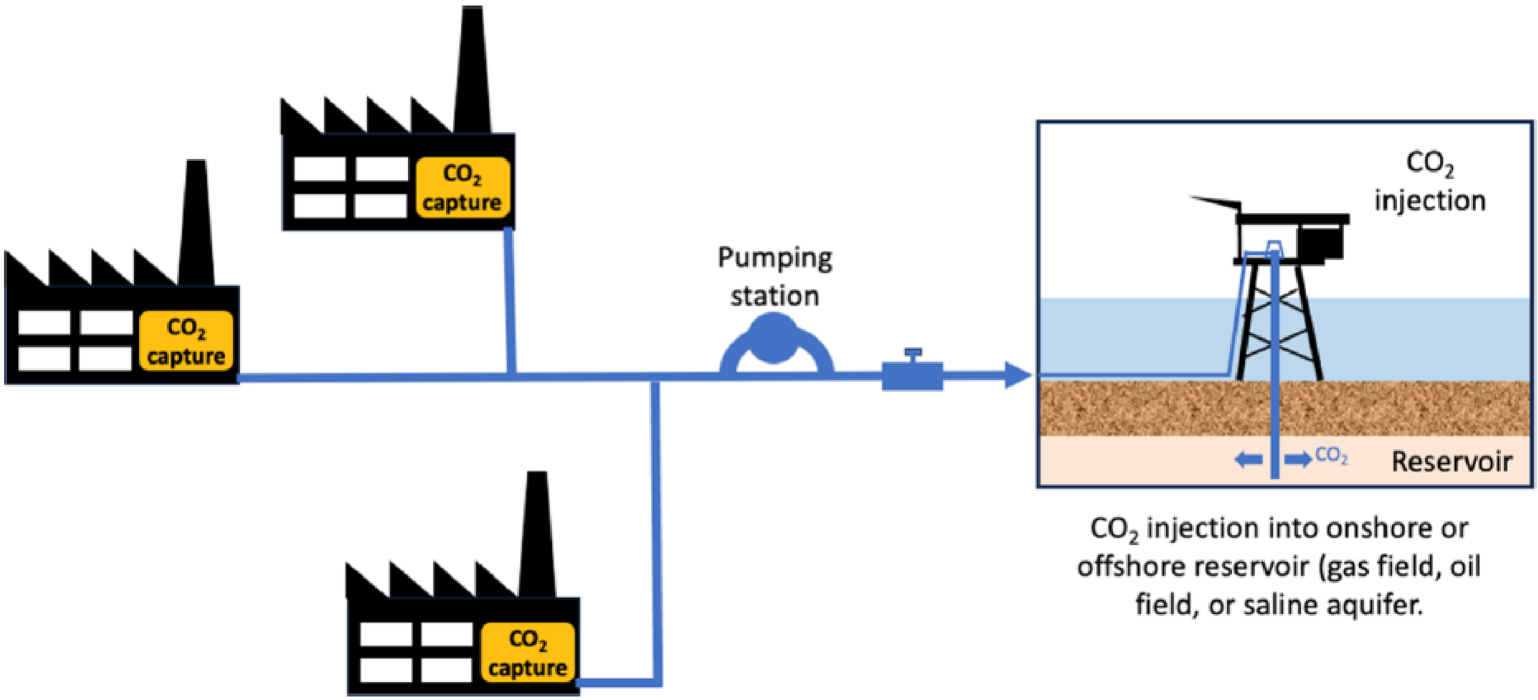




enhanced oil recovery and enhanced gas recovery are the only practical solutions

# Emitter-to-reservoir CCS concept

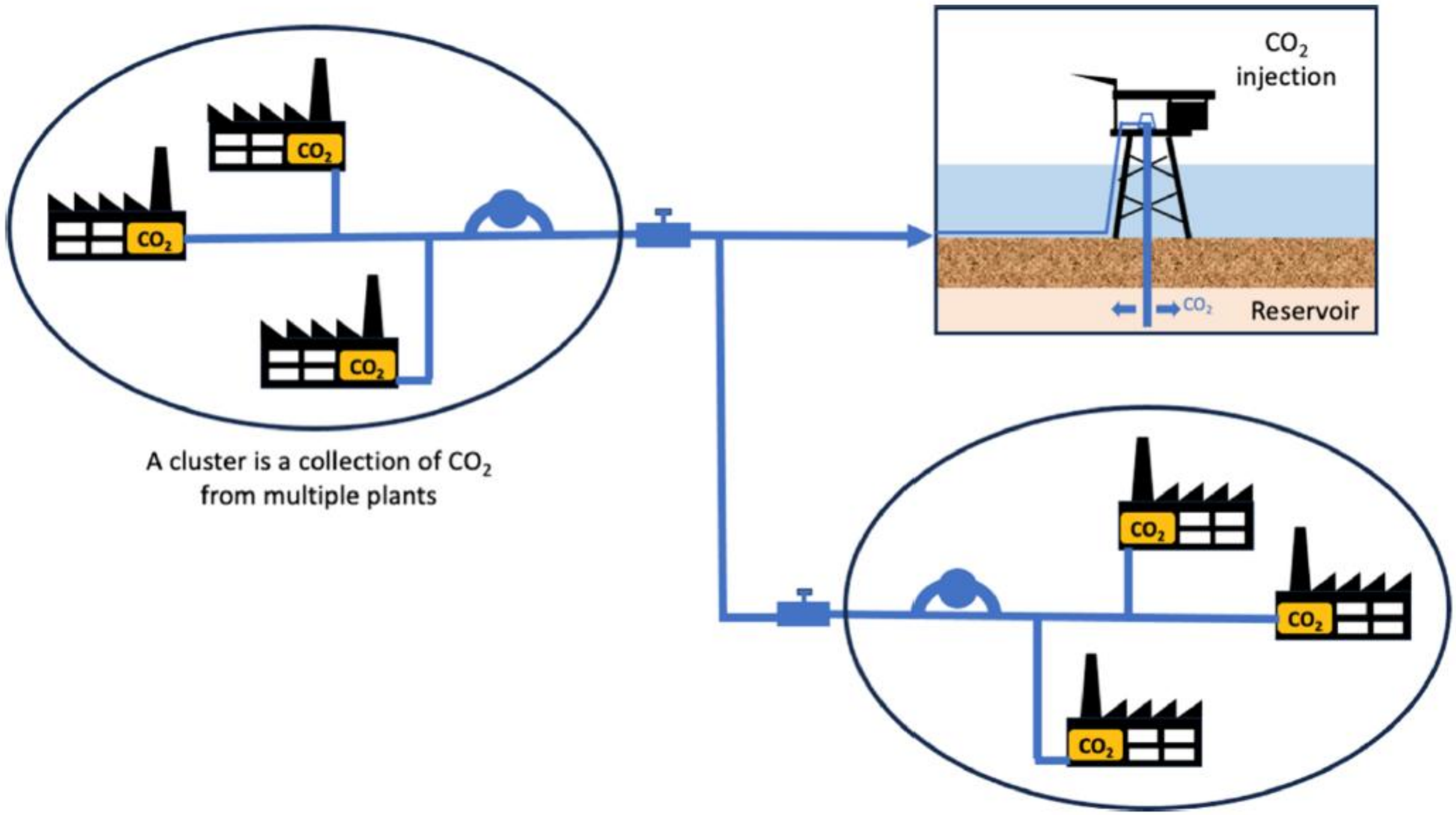


# Cluster-to-reservoir CCS concept

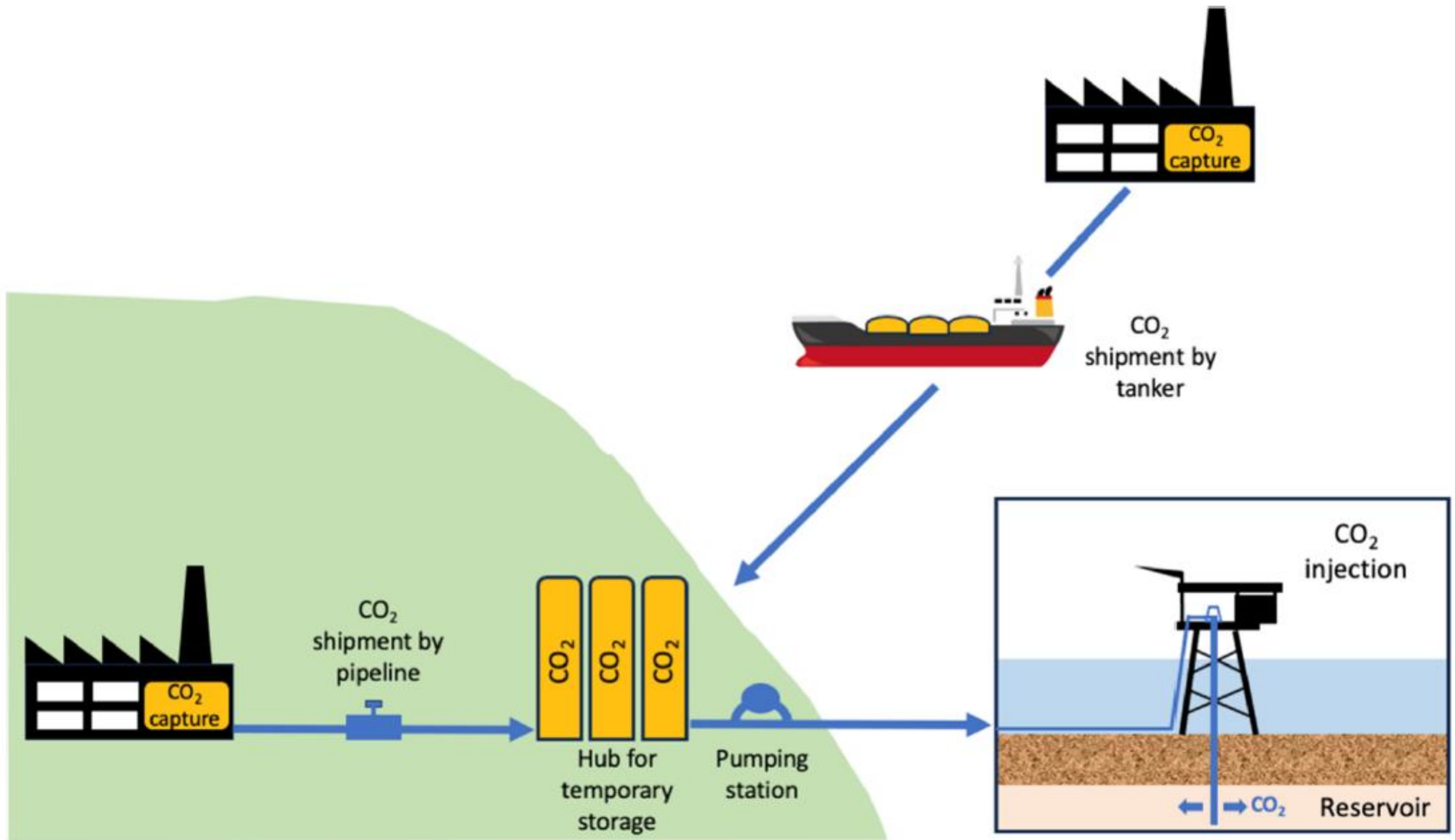



$CO_2$ injection into onshore or offshore reservoir (gas field, oil field, or saline aquifer.

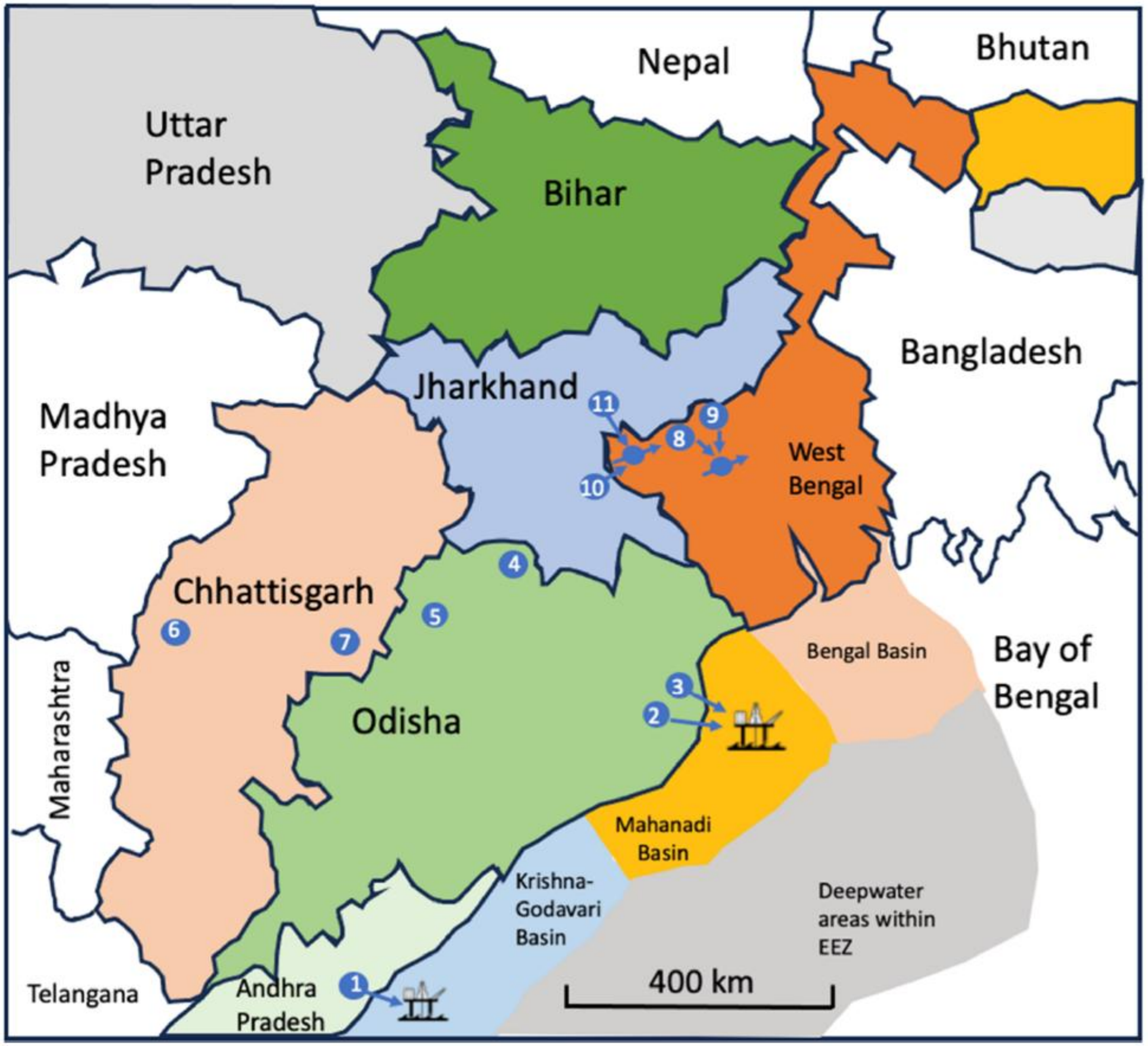
# Clusters-to-reservoir CCS concept



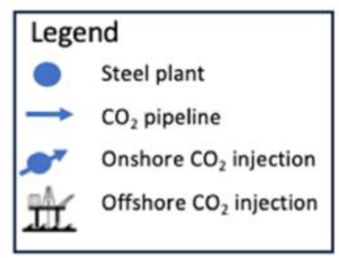
# Hub-to-reservoir CCS concept



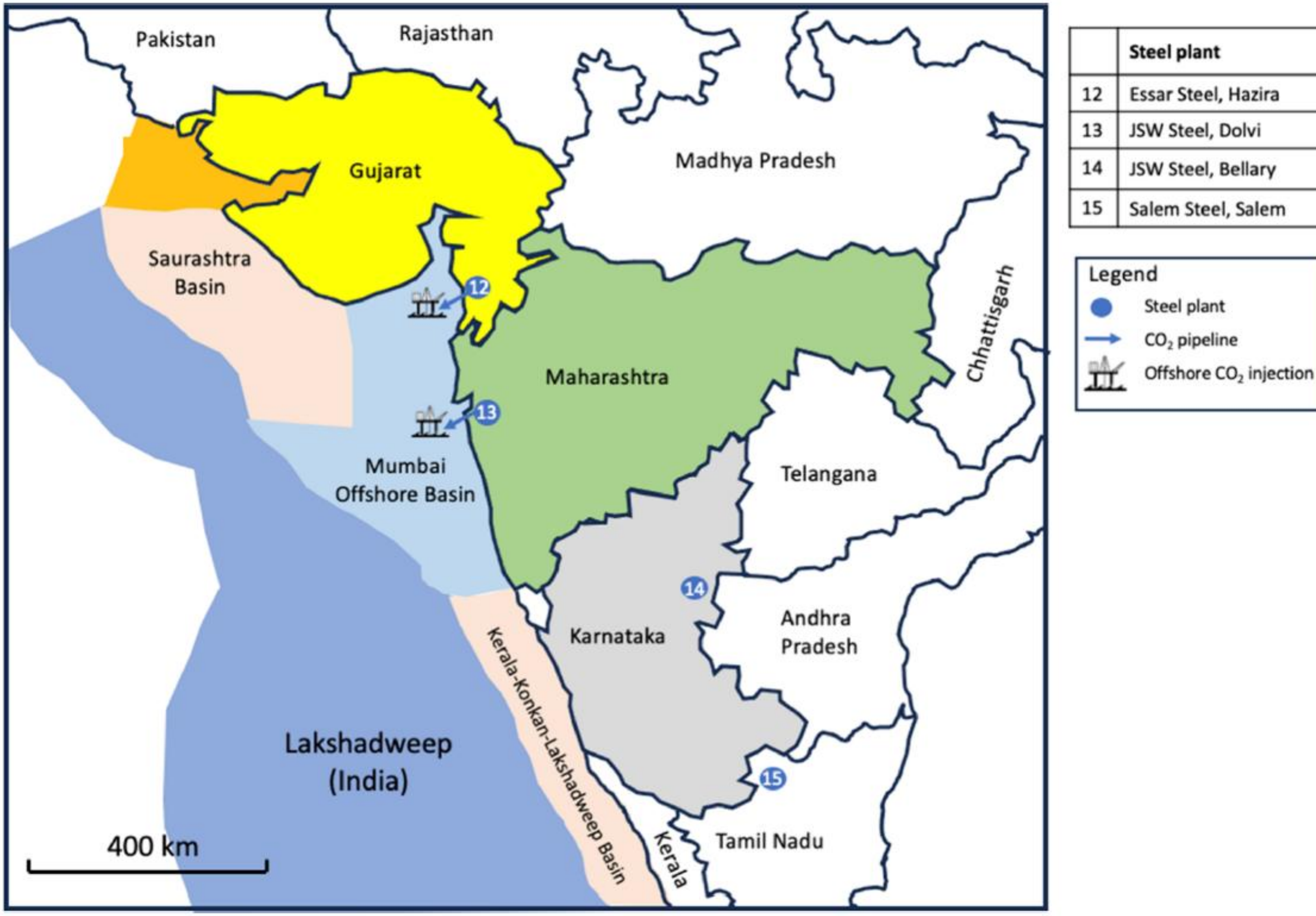
# $CO_2$ source-sink mapping




| | Steel plant |
|---|---|
| 1 | Vizag Steel |
| 2 | JSPL Steel, Angul |
| 3 | Tata Steel, Jajpur |
| 4 | Rourkela Steel Plant |
| 5 | JSW BPSL, Thelkoloi |
| 6 | Bhilai Steel |
| 7 | JSPL Steel, Ralgarh |
| 8 | IISCO Steel, Burnpur |
| 9 | Durgapur Steel Plant |
| 10 | Tata Steel Ltd, Jamshedpur |
| 11 | Bokaro Steel Plant |

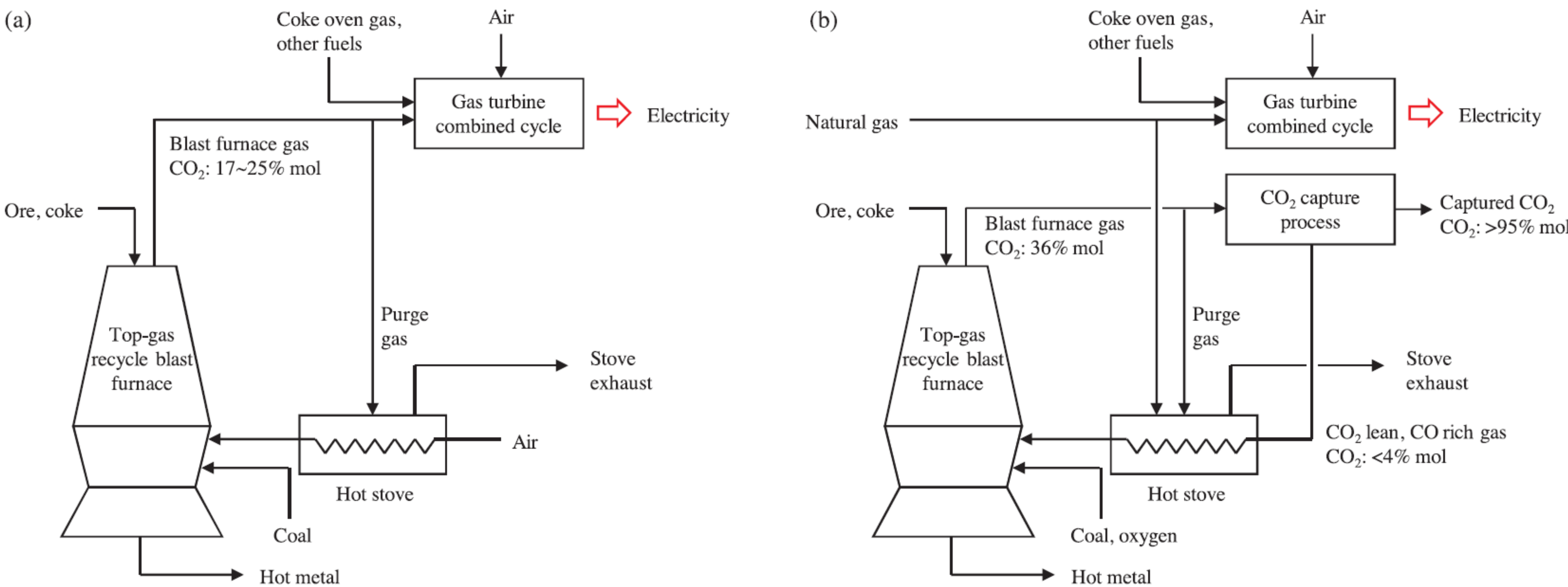


# $CO_2$ source-sink mapping

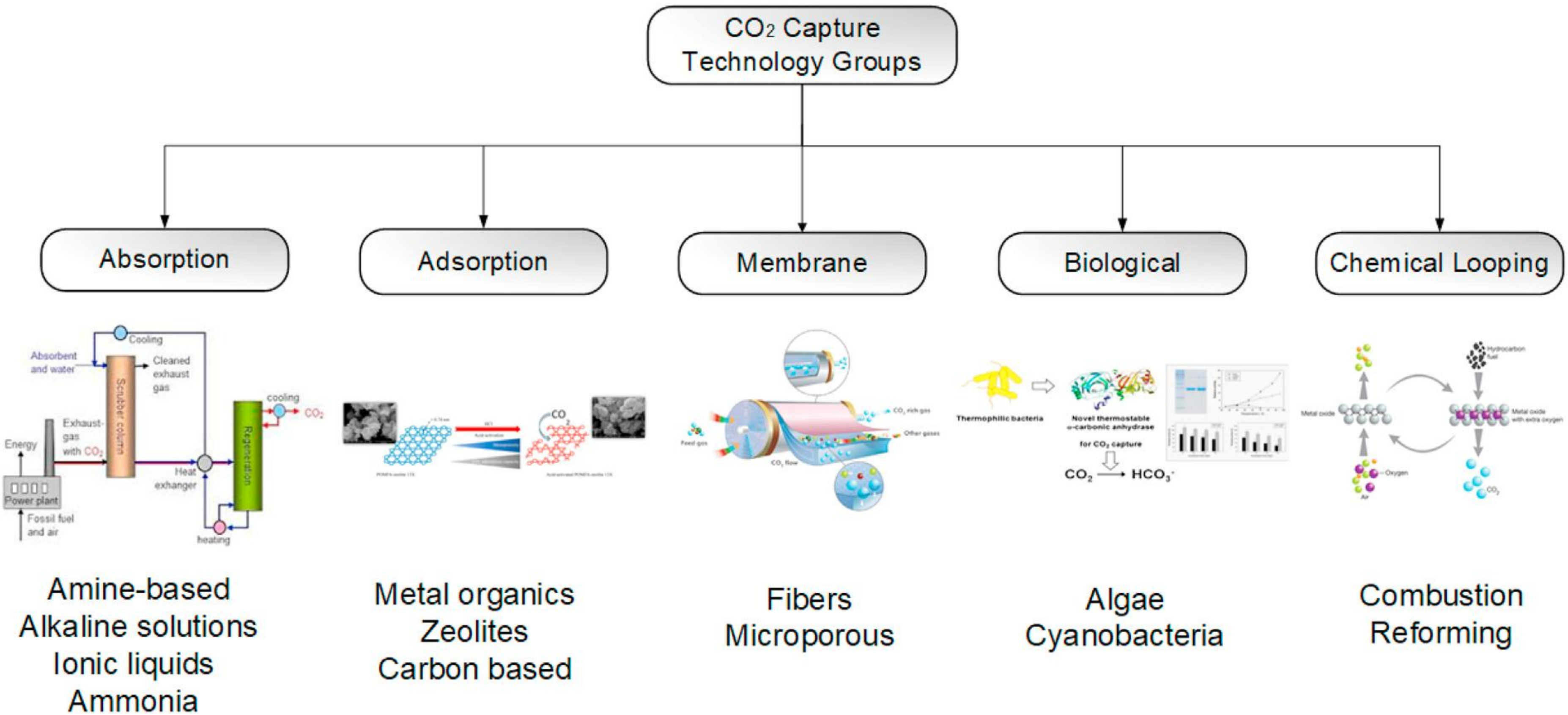



| | Steel plant |
|---|---|
| 12 | Essar Steel, Hazira |
| 13 | JSW Steel, Dolvi |
| 14 | JSW Steel, Bellary |
| 15 | Salem Steel, Salem |

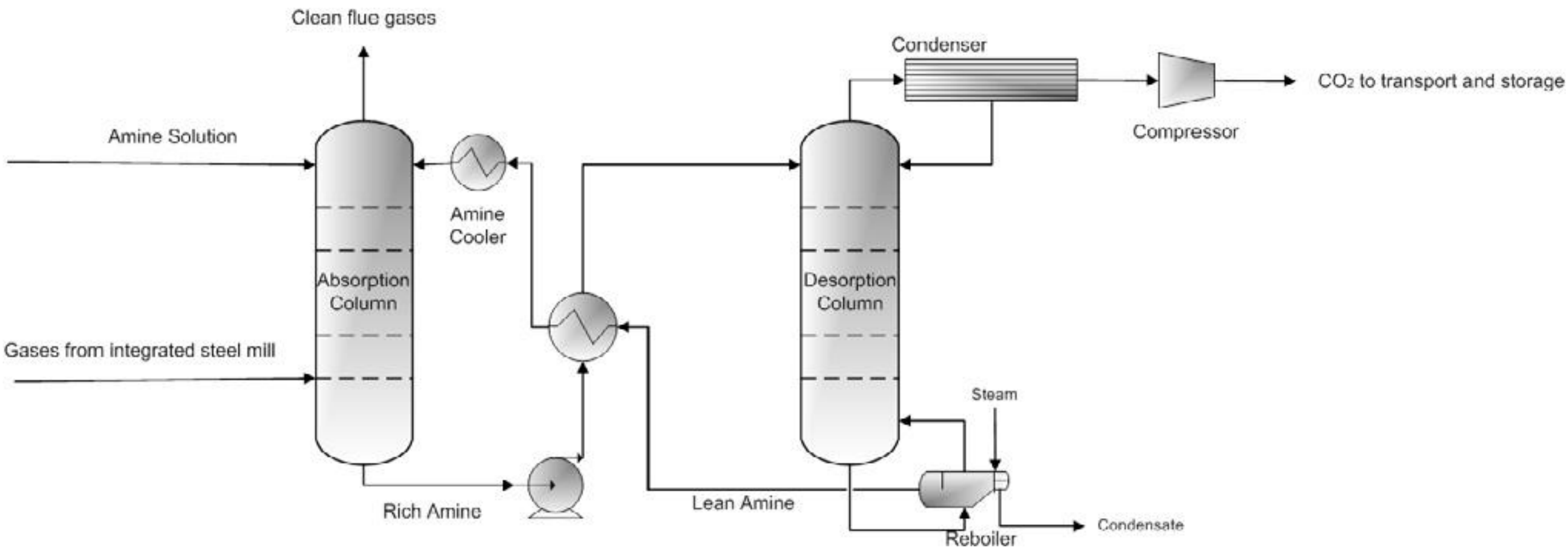
# Usual blast furnace and top gas recycle blast furnace

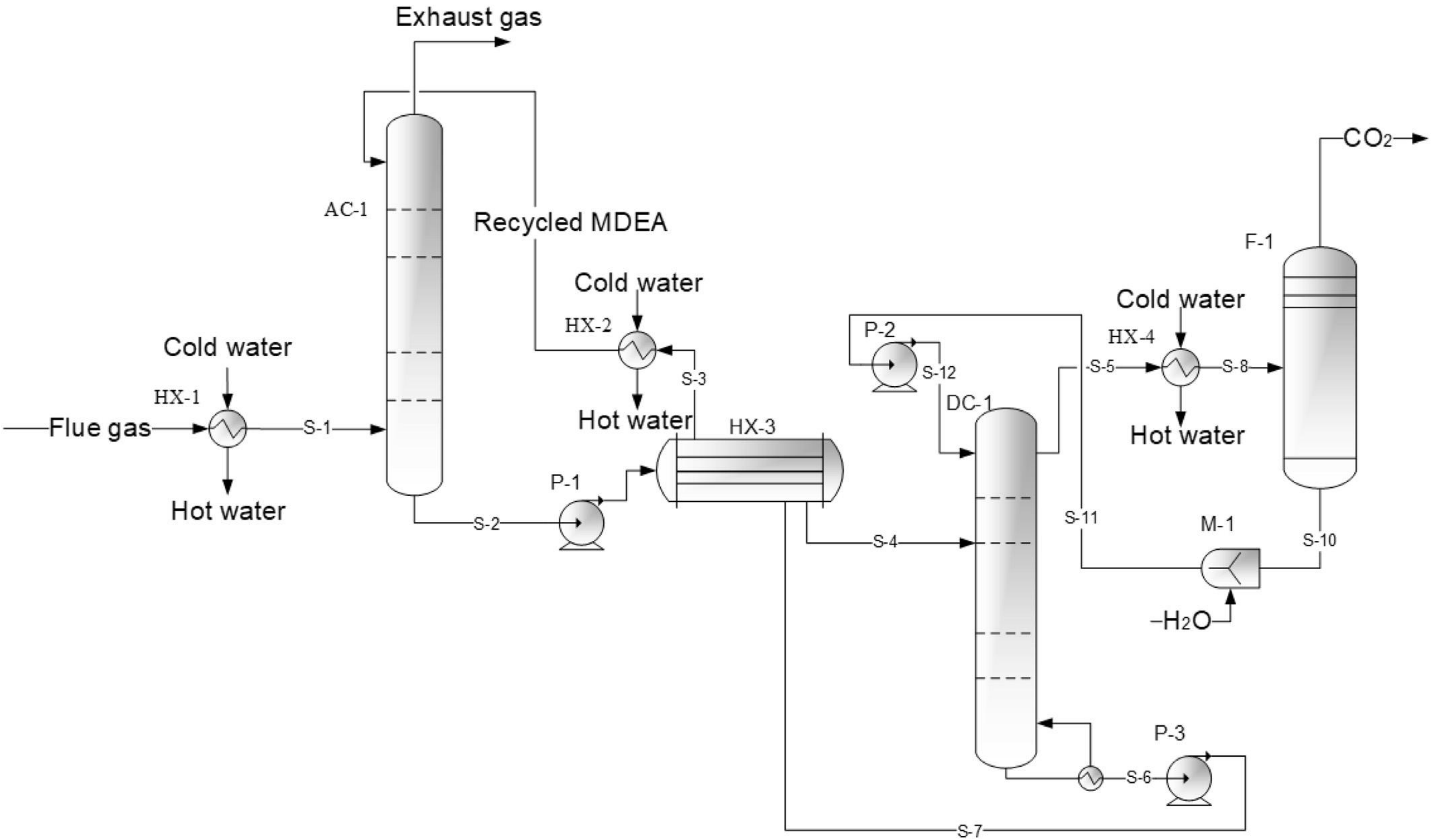


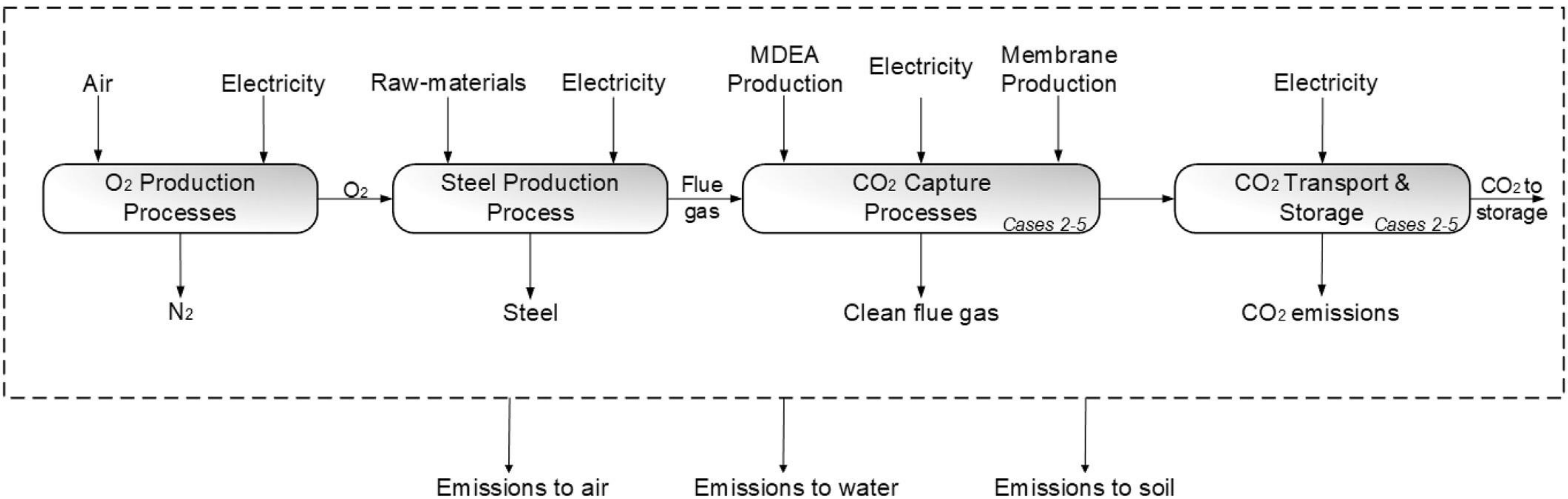
# Options for $CO_2$ capture



# Integrated steelmaking with $CO_2$ capture




Exhaust gas
AC-1
Recycled MDEA
Cold water
HX-2
Cold water
HX-1
Flue gas
S-1
Hot water
S-3
Hot water
HX-3
P-1
S-2
S-4
F-1
$CO_2$
Cold water
HX-4
P-2
S-12
S-5
S-8
DC-1
Hot water
S-11
M-1
S-10
$H_2O$
P-3
S-6
S-7



Air
Electricity
Raw-materials
Electricity
MDEA Production
Electricity
Membrane Production
Electricity
O2 Production Processes
O2
Steel Production Process
Flue gas
CO2 Capture Processes
Cases 2-5
CO2 Transport & Storage
Cases 2-5
CO2 to storage
N2
Steel
Clean flue gas
CO2 emissions
Emissions to air
Emissions to water
Emissions to soil

coarse 50 - secondary and tertiary amines (2.7 GJ/tonnes)    ULCOS - ultra low carbon